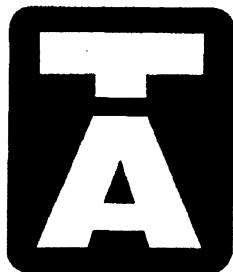
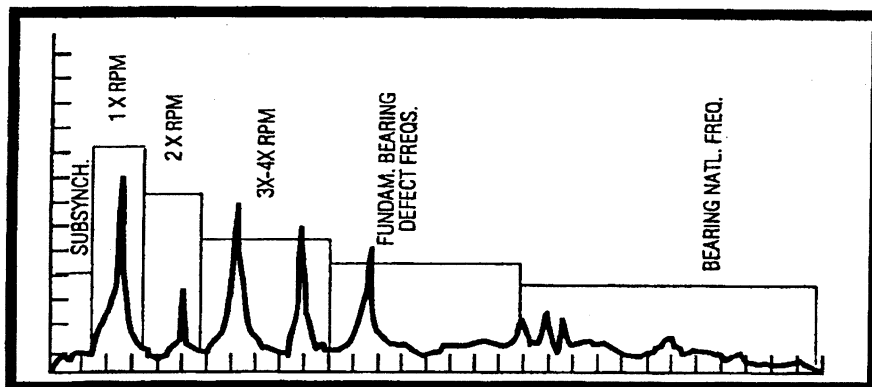


CHAPTER 10



ACTUAL CASE HISTORIES OF DIAGNOSTICS ON VARIOUS MACHINE TYPES



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SPECIALISTS IN PREDICTIVE MAINTENANCE, MACHINERY DIAGNOSTICS, AND VIBRATION REDUCTION

ACTUAL CASE HISTORIES OF VIBRATION DIAGNOSTICS ON VARIOUS MACHINE TYPES

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DETECTION AND CORRECTION OF FAULTS ON AIR HANDLER HAVING PROBLEMS DUE TO UNBALANCE, SHEAVE MISALIGNMENT, DEFECTIVE BELTS, RESONANCE AND A CRACKED FRAME SUPPORTING THE OUTBOARD FAN PILLOWBLOCK BEARING

I. BACKGROUND

The assistance of Technical Associates was requested to help resolve continuing failure problems on the belt driven air handler driven by a 200 HP motor pictured in Figure 1. Here, the motor operated at a speed of 1185 RPM while the fan speed was approximately 427 RPM. For several years, this fan had experienced failures, the most dramatic of which had been three instances where the fan shaft actually broke in half. In addition, both the motor and fan had experienced continuing bearing and belt failure problems.

Upon arrival at the plant site, Technical Associates inspected some of the failed components of this machine including two of the broken shafts. Close inspection of the failed shafts did not show a distinct fatigue failure pattern either due to bending or shear. However, the races particularly of the fan bearings tended to indicate misalignment with most of the loading concentrated on one side of the race. Following below will be a summary of the results found during the analysis, recommendations made and a look at the improvement in vibration spectra before and after corrective actions were taken.

II. RESULTS AND CONCLUSIONS

Vibration measurements were taken on each of the motor and fan bearing housings pictured in Figure 1. Plot 1 taken on the outboard motor bearing in the axial direction (Position 1A in Figure 1) showed high axial vibration with an overall level of .632 in/sec. Surprisingly, the motor running speed vibration at about 1192 CPM was only .090 in/sec. However, the dominant vibration on the motor was at 427 CPM with an excessive magnitude of .585 in/sec. Of course, the source of the 427 CPM vibration was the operating speed of the fan. Thus, this was a classic case often indicating sheave misalignment which will often generate excessive vibration at fan running speed in the axial direction when making measurements on the motor.

Next, another surprising spectrum is shown in Plot 2 which was taken in the axial direction on the sheave-side fan bearing at Position 3A (see Figure 1).

Here, the vibration at 427 fan speed was low, but there was a very pronounced frequency at only 300 CPM dominating the vibration. Since this was suspected to be a frequency generated by belt speed (or multiple), the client was requested to review the purchase order for the motor and fan sheaves to confirm their pitch diameters as well as confirming the length of the belts. After this information was given, the belt speed (RPM) was calculated to be about 149.5 RPM (see Note 1). Therefore, this revealed the source of the 300 CPM vibration to be 2X belt defect frequency. It is not

unusual for defects on belts to first generate a frequency at 2X belt speed (since the belt contacts two sheaves per revolution). However, what was unusual was the significant amplitude that resulted here at Position 3A in Plot 2. Therefore, the client was requested to stop the air handler to allow performance of impulse natural frequency tests. Such natural frequency testing was performed on each of the motor and fan bearing housings in horizontal, vertical and axial directions. Not surprisingly, when the test was performed on the sheave-side bearing in the axial direction, a pronounced axial natural frequency was found at 315 CPM as shown in Plot 3 (or only 5% away from the 300 CPM peak caused by 2X belt defect frequency). Therefore, even when defects would occur in the belts, their vibration would be greatly amplified by as much as 10 to 30 times higher due to resonance. And, if the belts suffered significant wear, resultant axial amplitudes would likely increase to highly unacceptable levels.

Next, the spectrum shown in Plot 4 indicated unbalance of the fan wheel as evidenced by a high amplitude .427 in/sec vibration at 427 RPM fan speed with the data taken on the same sheave-side bearing, but at Position 3H in the horizontal direction. Interestingly, note that none of the vibration at 2X belt defect frequency appeared whatsoever in the spectrum taken in this direction - another indication of the significant resonance problem this fan had in its axial direction. Phase analysis data was captured to ascertain the cause of the high 1X RPM fan vibration. This phase data is shown on Figure 1. Looking at the data taken on Positions 3 and 4, note that the phase difference in the horizontal direction between the outboard and inboard fan bearings (Positions 3 and 4) was approximately 110° while the vertical phase difference was approximately 130° between these two bearings. Since the phase difference was approximately the same in both the horizontal and vertical directions, and due to the high vibration 1X RPM, fan wheel unbalance was suspected.

NOTES:

$$1. \text{ Belt Speed} = \frac{\pi \times \text{Pulley RPM} \times \text{Pulley Pitch Diameter}}{\text{Belt Length}}$$

(If Fan Sheave Pitch Diameter is entered, then Fan Sheave RPM must be also; and vice versa for the motor sheave.)

III. RECOMMENDATIONS

- A. It was recommended that the belts be replaced and a close inspection be made of both the motor and fan sheaves to confirm they were in good condition. After replacing the belts (and sheaves if necessary), it was recommended to precision align the motor and fan sheaves with respect to one another.

B. It was also recommended that the frame of the fan be stiffened in the axial direction to raise its natural frequency above 315 CPM which would come very close to 2X belt speed when future belt defects did appear. However, it was cautioned that the natural frequency should not be raised to next coincide with either 3X or 4X belt speed. In other words, when stiffening attempts were made, an FFT analyzer was recommended to be on hand to ensure to what frequencies the frame natural frequency would be shifted.

C. After completion of recommendations A and B, it was recommended that a fan wheel balance should be performed in place to significantly lower the vibration at 1X fan speed below the .427 in/sec measured on May 18.

IV. RESULTS OF CORRECTIVE ACTIONS

Corrective actions were attempted by Technical Associates on the same test day (May 18) as well as after this test. First, a hydraulic jack was placed in the axial direction on the fan frame in an attempt to temporarily shift the axial natural frequency away from 2X belt speed (and below 3X belt speed). However, the frame itself was of insufficient cross section to provide the necessary restraint to noticeably shift the natural frequency. Therefore, it was recommended on May 18 that the center distance between the motor and fan sheaves be changed which would not shift the frame stiffness or natural frequency, but would move the belt speed away from coincidence with the axial natural frequency.

Follow-up tests were performed on June 22 after the belts had been replaced and the motor had been moved further away from the fan, thereby increasing the center distance. This increase in center distance lowered the belt speed from about 150 RPM down to approximately 125 RPM. Therefore, now neither 2X belt speed (250 CPM) nor 3X belt speed (375 CPM) would be close to the fan frame axial resonance of about 315 CPM. Next, a fan wheel balance was performed in-place by Technical Associates on June 22. Plots 5 and 6 show the results of this balancing exercise on the inboard (Pos. 3H) and outboard (Pos. 4H) fan bearing. Even though the fan wheel showed noticeable wobble when it was slow-rolled, Plot 5 taken on Position 3H showed the field balance lowered vibration at fan speed from .444 in/sec down to .052 in/sec; while Plot 6 taken on Position 4H showed that it lowered the 1X RPM level from .365 in/sec down to only .052 in/sec.

However, please note the peak at 2X fan running speed in Plot 6 taken on Position 4H on the outboard fan bearing. Note that its level did not decrease (in fact, slightly increased to .241 in/sec) after the balance exercise. An examination of the frame supporting the outboard fan pillowblock bearing revealed a crack in the frame itself. This was pointed out to the client personnel who were advised to either replace and/or weld this frame.

V. LATER FOLLOW-UP AFTER WELDING OF THE OUTBOARD FAN PILLOWBLOCK FRAME

Plot 7 shows the spectrum taken after the frame supporting the outboard pillowblock fan bearing was welded. It lowered the vibration at 2X RPM from .241 in/sec down to .078 in/sec. Note that this spectrum was taken some time after the original corrective actions and that the vibration at 1X fan speed had now increased somewhat up to .193 in/sec. A check with the client showed that this was likely to buildup on the fan wheel. Still, a level less than .20 in/sec at 1X RPM on this fan wheel was quite acceptable and the noticeable drop in 2X RPM vibration was beneficial.



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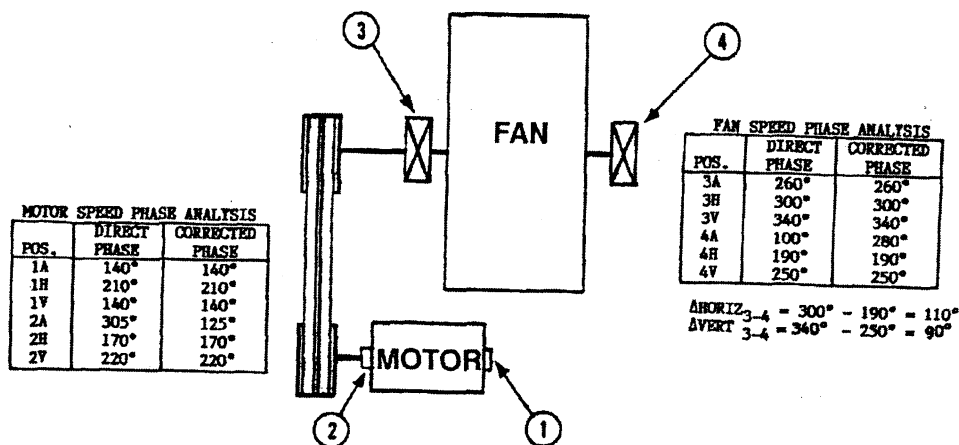
UNBALANCE DETECTION & CORRECTION BY A REPUTABLE FIRM

MACHINERY DATA
GUIDE SHEET

RELIABILITY SURVEY

predictive maintenance program

(Page ___ of ___)



MACHINE IDENT: L-12 AIR HANDLER

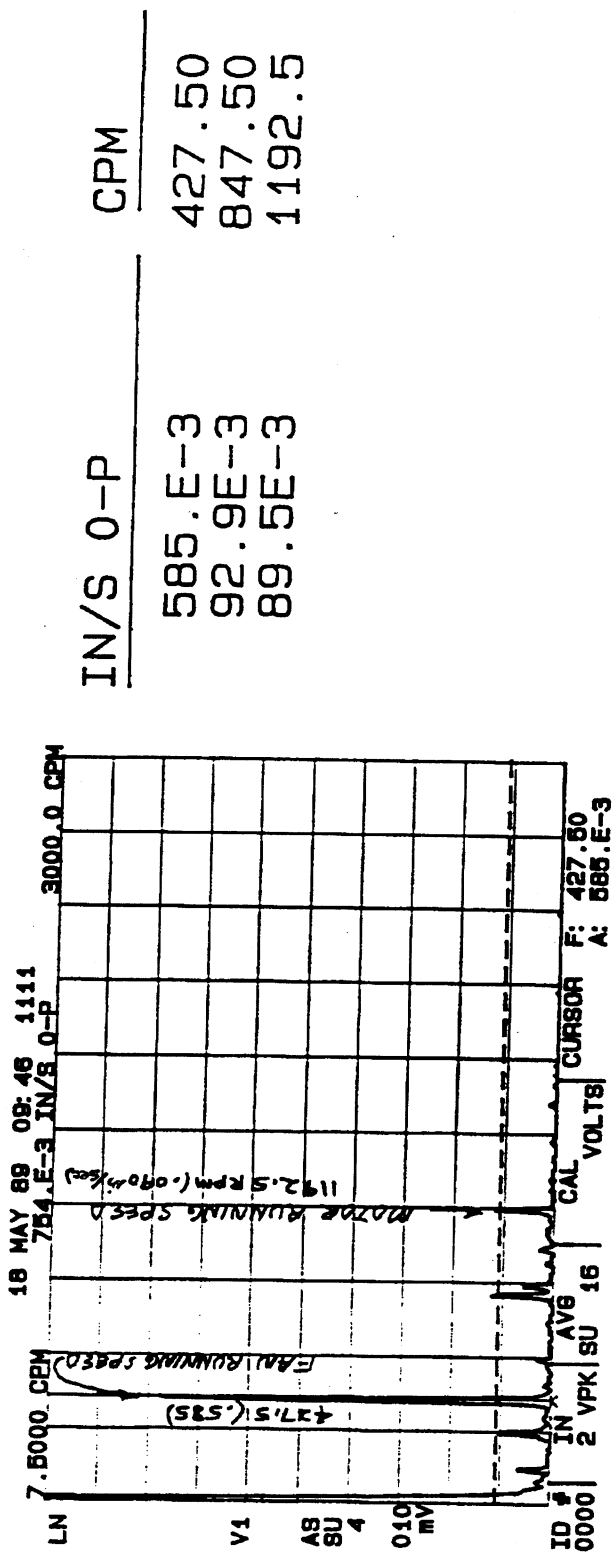
AREA: _____

ROUTE: _____

DRIVER	INTERMEDIATE	DRIVEN
RPM min : <u>1185</u>	RPM in : _____	RPM min : <u>427.5</u>
max : _____	out : _____	max : _____
MFGR : _____	RATIO: _____	MFGR. : _____
MODEL : _____	MODEL : _____	MODEL : _____
SER NO. : _____	SER NO. : _____	SER NO. : _____
V _____ P _____ A _____	RATING : _____	RATING : _____
SHEAVE : _____ DIA.	BELT # : _____	#BLADES : _____
BRG#: _____ (I)	BRG#: _____ (I)	BRG#: <u>NU322</u> (I)
BRG#: _____ (O)	BRG#: _____ (O)	BRG#: <u>6320</u> (O)
H.P.: <u>200</u>	GEAR T: _____ CT.	GEAR T: _____ CT.
OTHER: _____	OTHER: _____	OTHER: _____

R-112492-1

FIGURE 1



OVERALL = .632 in/sec

PEAK PICK ON TRACE 1

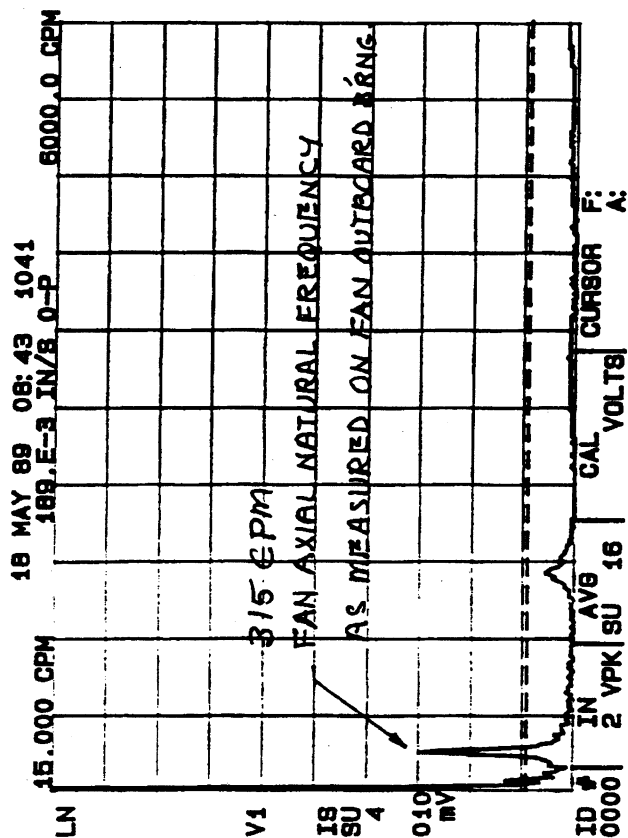
24-MAY-89

**MOTOR VIBRATION
AXIALLY ON OUTBOARD END
POSITION 1A**

13:17:04

PLOT 1

L-12AH



PEAK PICK ON TRACE 1

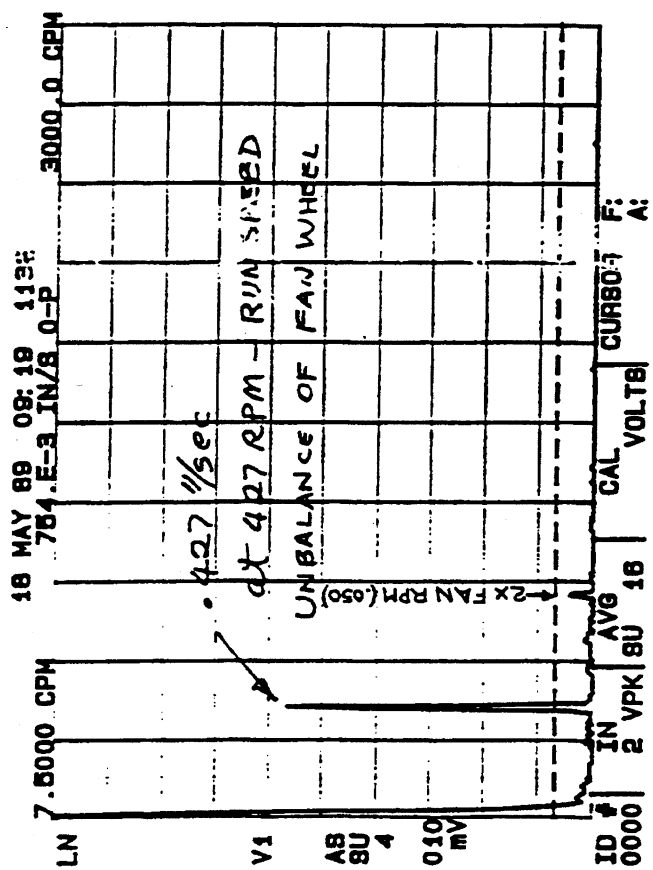
IN/S 0-P	CPM
25.0E-3	90.000
57.0E-3	315.00

24-MAY-89

IMPULSE TEST ON FAN OUTBOARD BEARING
POSITION 3A - IMPULSE IN AXIAL DIRECTION
PLOT 3

12: 36: 05

L-12AH



IN/S 0-P	CPM
427.E-3	427.50

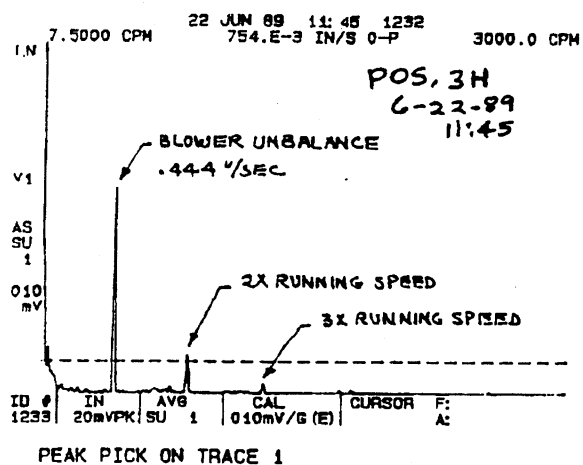
PEAK PICK ON TRACE 1

24-MAY-89

12: 55: 56

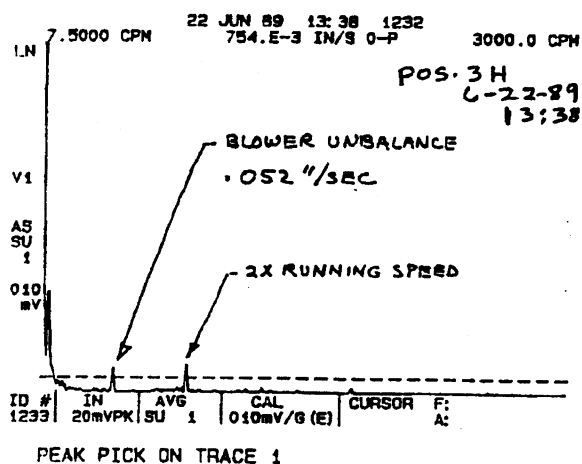
INBOARD FAN BEARING - HORIZONTAL
POSITION 3H
PLOT 4

L-12 AH



IN/S 0-P	CPM
99.8E-3	45.000
444.E-3	427.50
82.6E-3	855.00

BEFORE
BALANCING



IN/S 0-P	CPM
217.E-3	52.500
51.6E-3	427.50
58.5E-3	855.00

AFTER
BALANCING

22-JUN-89

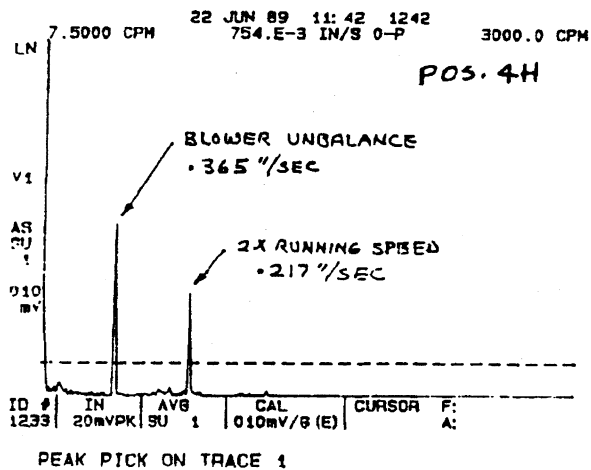
15:34:00

UPPER PLOT = BEFORE BALANCING
.444"/sec

L-12 FAN

LOWER PLOT = AFTER BALANCING
.052"/sec

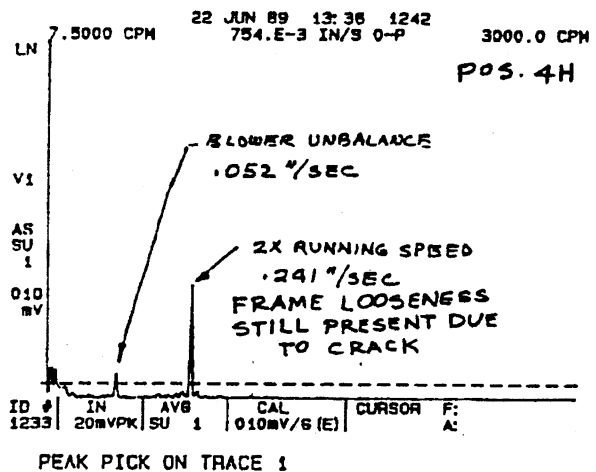
PLOT 5



10

IN/S 0-P	CPM
365.E-3	427.50
217.E-3	855.00

BEFORE
BALANCING



IN/S 0-P	CPM
65.4E-3	60.000
62.0E-3	82.500
51.6E-3	427.50
241.E-3	855.00

AFTER
BALANCING

22-JUN-89

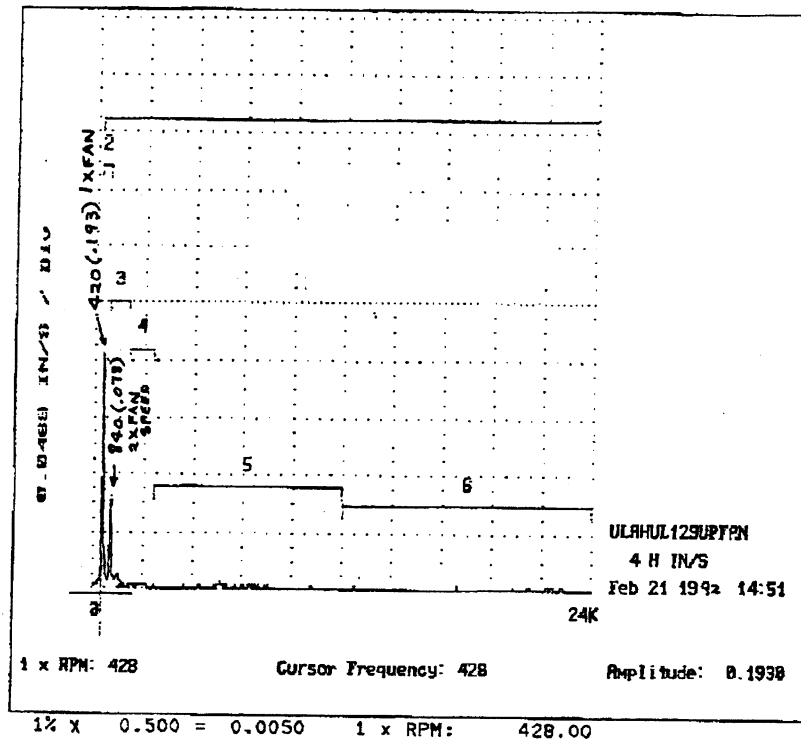
15:31:17

UPPER PLOT = BEFORE BALANCING
.365"/sec

LOWER PLOT = AFTER BALANCING
.052"/sec

L-12 FAN

PLOT 6



Machine	Pos	Dir	Units	Date
ULAHUL12SUPFAN	4	H	IN/S	Feb 21 1980 14:51
Frequency	1 x RPM	Amplitude		
420	0.98	0.1930		
840	1.96	0.0775		
1260	2.94	0.0125		
6180	14.44	0.0060		

AFTER WELDING FRAME
SUPPORTING OUTBOARD FAN
PILLOWBLOCK BEARING

PLOT 7

T - Threshold adjust Q - Quit

DETECTION & CORRECTION OF UNBALANCE ON AN AIR CONDITIONER SUPPLY FAN

I. BACKGROUND

An HVAC contractor had been having vibration and associated noise difficulties with one of their customer's HVAC circulating fans. The contractor had changed bearings, adjusted belts, etc., without any success in improving vibration or noise levels and requested that Technical Associates perform a diagnostic investigation to determine the problem source.

II. RESULTS

The unit was belt driven as shown in Figure 1 with one (1) center mounted squirrel cage fan and one (1) overhung squirrel cage fan. The motor rotated at 1738 RPM and the fan at 738 RPM. All the spectra collected were dominated by a 720 CPM vibration with the highest amplitudes (axial and horizontal) at Position 4 on the fan shaft. It was determined that fan imbalance was the problem, and since Position 4 had the highest amplitude, balancing the overhung fan would produce the best results. What followed was a successful single plane balance with measurements from the Position 4 bearing. The weights were added at the inboard end of the overhung fan wheel on its backing plate close to bearing 4. Vibration levels at Position 4 fell from 1.74 in/sec at 720 CPM (fan rotational speed) to .035 in/sec at 720 CPM after the balance as seen by the "Before" and "After" spectra of Figure 2. The axial vibration at Position 4 also improved from 1.21 in/sec "Before" the balance to .448 in/sec "After" as seen in the Figure 3 spectra. The remaining axial response was attributed to:

1. Inadequate axial support
2. The variable pitch motor sheave
3. Sheave misalignment.

Noise levels also fell with the improved vibration.

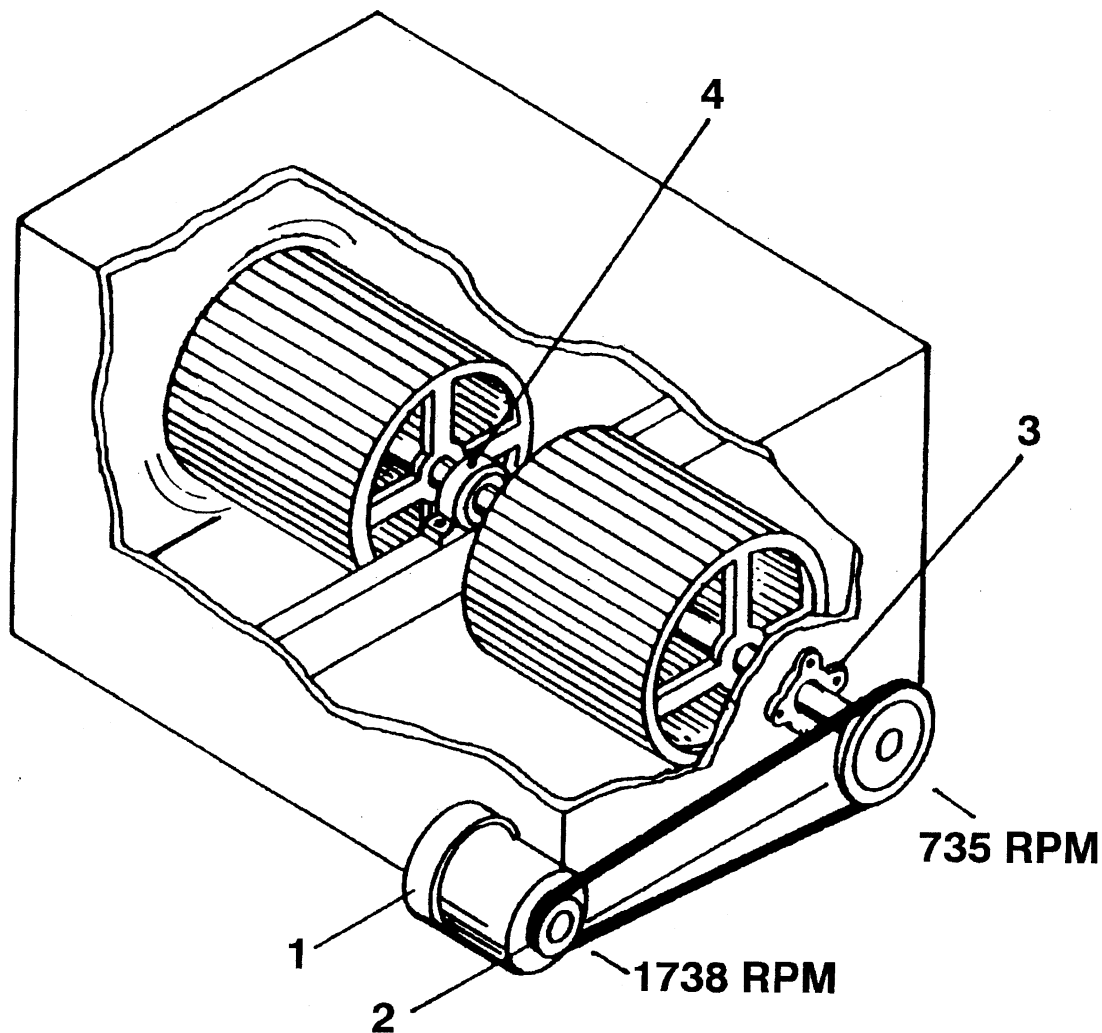
III. CONCLUSIONS

1. The dominant problem was fan imbalance (168 grams of correction weights added to the outboard fan reduced unbalance to acceptable levels).
2. The axial response that remained after the balance was likely a result of:
 - a. Structural flexing at the bearing supports
 - b. The variable pitch motor sheave
 - c. Sheave/belt misalignment.

3. Noise levels improved along with the vibration.
4. The simple single plane balance procedure worked well in correcting a significant imbalance in a complicated fan arrangement.

IV. RECOMMENDATIONS

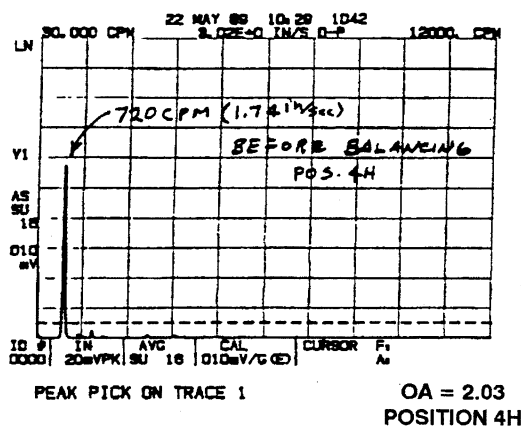
1. Replace the motor's variable pitch sheave with a fixed diameter sheave of the proper size.
2. Align and tension the belts to specification.



SQUIRREL CAGE FAN ARRANGEMENT
FIGURE 1

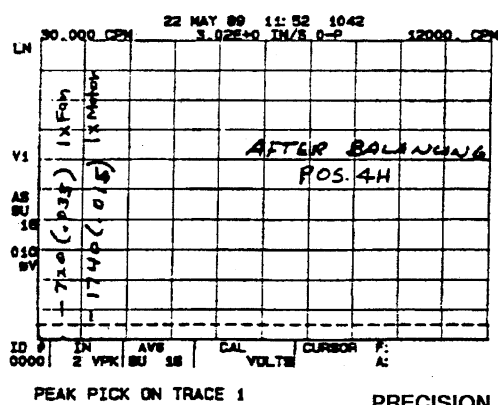
CF10

R-9704-1



23-MAY-89

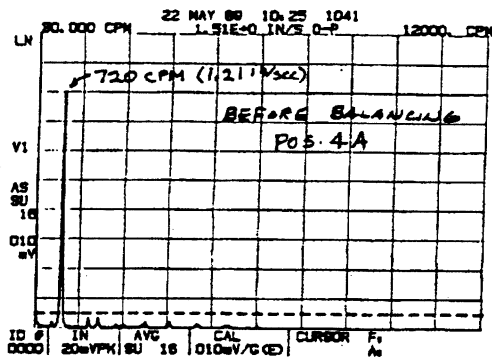
16:33:40



24-MAY-89

09:26:19

POSITION 4
HORIZONTAL "BEFORE" AND "AFTER" BALANCE SPECTRA
FIGURE 2

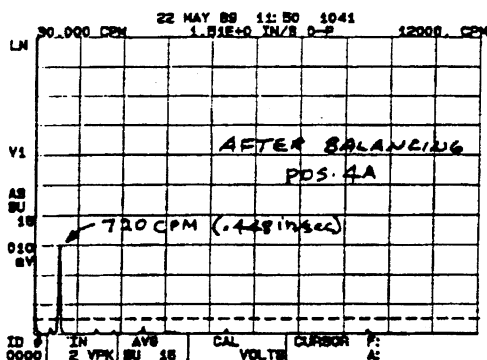


IN/S 0-P	CPM
1.21	720.00

PEAK PICK ON TRACE 1
OA = 1.42
POSITION 4A

23-MAY-89

16:32:14



IN/S 0-P	CPM
448.E-3	720.00

PEAK PICK ON TRACE 1

- 1 INADEQUATE AXIAL SUPPORT BY FRAME (Esp. at IB Fan Brg.)
- 2 VARIABLE PITCH MOTOR SHEAVE IS PROBLEM
- 3 SHEAVES HAVE SOME MISALIGNMENT

24-MAY-89

RECOMMEND FIXED SHEAVE

09:25:34

POSITION 4
AXIAL "BEFORE" AND "AFTER" BALANCE SPECTRA
FIGURE 3

EFFECT OF COUPLING MISALIGNMENT ON VIBRATION OF A WATER SCRUBBER PUMP

I. BACKGROUND

During a July 13, 1989 predictive maintenance vibration survey by Technical Associates, a horizontally mounted centrifugal water pump (1-1 Scrubber Pump) was detected as having a sudden axial vibration increase. See Figure 1 for the machine arrangement. The increase is shown in the trend graphs and waterfall plots of Figures 2, 3 and 4 taken on Positions 1A and 2A as shown. Notice that the waterfall spectra show the dominant vibration was at 1X RPM (nominal 1800 RPM). Since the motor was having other problems, our client decided to replace the motor instead of just aligning it. A new coupling was unavailable at that time, so the old coupling was reused. The next survey of 9/7/89 had reduced axial measurements indicating the unit was well aligned as seen by the trends and waterfall plots of Figures 2 and 4. The axial vibration remained stable until the October 5, 1989 survey. This survey found the axial motor vibration had again increased to levels in excess of 1.0 in/sec as seen in Figures 2, 3 and 4. This time, the problem was found to be caused by excessive wear of the "used" coupling. After the coupling was replaced and the unit aligned, vibration levels again fell to acceptable levels as seen in the 11/9/89 measurements of Figure 5 and the trends and spectra of Figures 2 and 3.

II. CONCLUSIONS

- A. 1-1 Scrubber Pump's first misalignment condition resulted from a motor problem and the second from a coupling problem.
- B. Vibration measurements after the final alignment indicate the unit was well aligned.

III. RECOMMENDATIONS

- A. Recommend continuing the periodic vibration measurements on a monthly basis.

IV. RESULTS OF CORRECTIVE ACTIONS

- A. The final alignment with the new coupling has remained stable, at least through November, 1991 as seen in the Trend Graphs of Figures 6 and 7.



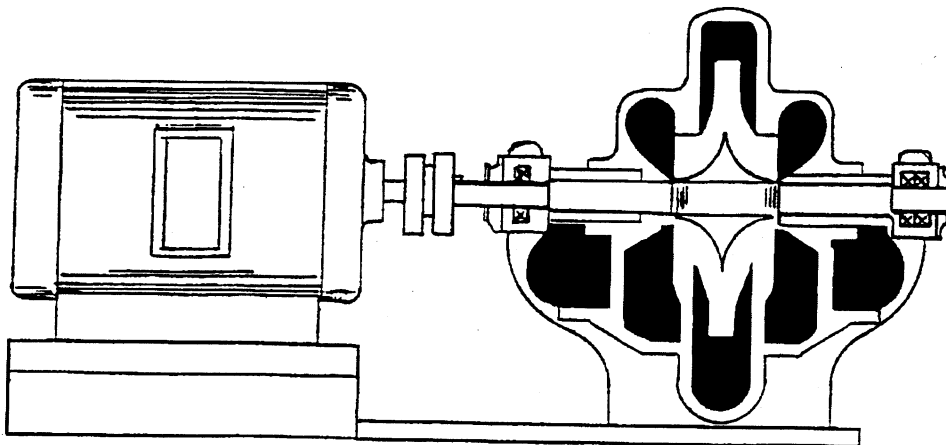
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MACHINERY DATA GUIDE SHEET

RELIABILITY SURVEY

predictive maintenance program

(Page ___ of ___)



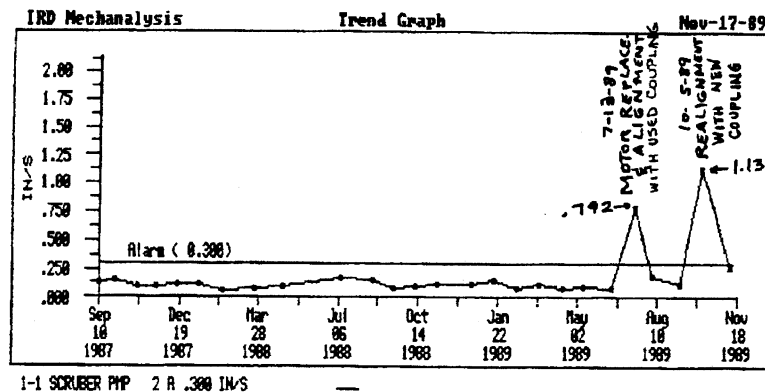
MACHINE IDENT: 1-1 SCRUBBER PUMP

AREA: _____

ROUTE: 7

DRIVER	INTERMEDIATE	DRIVEN
RPM min : _____	RPM in : _____	RPM min : _____
max : <u>1770</u>	out : _____	max : _____
MFGR : <u>TOSHIBA</u>	RATIO: _____	MFGR. : <u>WORTHINGTON</u>
MODEL : <u>B2004FKF4BD</u>	MODEL : _____	MODEL : <u>10LR15</u>
SER NO. : <u>4903316</u>	SER NO. : _____	SER NO. : <u>60-740830</u>
<u>V 460 P 3 A 230</u>	RATING : _____	RATING : _____
SHEAVE : _____ DIA.	BELT # : _____	#BLADES : _____
BRG#: _____ (I)	BRG#: _____ (I)	BRG#: _____ (I)
BRG#: _____ (O)	BRG# _____ (O)	BRG#: _____ (O)
H.P.: <u>200</u>	GEAR T: _____ CT.	GEAR T: _____ CT.
OTHER: _____	OTHER: _____	OTHER: _____

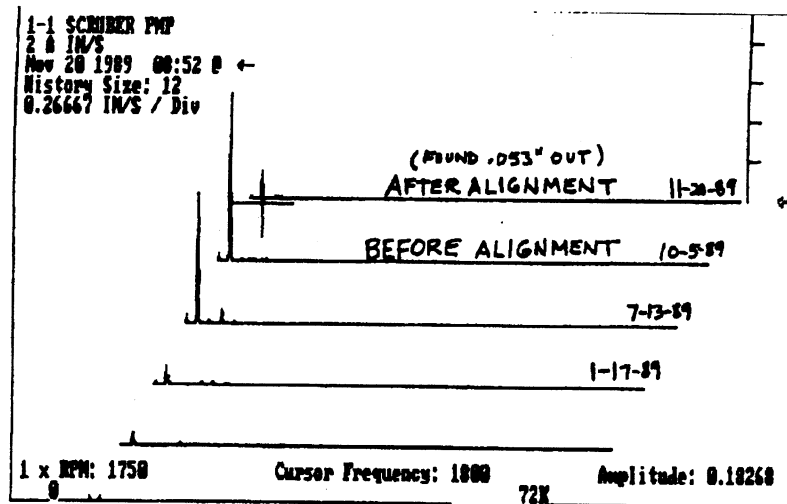
FIGURE 1



IRD Mechanalysis Data Analysis System

Machine	1-1 SCRUBER PMP	Pos	2	Dir	A	Units	IN/S	Alarm	.300
Ampl	Insp	Date	Time	Ampl	Insp	Date	Time		
	Code	mm/dd/yy			Code	mm/dd/yy			
.134	0	09-10-87	0:00	.114	0	11-07-88	12:16		
.159	0	09-30-87	0:00	.122	0	12-20-88	12:46		
.099	0	10-28-87	0:00	.147	0	01-17-89	11:05		
.099	0	11-20-87	0:00	.075	0	02-14-89	10:47		
.114	0	12-16-87	0:00	.117	0	03-14-89	10:44		
.114	0	01-11-88	0:00	.078	0	04-13-89	12:00		
.066	0	02-11-88	0:00	.103	0	05-09-89	9:33		
.084	0	03-23-88	0:00	.078	0	06-13-89	12:57		
.094	0	04-27-88	0:00	.792	0	07-13-89	11:34		
.164	0	07-09-88	0:00	.190	0	08-03-89	13:10		
.155	0	08-18-88	12:43	.116	0	09-07-89	10:49		
.085	0	09-13-88	15:12	1.13	0	10-05-89	10:51		
.099	0	10-11-88	12:46	.258	0	11-09-89	12:20		

FIGURE 2



2% x 2.0000 = 0.04000 1 x RPM: 1750.00

Machine	Pos	Dir	Units	Date
1-1 SCRUBER PMP	2	A	IN/S	Nov 20 1989 08:52

Frequency	1 x RPM	Amplitude
1800	1.03	0.18268

2% x 2.0000 = 0.04000 1 x RPM: 1750.00

Machine	Pos	Dir	Units	Date
1-1 SCRUBER PMP	.2	A	IN/S	Oct 05 1989 10:51

Frequency	1 x RPM	Amplitude
180	0.10	0.04988
1800	1.03	1.09992

2% x 2.0000 = 0.04000 1 x RPM: 1750.00

Machine	Pos	Dir	Units	Date
1-1 SCRUBER PMP	2	A	IN/S	Jul 13 1989 11:34

Frequency	1 x RPM	Amplitude
180	0.10	0.04988
1800	1.03	0.82986
5400	3.09	0.08483

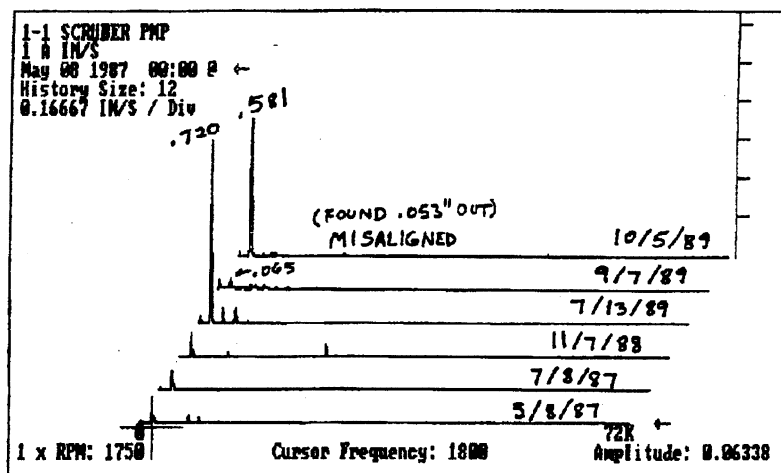
2% x 2.0000 = 0.04000 1 x RPM: 1750.00

Machine	Pos	Dir	Units	Date
1-1 SCRUBER PMP	2	A	IN/S	Jan 17 1989 11:05

Frequency	1 x RPM	Amplitude
1800	1.03	0.10898

FIGURE 3

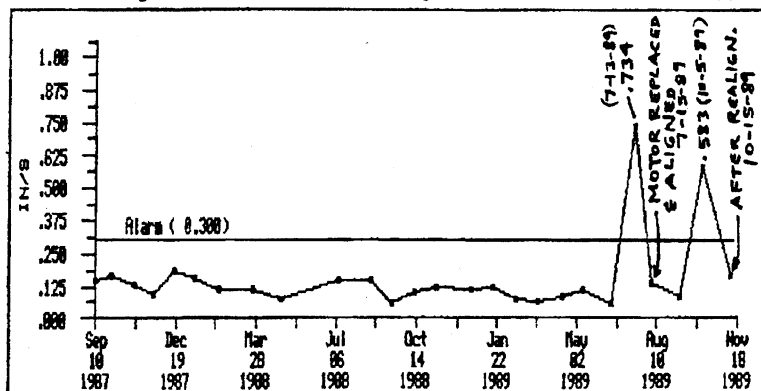
EFFECT ON VIBRATION OF ALIGNMENT



IRD Mechanalysis

Trend Graph

Nov-17-89



1-1 SCRUBER PMP 1 A .300 IN/S

IRD Mechanalysis Data Analysis System

Machine	1-1 SCRUBER PMP	Pos	1	Dir	A	Units	IN/S	Alarm	.300
Ampl	Insp	Date	Time	Ampl	Insp	Date	Time		
	Code	mm/dd/yy			Code	mm/dd/yy			
.144	0	09-10-87	0:00	.124	0	11-07-88	12:15		
.164	0	09-30-87	0:00	.111	0	12-20-88	12:45		
.129	0	10-28-87	0:00	.117	0	01-17-89	11:04		
.094	0	11-20-87	0:00	.077	0	02-14-89	10:47		
.184	0	12-16-87	0:00	.068	1	03-14-89	10:43		
.159	0	01-11-88	0:00	.083	0	04-13-89	12:00		
.109	0	02-11-88	0:00	.114	0	05-09-89	9:33		
.109	0	03-23-88	0:00	.061	0	06-13-89	12:57		
.079	0	04-27-88	0:00	.734	0	07-13-89	11:33		
.144	0	07-09-88	0:00	.140	0	08-03-89	13:09		
.145	0	08-18-88	12:42	.085	0	09-07-89	10:49		
.060	0	09-13-88	15:11	.583	0	10-05-89	10:49		
.099	0	10-11-88	12:45	.171	0	11-09-89	12:19		

FIGURE 4

Machine Report (Latest Data) for: 1-1 SCRUBER FMP

Date	Time	Pos	Dir	Ampl	Units	Change	Alarm	Inspection
Nov-09-89	12:19	1	A	.171	IN/S	-71%	.300	
Nov-09-89	12:19	1	H	.079	IN/S	-64%	.300	
Nov-09-89	12:19	1	H	.262	g/SE	134%	.600	←
Nov-09-89	12:19	1	V	.071	IN/S	-49%	.300	
Nov-09-89	12:20	2	A	.258	IN/S	-77%	.300	
Nov-09-89	12:20	2	H	.061	IN/S	-73%	.300	
Nov-09-89	12:20	2	H	.175	g/SE	-8%	.600	←
Nov-09-89	12:20	2	V	.174	IN/S	-61%	.300	
Nov-09-89	12:20	3	A	.202	IN/S	2%	.300	
Nov-09-89	12:20	3	H	.116	IN/S	10%	.300	
Nov-09-89	12:20	3	H	.228	g/SE	24%	.600	←
Nov-09-89	12:20	3	V	.156	IN/S	47%	.300	
Nov-09-89	12:21	4	A	.170	IN/S	-1%	.300	
Nov-09-89	12:21	4	H	.112	IN/S	-7%	.300	
Nov-09-89	12:21	4	H	.127	g/SE	7%	.600	←
Nov-09-89	12:21	4	V	.118	IN/S	-2%	.300	

Arrows Show Little Improvement
in Spike Energy as Expected

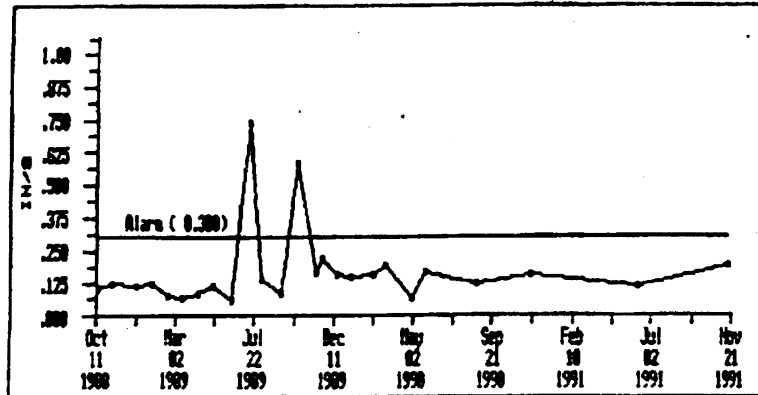
16 lines this report

FIGURE 5

IRD Mechanalysis

Trend Graph

Jan/84/92



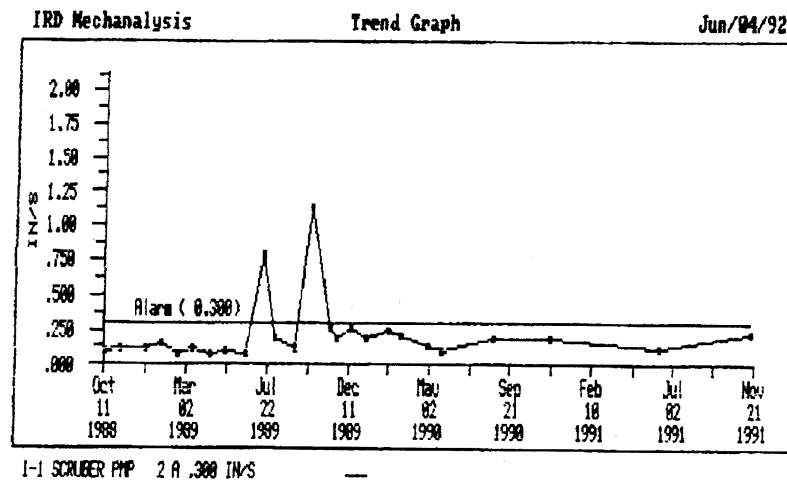
1-1 SCRUBER PMF 1A .300 IN/S

IRD Mechanalysis Data Analysis System

Machine 1-1 SCRUBER PMF Pos 1 Dir A Units IN/S Alarm .300

Ampl	Insp Code	Date mm/dd/yy	Time	Ampl	Insp Code	Date mm/dd/yy	Time
.099	0	10/11/88	12:45	.171	0	11/09/89	12:19
.124	0	11/07/88	12:15	.218	0	11/20/89	8:50
.111	0	12/20/88	12:45	.160	0	12/15/89	10:08
.117	0	01/17/89	11:04	.151	0	01/11/90	9:01
.077	0	02/14/89	10:47	.154	0	02/20/90	8:21
.068	1	03/14/89	10:43	.194	0	03/13/90	8:19
.083	0	04/13/89	12:00	.069	0	04/30/90	7:43
.114	0	05/09/89	9:33	.168	0	05/24/90	10:36
.061	0	06/13/89	12:57	.168	0	05/24/90	10:36
.734	0	07/13/89	11:33	.125	0	08/23/90	11:34
.140	0	08/03/89	13:09	.156	0	11/29/90	10:59
.085	0	09/07/89	10:49	.110	0	06/06/91	10:51
.583	0	10/05/89	10:49	.190	0	11/14/91	9:04

FIGURE 6

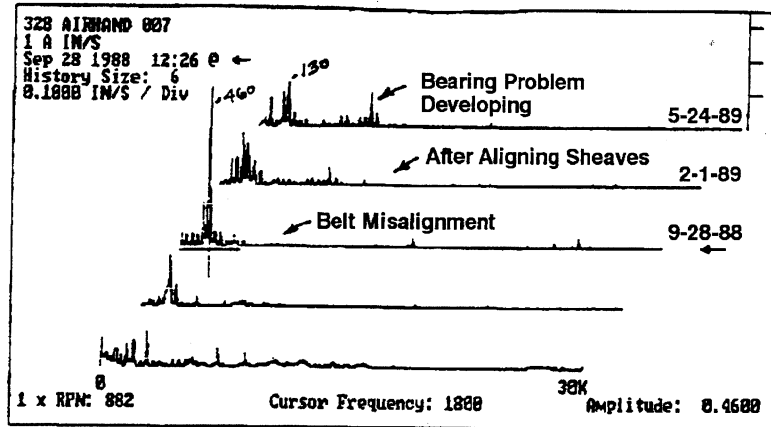


IRD Mechanalysis Data Analysis System

Machine	1-1 SCRUBER PMP	Pos	2	Dir	A	Units	IN/S	Alarm	.300
Ampl	Insp	Date	Time	Ampl	Insp	Date	Time		
	Code	mm/dd/yy			Code	mm/dd/yy			
.099	0	10/11/88	12:46	.258	0	11/09/89	12:20		
.114	0	11/07/88	12:16	.187	0	11/20/89	8:52		
.122	0	12/20/88	12:46	.257	0	12/15/89	10:08		
.147	0	01/17/89	11:05	.187	0	01/11/90	9:01		
.075	0	02/14/89	10:47	.244	0	02/20/90	8:22		
.117	0	03/14/89	10:44	.199	0	03/13/90	8:20		
.078	0	04/13/89	12:00	.134	0	04/30/90	7:45		
.103	0	05/09/89	9:33	.103	0	05/24/90	10:37		
.078	0	06/13/89	12:57	.103	0	05/24/90	10:37		
.792	0	07/13/89	11:34	.194	0	08/23/90	11:35		
.190	0	08/03/89	13:10	.194	0	11/29/90	10:59		
.116	0	09/07/89	10:49	.113	0	06/06/91	10:32		
1.13	0	10/05/89	10:51	.228	0	11/14/91	9:04		

FIGURE 7

DETECTION & CORRECTION OF SHEAVE MISALIGNMENT



5% x 1.000 = 0.0500 1 x RPM: 882.00

Machine	Pos	Dir	Units	Date
328 AIRHAND 007	1	A	IN/S	May 24 1989 12:16
Frequency	1 x RPM	Amplitude		
750	0.85	0.0824		
1425	1.62	0.0975		
1800	2.04	0.1299		
6825	7.74	0.1025		

5% x 1.000 = 0.0500 1 x RPM: 882.00

Machine	Pos	Dir	Units	Date
328 AIRHAND 007	1	A	IN/S	Feb 01 1989 13:06
Frequency	1 x RPM	Amplitude		
750	0.85	0.0750		
1050	1.19	0.0824		
1425	1.62	0.1525		
1725	1.96	0.1174		
2175	2.47	0.0699		

003% x 1.000 = 0.0300 1 x RPM: 882.00

Machine	Pos	Dir	Units	Date
328 AIRHAND 007	1	A	IN/S	Sep 28 1988 12:26
Frequency	1 x RPM	Amplitude		
75	0.09	0.0349		
375	0.43	0.0300		
675	0.77	0.0349		
1050	1.19	0.0300		
1425	1.62	0.1199		
1800	2.04	0.4600		
2175	2.47	0.0400		
2475	2.81	0.0349		
3375	3.83	0.0300		

FIGURE 1

IMPROVEMENT OF VIBRATION FOLLOWING ALIGNMENT & BALANCING OF A
SPENCER VACUUM PUMP

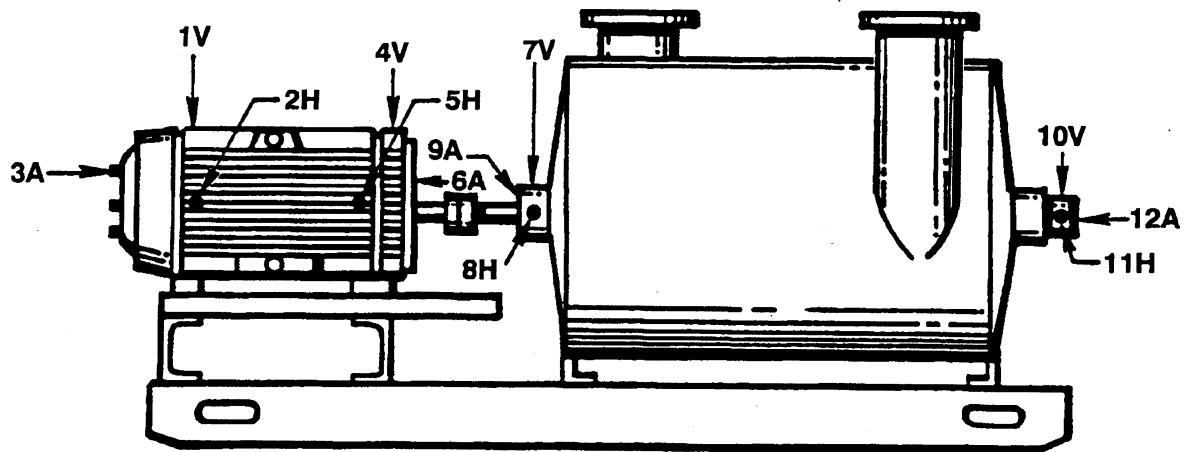
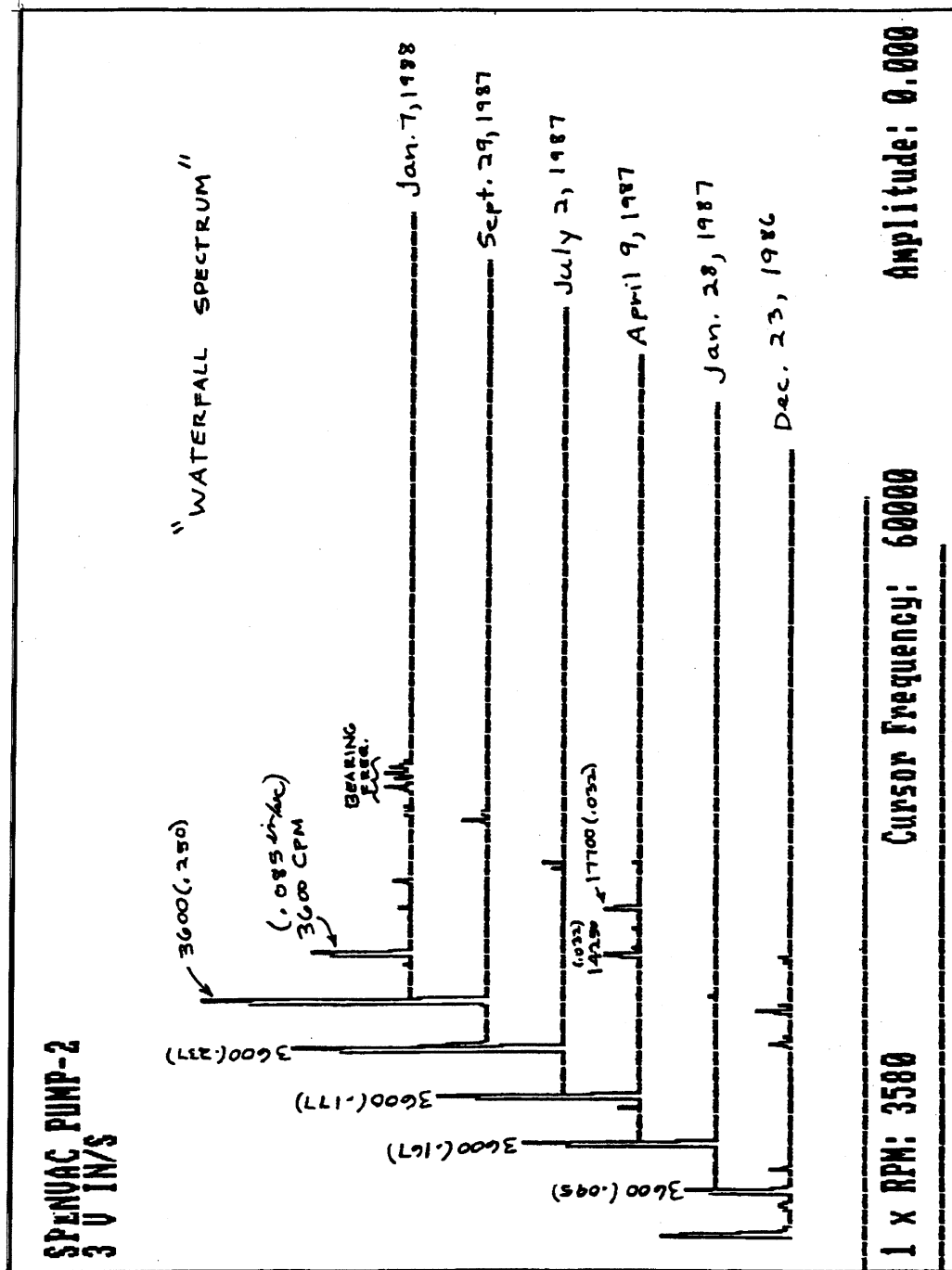
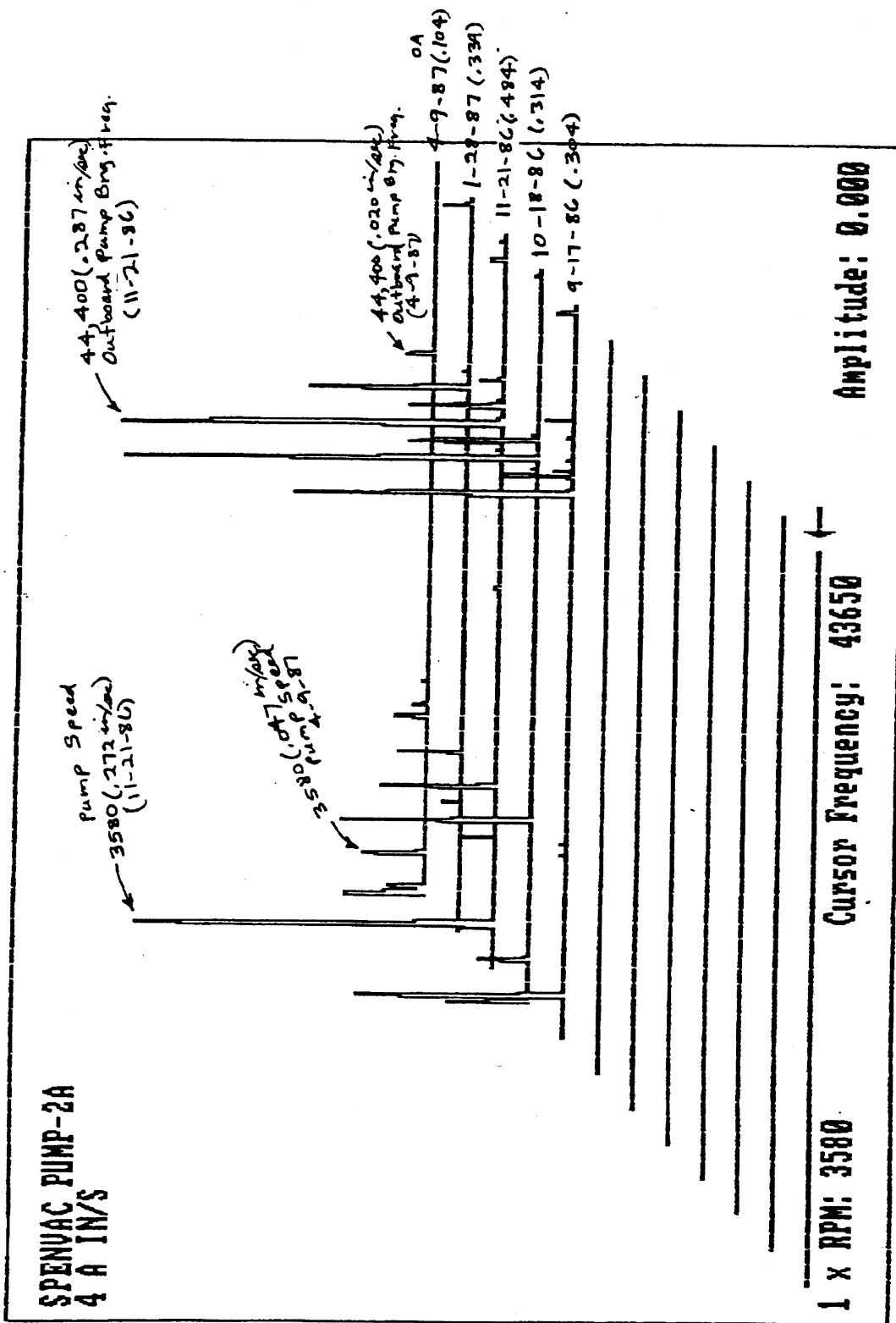


FIGURE 1



IMPROVEMENT OF MACHINE VIBRATION SIGNATURE
FOLLOWING ALIGNMENT OF MOTOR AND VACUUM PUMP SHAFTS
FIGURE 2



SAMPLE 0. REDUCTION IN VIBRATION FOLLOWING BEARING REPLACEMENT
AND BALANCING OF VACUUM PUMP (USING IN-PLACE BALANCE WHEELS)

FIGURE 3

CORRECTION OF INCINERATOR FAN WITH A LOOSE BASE AND UNBALANCE (LOOSENESS TYPE A-1X RPM)

I. BACKGROUND

A client's Predictive Maintenance Program measurements detected high vibration levels at their McGill incinerator fan shown in Figure 1. They diagnosed the problem as fan unbalance and requested that Technical Associates "balance" the fan. The customer did not have balancing capabilities and previous attempts by other outside vendors had not produced satisfactory results. The fan was integrally mounted, overhung fashion, onto the motor shaft as illustrated in Figure 1. The motor rotated at 3574 RPM.

II. RESULTS

Vibration spectral and phase diagnostics were performed before making any balance attempts. The resulting vibration measurements showed the overall vertical vibration at Position 2V had the highest amplitude at .452 in/sec as can be seen in the "BEFORE" column of Table I. Looking at Figure 1 and Table II, vertical phase measurements at Positions 1, 3, 4 and 5 revealed in-phase motion between Positions 1 and 3, in-phase motion between Positions 3 and 4, but a major phase change of 145° between Positions 4 and 5 (that is, between the motor baseplate and the concrete foundation). The horizontal overall vibration at Position 2H was only .228 in/sec. Spectra for motor Positions 2H and 2V were dominated by 1X RPM vibration as seen in the "BEFORE" spectra of Figures 2A and 3A indicating the likelihood of unbalance existed (.028 in/sec at 2H while 2V had .430 in/sec at 1X RPM). However, since a vertical vibration much higher than a horizontal vibration at 1X RPM is often indicative of a loose/weak/broken base, and because of the vertical phase anomaly, a visual inspection was performed looking for this possibility. The inspection found the anchor bolt nut at Position 6 on Figure 1 was rusted (frozen) onto the anchor bolt threads in a "backed off" condition. Thus at Position 6, the base was able to vibrate freely in a vertical direction. After the nut was freed by heating, it was tightened securely against the base flange. Figures 4A and 4B show the 1X RPM vertical vibration at Position 3 on the motor base where 1X RPM vibration fell from .255 in/sec to .108 in/sec "AFTER" tightening (a 58% reduction before any balancing was performed). This was a significant improvement. Without tightening this nut, the ensuing balance would likely have been difficult and unfruitful.

The fan balance then proceeded in a normal fashion from this point on, reducing vibration levels as noted in the "AFTER" column of Table I. Position 2H overall vibration decreased to .099 in/sec from .228 in/sec and the 2V vibration to .144 in/sec from .452 in/sec. The "AFTER" balancing spectra for Positions 2H and 2V are shown in Figures 2B and 3B, respectively. Notice in the Position 2H and 2V spectra of Figures 2B and 3B that little effect was seen at 7200 CPM as a result of the bolt tightening and fan wheel balance. This was due to the 7200 CPM frequency peak being composed of both a 2X RPM component at 7148.7 CPM and a 2X electrical line frequency (2X 3600 CPM) component at 7200 CPM as seen in the "zoom" spectra of Figure 5 taken on Position 2V before the bolt tightening and balance. Therefore, in this case, tightening loose bolts and balancing did not lower vibration from an electrical source at 7200 CPM.

III. CONCLUSIONS

- A. The "unbalance" problem was actually a combination of looseness and unbalance.
- B. Though the "unbalance" vibration appeared to lessen when an anchor bolt was tightened, it was actually the looseness that was being reduced, and not the unbalance.
- C. Tightening bolts and balancing did not correct slight electrical problems.

IV. RECOMMENDATIONS

- A. Investigate all "unbalance" problems prior to any balance attempts. Often, additional conditions will be detected that must be corrected prior to balancing. Balancing is usually the last corrective measure taken.
- B. It was also recommended that the slight electrical problem likely due to slight stator problems be trended in future spectra taken on this machine.

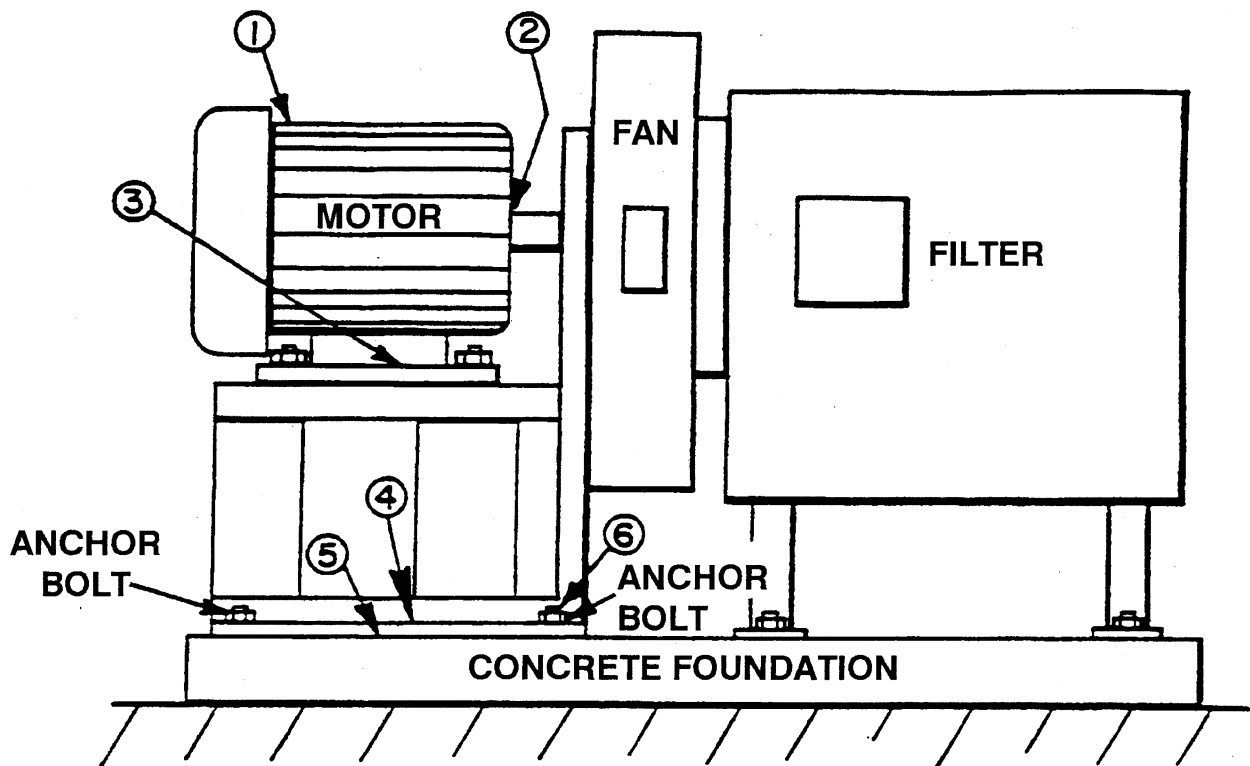


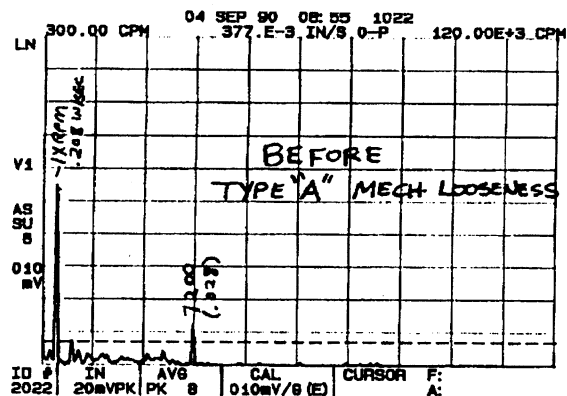
FIGURE 1

TABLE I. AMPLITUDES BEFORE AND AFTER BALANCING INCINERATOR FAN

POSITION	PHASE MEAS.	OVERALL (IN/SEC)			1 X RPM (IN/SEC)		
		BEFORE	AFTER	% CHANGE	BEFORE	AFTER	% CHANGE
1A	20°	.193	.140	- 27%	.155	.090	- 42%
1H	5°	.410	.196	- 52%	.392	.172	- 56%
1V	85°	.203	.137	- 33%	.145	.072	- 50%
2A	30°	.324	.148	- 54%	.306	.126	- 59%
2H	10°	.228	.099	- 57%	.208	.057	- 73%
2V	95°	.452	.144	- 68%	.430	.103	- 76%

TABLE II. VERTICAL PHASE ANALYSIS BEFORE CORRECTIVE ACTIONS

COMPONENT DESCRIPTION	POS. NO.	VERTICAL PHASE	EVALUATION
Top of Motor	1V	85°	Reference Phase
Motor Support Frame	3V	95°	OK (In-Phase)
Motor Baseplate	4V	110°	OK (In-Phase)
Concrete Foundation	5V	255°	145° Phase Change (Excessive)



IN/S 0-P	CPM
208.E-3-1X	3600.0
27.5E-3	7200.0
46.5E-3	35700.

PEAK PICK ON TRACE 1

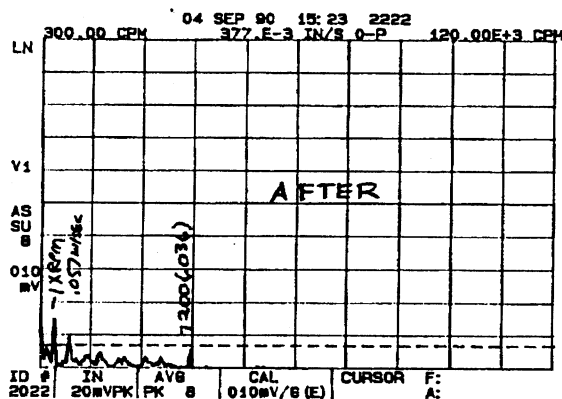
O/A .228 in/sec
POS. 2 HOR-BEFORE TIGHTENING BOLTS AND BALANCING

05-SEP-90

FIGURE 3A

09: 38: 08

MCGILL FAN



IN/S 0-P	CPM
56.8E-3-1X	3600.0
36.1E-3	7200.0

PEAK PICK ON TRACE 1

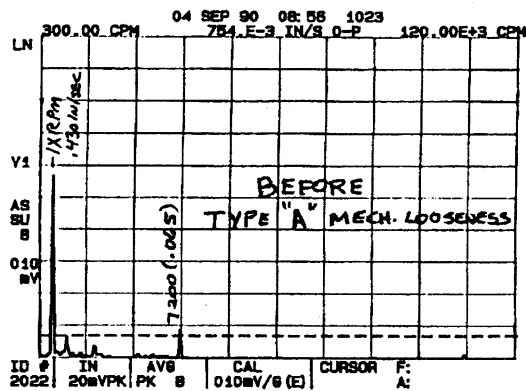
O/A .099 in/sec
POS. 2 HOR-AFTER TIGHTENING BOLTS AND BALANCING

05-SEP-90

FIGURE 3B

09: 58: 58

MCGILL FAN



IN/S 0-P	CPM
430.E-3-1X	3600.0
65.4E-3	35700.

PEAK PICK ON TRACE 1 POS 2 VERT-BEFORE TIGHTENING BOLTS AND BALANCING

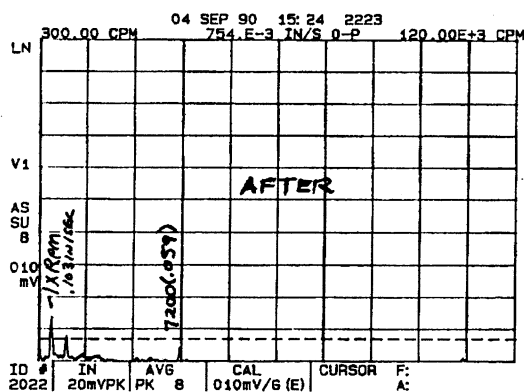
O/A .452 in/sec

05-SEP-90

FIGURE 4A

09: 39: 25

MCGILL FAN



IN/S 0-P	CPM
103.E-3-1X	3600.0
58.5E-3	7200.0

PEAK PICK ON TRACE 1 POS .2 VERT-AFTER TIGHTENING BOLTS AND BALANCING

O/A .144 in/sec

05-SEP-90

FIGURE 4B

10: 42: 51

MCGILL FAN



O/A .275 in/sec
POS 3 VERT-BEFORE BOLTS TIGHTENED AND
BEFORE BALANCING

FIGURE 5A

Position 3-V
MCGILL FAN



O/A .135 in/sec
POS 3 VERT-AFTER BOLTS TIGHTENED AND
BEFORE BALANCING

FIGURE 5B

Position 3-V
MCGILL FAN

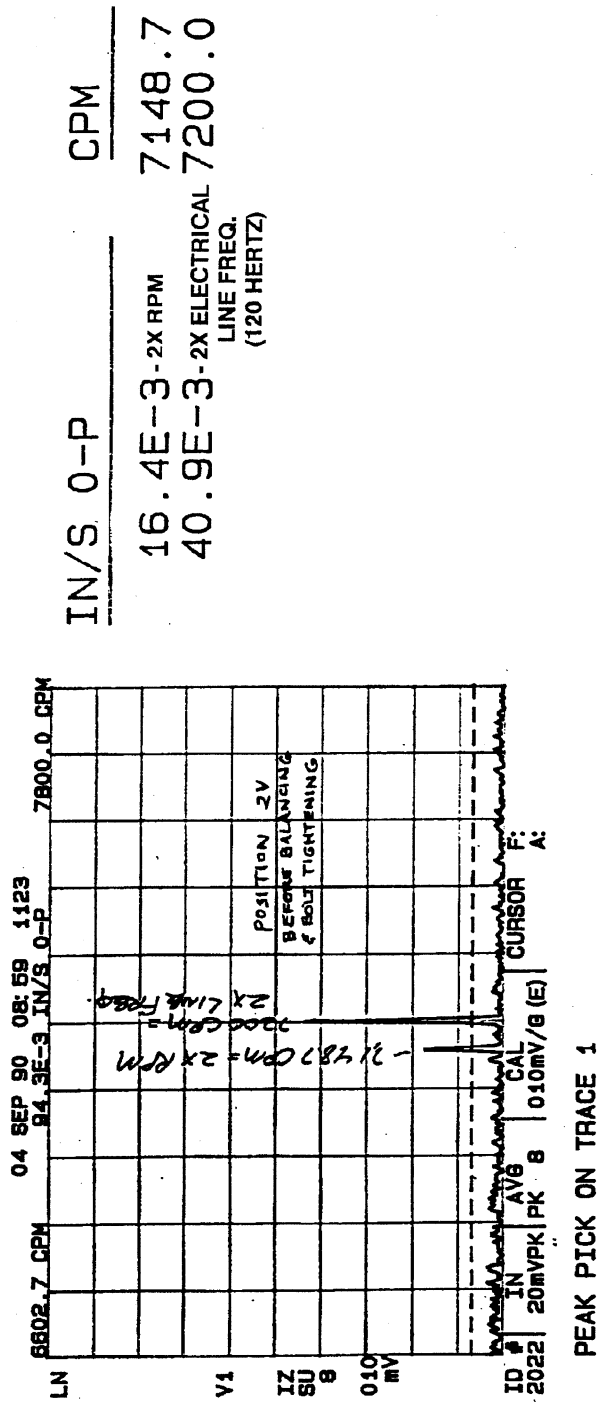


FIGURE 6

05-SEP-90

09: 41: 51

"ZOOM" SPECTRUM AT
POSITION 2V SEPARATING
2X RPM (7,148.7 CPM) FROM 2X ELECTRICAL
LINE FREQUENCY (2X 3600 CPM = 7200 CPM)

POSITION 2-V
MCGILL FAN

UNBALANCE AND LOOSENESS PROBLEMS ON AN IN-LINE INDUCED DRAFT FAN

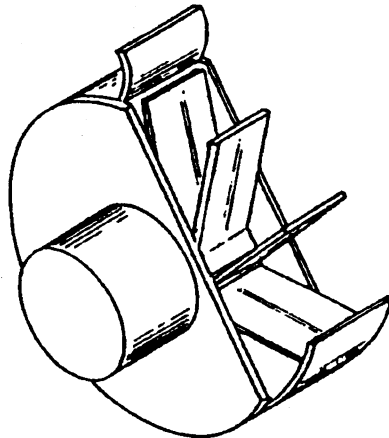
(LOOSENESS TYPE B FROM DIAGNOSTIC CHART)

I. BACKGROUND

A manufacturer of in-line induced draft fans used in HVAC ducts wanted to see if vibration levels in their production models could be reduced through balancing. The fan had a small paddle wheel impeller mounted directly on the shaft of a fractional horsepower motor in the configuration illustrated in Figure 1. The unit is mounted to ductwork when in use.

II. RESULTS

The initial fan vibration spectra showed an unbalance at 1X RPM of .250 in/sec along with .052 in/sec of looseness at 2X RPM as seen in Figure 2. The overall vibration was .256 in/sec at this time. After the addition of 1.2 grams of balance correction weight, the unbalance was successfully reduced to .053 in/sec - a 79% reduction. Figure 3 pictures the "After" spectra. The "After" looseness at 2X RPM was .089 in/sec, approximately the same as before the balance. The overall vibration was now .108 in/sec - a 58% reduction. The balance was a success. The looseness problem manifested at 2X RPM was relatively unaffected.



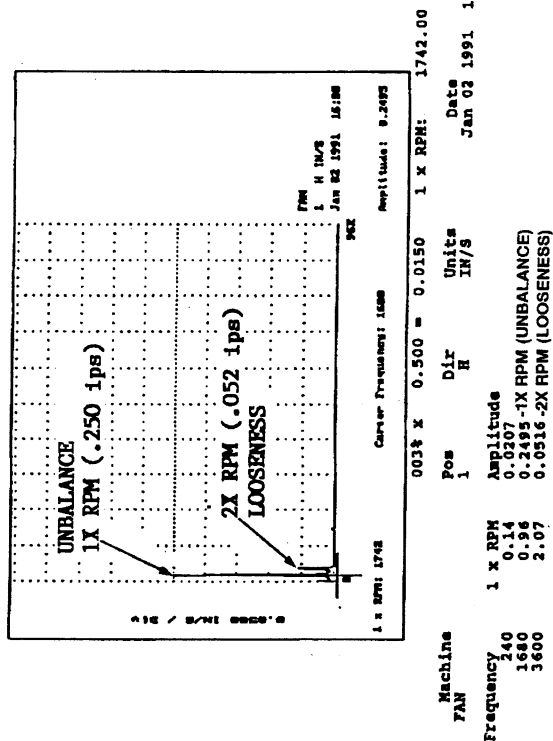
**FIGURE 1
IN-LINE INDUCED DRAFT FAN**

III. CONCLUSIONS

- A. Correcting unbalance does not remove a preexisting looseness condition, or practically any other position for that matter. Balancing may improve other existing conditions since it reduces the amount of vibration energy but, it does not solve them. Even at the supposed "unbalance" frequency of 1X RPM, other conditions may exist that will limit the improvements achievable through balancing. Balancing is done only after all other conditions (i.e., misalignment, looseness, etc.) are addressed.
- B. The overall vibration in this machine did not improve as much as did the 1X RPM vibration. This results from the premise that correcting unbalance does not correct other conditions that also affect the overall vibration levels. Further reduction in the overall vibration level would require correcting the looseness condition.
- C. The final balance weight was 1.2 grams, about half the weight of a dime or two No. 1 paper clips. This is an indication of the system's sensitivity. In this case, it may not take long for dirt to build up a 1.2 gram unbalance. Extreme sensitivity often results from looseness, resonance or a weak mounting structure. It is desirable to reduce the sensitivity levels in extremely sensitive machines.

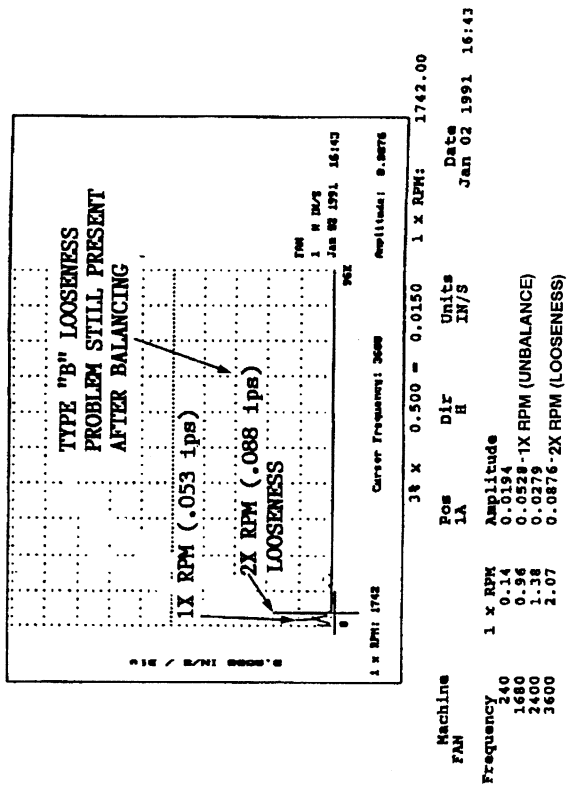
IV. RECOMMENDATIONS

- A. It was recommended that the source of the 2X RPM looseness vibration be identified. The client was told to look closely for possible cracks in the frame, loose bolts or possibly even misaligned (cocked) bearings. Correction of these problems would likely greatly reduce vibration at 2X RPM.
- B. A balancing stand was recommended for the plant to enable them to balance the fan wheels right off the production line. A balance tolerance of ISO G-2.5 was recommended to provide satisfactory balance with vibration at 1X RPM likely below approximately .075 in/sec peak velocity.



T - Threshold adjust Q - Quit

FIGURE 2



T - Threshold adjust Q - Quit

FIGURE 3

DETECTION OF FAN WHEEL ROTOR LOOSENESS ON A NUVA FEEDER VENT FAN

I. BACKGROUND

A utility company contracted Technical Associates to establish and install a predictive maintenance vibration program at each of their fossil fuel plants. Additionally, the client wanted Technical Associates to acquire the baseline measurements and issue a **Baseline Report**. This report would include the diagnoses of all problems found ranking these problems in order of severity, a condition evaluation of each machine tested, a drawing of each machine evaluated showing the locations where measurements were acquired and a plant layout showing the location of each machine which had been baselined.

II. DISCUSSION OF RESULTS

There were many machines baselined and many problems were found. One particular problem was found on the Nuva Feeder Vent Fan 68N as seen in the machine sketch Figure 1. The fan had an overall vibration of .460 in/sec (which caused it to exceed its alarm level of .375 in/sec) at Position 4 horizontal as seen in the Machine Report of Figure 2. The Position 4H spectrum, Figure 3, showed some well defined multiple harmonics of running speed with a large 1X RPM amplitude of .340 in/sec. Technical Associates defines this spectral appearance as a Type "C" looseness problem, often attributed to a poor bearing fit on the shaft or in its housing (pillowblock in this case). However, it should be pointed out this same signature can result from a bearing turning on a shaft (a Stage 4 bearing failure condition).

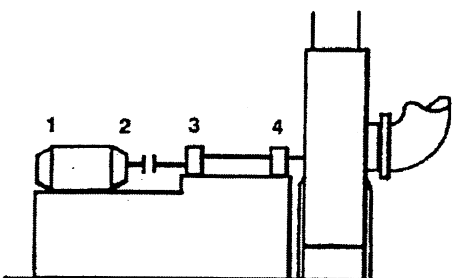
This Type "C" looseness is often the result of a replacement anti-friction bearing being installed on a worn (or undersized) shaft, or in a "wallowed-out" (oversized) bearing housing. A Stage 4 bearing catastrophic failure can occur due to total degradation of the bearing components (i.e., rolling elements and races) finally resulting in an unstable mechanical looseness. Neither the Type "C" nor Stage 4 looseness provide sufficient constraints to keep a shaft on its center of rotation which results in amplification of even a small imbalance.

III. CONCLUSIONS

- A. The Position 4 bearing had a mechanical looseness condition that could be a result of bearing fit, bearing wear, bearing turning on a shaft, broken structures and/or loosened fasteners.
- B. The unbalance may not have been a true unbalance, but may have resulted from the looseness condition aggravating and amplifying an otherwise insignificant unbalance.

IV. RECOMMENDATIONS

- A. Correct the looseness condition.
- B. Make a careful field check to ensure neither fan bearing was turning on the shaft.
- C. After correcting the looseness condition, reevaluate the imbalance and balance if necessary.

	Technical Associates of Charlotte (704) 333-8011	MACHINERY DATA GUIDE SHEET
RELIABILITY SURVEY predictive maintenance program (Page ___ of ___)		
67 N 67 S 68 N 68 S	DETECTION OF LOOSENESS ON A NUVA FEEDER VENT FAN	
		
MACHINE IDENT: <u>NUVA FEEDER VENT FAN</u>		AREA: _____ ROUTE: _____
DRIVER	INTERMEDIATE	DRIVEN
RPM min : <u>1750</u>	RPM in : _____	RPM min : <u>1750</u>
max : _____	out : _____	max : _____
MFGR : _____	RATIO: _____	MFGR. : _____
MODEL : _____	MODEL : _____	MODEL : _____
SER NO. : _____	SER NO. : _____	SER NO. : _____
V _____ P _____ A _____	RATING : _____	RATING : _____
SHEAVE : _____ DIA.	BELT # : _____	#BLADES : _____
BRG#: <u>A</u> (1)	BRG#: _____ (1)	BRG#: <u>A</u> (1)
BRG#: _____ (0)	BRG#: _____ (0)	BRG#: _____ (0)
H.P.: <u>10</u>	GEAR T: _____ CT.	GEAR T: _____ CT.
OTHER: _____	OTHER: _____	OTHER: _____

DCB -17H

CF3

R-1092-1

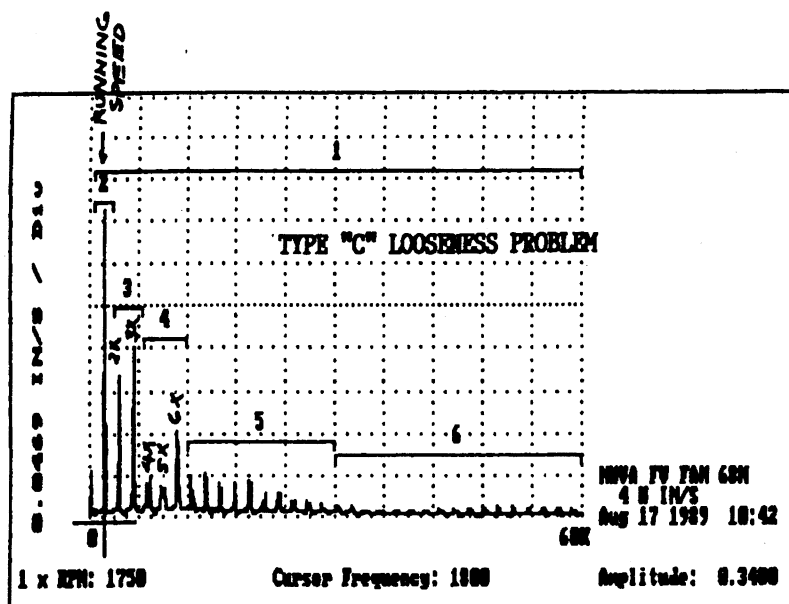
IRD Mechanalysis Data Analysis System
Dec-04-89

Machine Report (Latest Data) for: NUVA FV FAN 68N

Date	Time	Pos	Dir	Ampl	Units	Change	Alarm	Inspection
Aug-17-89	10:39	1	H	.190	IN/S	-----X	.375	
Aug-17-89	10:39	1	H	.042	g/SE	-----X	.750	
Aug-17-89	10:39	1	O	.132	IN/S	-----X	.375	
Aug-17-89	10:40	1	V	.092	IN/S	-----X	.375	
Aug-17-89	10:40	2	H	.130	IN/S	-----X	.375	
Aug-17-89	10:40	2	H	.256	g/SE	-----X	.750	
Aug-17-89	10:40	2	V	.089	IN/S	-----X	.375	
Aug-17-89	10:41	3	A	.309	IN/S	-----X	.375	
Aug-17-89	10:41	3	H	.132	IN/S	-----X	.375	
Aug-17-89	10:41	3	H	0.72	g/SE	-----X	.750	
Aug-17-89	10:41	3	V	.177	IN/S	-----X	.375	
Aug-17-89	10:42	4	A	.296	IN/S	-----X	.375	
Aug-17-89	10:42	4	H	*.460	IN/S	-----X	.375	
Aug-17-89	10:42	4	H	.229	g/SE	-----X	.750	
Aug-17-89	10:43	4	V	.240	IN/S	-----X	.375	

15 lines this report

FIGURE-2



002% x 0.500 = 0.0100 1 x RPM: 1750.00

Machine	Pos	Dir	Units	Date
NUVA FV FAN 68N	4	H	IN/S	Aug 17 1989 10:42

Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude
150	0.09	0.0525	19800	11.31	0.0375
1800	1.03	0.3400	21300	12.17	0.0175
3600	2.06	0.1575	21600	12.34	0.0275
5400	3.09	0.1875	23100	13.20	0.0275
6900	3.94	0.0375	23400	13.37	0.0275
7350	4.20	0.0450	24750	14.14	0.0175
8700	4.97	0.0350	25200	14.40	0.0175
9150	5.23	0.0325	26550	15.17	0.0175
10650	6.09	0.0975	27000	15.43	0.0150
12300	7.03	0.0475	28350	16.20	0.0150
12600	7.20	0.0300	30150	17.23	0.0125
14100	8.06	0.0500	30450	17.40	0.0125
14400	8.23	0.0425	32100	18.34	0.0125
15900	9.09	0.0375	48000	27.43	0.0125
16200	9.26	0.0325	49800	28.46	0.0125
17850	10.20	0.0375	51600	29.49	0.0125
19500	11.14	0.0400			

T - Threshold adjust Q - Quit

FIGURE 3

DETECTION OF SERIOUS WEAR OF A CENTRIFUGAL COMPRESSOR BULLGEAR SKF 6213 BALL BEARING

I. BACKGROUND

An annual contract was awarded Technical Associates beginning January, 1984 encompassing 52 machines including several Ingersoll-Rand Centac centrifugal air compressors. The primary instrument used at that time was a Hewlett Packard model 3561A real-time analyzer. Measurements were taken at 4-week intervals on each of the machines. Figures 1 thru 3 include drawings showing one of the Ingersoll-Rand Centac compressors (in this case, a model C-21 having an output capacity of 2100 ICFM). Figure 1 shows that measurements were taken on each of the motor bearing housings as well as on the bullgear (Position 3) and each of the four impeller rotor bearing housings (Positions 4 thru 7). Note the pictorial view of the bullgear and each of its mating pinions on Figure 1. The bullgear operated at about 3580 RPM (59.67 Hz), whereas the impeller rotor speeds ranged from 29,325 RPM (488.75 Hz) for the 1st Stage up to 49,270 RPM (821.13 Hz) in the case of the 4th Stage.

The problem with these machines is that once a defect occurs on either a bullgear or on one of its mating pinions, it can propagate rapidly from one component to another. Note that the cost for rebuilt rotors at that time (1984) ranged from about \$8,000 for the 4th Stage up to around \$26,000 for the 1st Stage. Thus, it was most important to detect problems in initial stages and take corrective actions before the problems were allowed to transmit from one component to another.

Figure 2 shows a typical cross section of one of the impeller rotors, each of which were outfitted with proximity probes which were serving permanent vibration monitors. The Alarm level for these machines was set at .70 mil (Pk-Pk) whereas the Trip level was at 1.0 mil (as per the manufacturer).

It is important to point out that each of three types of measurements were taken on these machines including:

- A. Vibration Displacement Measurements Directly from Proximity Probe Monitors** - a cable is connected directly to the outputs of each of the four proximity probes serving each of the four impeller stages. This cable is then connected directly to the HP 3561A analyzer allowing the capture not only of overall displacement (which is displayed by the monitor), but also enable the generation of complete vibration spectra of the shaft vibration at each of these stages.
- B. Vibration Velocity Measurements from each Bearing Housing** - a standard accelerometer having a frequency range of 1.0 to 10,000 Hz (60 - 600,000 CPM) is mounted on each of the bearing housings in the locations shown in Figure 1. A number of different frequency range spectra are captured on the various bearing housings, depending on the speed of the shaft at that location. The acceleration signal from the accelerometer is integrated to velocity within the HP 3561A analyzer (from peak acceleration to peak velocity).

C. **High Frequency Acceleration Measurement from Compressor Bearing Housings** - a special, high frequency accelerometer is employed along with signal conditioning instrumentation in order to capture measurements up to at least 2 X gear mesh frequency (2 GMF). Table I shows that the fundamental gear mesh frequency (GMF) is 20,528 Hz (1,231,700 CPM). Because measurements need to be taken up to 2 and 3 X gear mesh frequency (2 GMF and 3 GMF), this puts great demands on the transducer system as well as the mounting of the accelerometer onto the housing as well. Great attention to each of these items has enabled measurements with repeatable amplitudes up thru 3 GMF (61,585 Hz) although the absolute value of the amplitude itself at this high frequency might be in question. That is, if the accelerometer is mounted and measurements taken several different times, it is possible to get good repeatable acceleration measurements. However, if the system displays 10 g acceleration, the true level might be 7 or 8 g, or on the other hand, it might be 12 to 14 g. In any case, due to the good repeatability of the measurement, data taken from one month to the next is successfully trended. In fact, these same procedures were used on approximately 30 different Centac compressors on a monthly basis during 1984 by Technical Associates and have been used on several hundred additional Centacs since that time.

II. ANALYSIS RESULTS

Figure 3 includes a section view showing where measurements are taken in the axial direction on the bullgear (Position 3A), 2nd Stage (Position 5A) and 3rd Stage (Position 6A). Note the location of each of the bullgear ball bearings in Figure 3. The SKF 6213 bearing is inboard (closest to the coupling) whereas the SKF 7213 is somewhat remote from the measurement location. Note that the vibration signal from the SKF 6213 bearing transmits cleanly from the housing through the casing into the casing bolt on which the accelerometer is mounted. On the other hand, the vibration from the SKF 7213 bearing must travel along the shaft and then through either the SKF 6213 bearing and/or machine casing. Therefore, a number of experiments over the years on these machines have shown suppressed amplitudes which must be taken into account that come from bearing defect frequencies from the SKF 7213 bearing relative to those of the SKF 6213.

Table I provides many of the anticipated vibration frequencies generated by components of the compressor. These include the gear mesh frequency (GMF), the blade pass frequencies of each of the 4 Stages (BPF), each of the impeller rotor speeds (F_1 thru F_4), the main oil pump frequency (F_{PUMP}), the SKF 7213 bearing defect frequencies, the SKF 6213 bearing defect frequencies, and finally the bullgear and motor running speed frequency (F_M). Note that this table includes the first 10 harmonics of each of these frequencies (1F thru 10F).

Figure 4 shows one of these spectra captured approximately 9 months after initial baselines were captured. Note that several frequencies appear on this spectrum including the 2nd Stage running speed peak (F_2), the 3rd Stage speed (F_3) and the 4th Stage speed (F_4). Also showing up was a high amplitude peak at approximately 352 Hz. Referring back to Table I, note that the 352 Hz frequency corresponds to the inner race defect frequency of the SKF 6213 bearing (6213 BPFI).

Importantly, when initial baseline signatures were captured in January, 1984, this 352 Hz peak appeared with an amplitude of .216 in/sec and did not have bullgear running speed sidebands to its left and right (which later did appear as shown in Figure 4). Due to the baseline amplitude of over .20 in/sec which is quite high for a bearing defect frequency, Technical Associates recommended replacement of the bullgear bearing immediately before the problem, then isolated only with the bullgear bearing, caused problems with the mating components. Due to great demands on this compressor and its support of critical process machinery, the plant decided it could not shut down the machine to accomplish the bearing replacement at this time. Therefore, the decision was made to simply trend future spectra on this machine and try to schedule this replacement along with the production department ASAP. Three months later, the amplitude of this frequency began climbing and went up to about .252 in/sec as of April, 1984. At this point, small amplitude sideband frequencies at bullgear speed began to appear to the left and right of this bearing frequency. Once again, Technical Associates recommended bullgear bearing replacement even though the other measurements on this machine did not indicate any damage to other machine components yet (for example, acceleration measurements still showed that no gear mesh frequencies exceeded about 7 g which is normal for these machines at these GMF frequencies and harmonics). Still the plant felt it could not schedule the bearing replacement.

By June, amplitudes at BPFI (352 Hz) increased from .252 in/sec in April up to .305 in/sec and sidebands at 292 and 410 Hz were now well developed with amplitudes of almost .05 in/sec each. Because amplitudes at gear mesh frequencies had now begun to increase from about 7 g in both January and April up to approximately 12 g now in June, Technical Associates recommended immediate bullgear bearing replacement before significant damage was done to the impeller rotors. However, once again, the plant production department would not allow the machine to be brought down for maintenance.

Therefore, signatures later captured on September 28 showed extensive problems throughout the compressor. Figure 4 shows that the BPFI amplitude was now up to .402 in/sec and had well developed running speed sidebands at 292.5 Hz and 410.0 Hz. In addition, it now showed a small second harmonic inner race frequency at 705 Hz. Of even greater concern were the acceleration spectra captured on the bullgear and pinion housings which are shown in Figures 6 and 7. Figure 6 was a wideband plot from 0 to 100,000 Hz (6,000,000 CPM) and showed that levels at twice gear mesh frequency (2 GMF) had now climbed from only 7 g at baseline up to 42.1 g. Then, looking at the zoom spectrum shown in Figure 7, note that 2 GMF was now surrounded by well developed sidebands at 1st Stage speed indicating significant wear of the 1st Stage pinion. Subsequent measurements on the other stages indicated problems with at least three of the pinions as well as the bullgear itself.

III. RECOMMENDATIONS MADE

As of the September 28 data, Technical Associates recommended replacement of each of the bullgear bearings. Also, it was recommended that each of the gears be closely inspected for suspected wear.

IV. CORRECTIVE ACTIONS TAKEN

When inspection of the gears was made, significant wear was found on the bullgear as well as the 1st, 3rd and 4th Stage pinions. As a result, each of these impeller rotor pinion assemblies had to be replaced as well as the bullgear and its bearings. Figures 5 and 8 were captured after corrective actions were taken. Figure 5 shows that the velocity spectrum showed a dramatic drop in inner race frequency from .402 in/sec down to .015 in/sec after replacement of the SKF 6213 bearing. Figure 8 shows that acceleration levels at 2 GMF dropped from 42.1 g before to 7.62 g after replacement of 4 of the 5 gears. One of the hard lessons learned from this case history was that corrective actions should be taken on these centrifugal compressors when vibration spectra indicate if this is at all possible. What would have been less than a \$10,000 repair cost had actions been taken right after the January baseline ended up being repair costs of slightly over \$100,000 since the problem was allowed to propagate to so many other components. Fortunately, after this case history, the plant has paid close attention to vibration spectra and successfully detected each of 3 potentially serious incidents that could have resulted in similar costly failures. However, on each of these occasions, the maintenance department coordinated with the production department and took actions which prevented any of the failure scenario costs from exceeding \$20,000.

CC4

R-9210-1

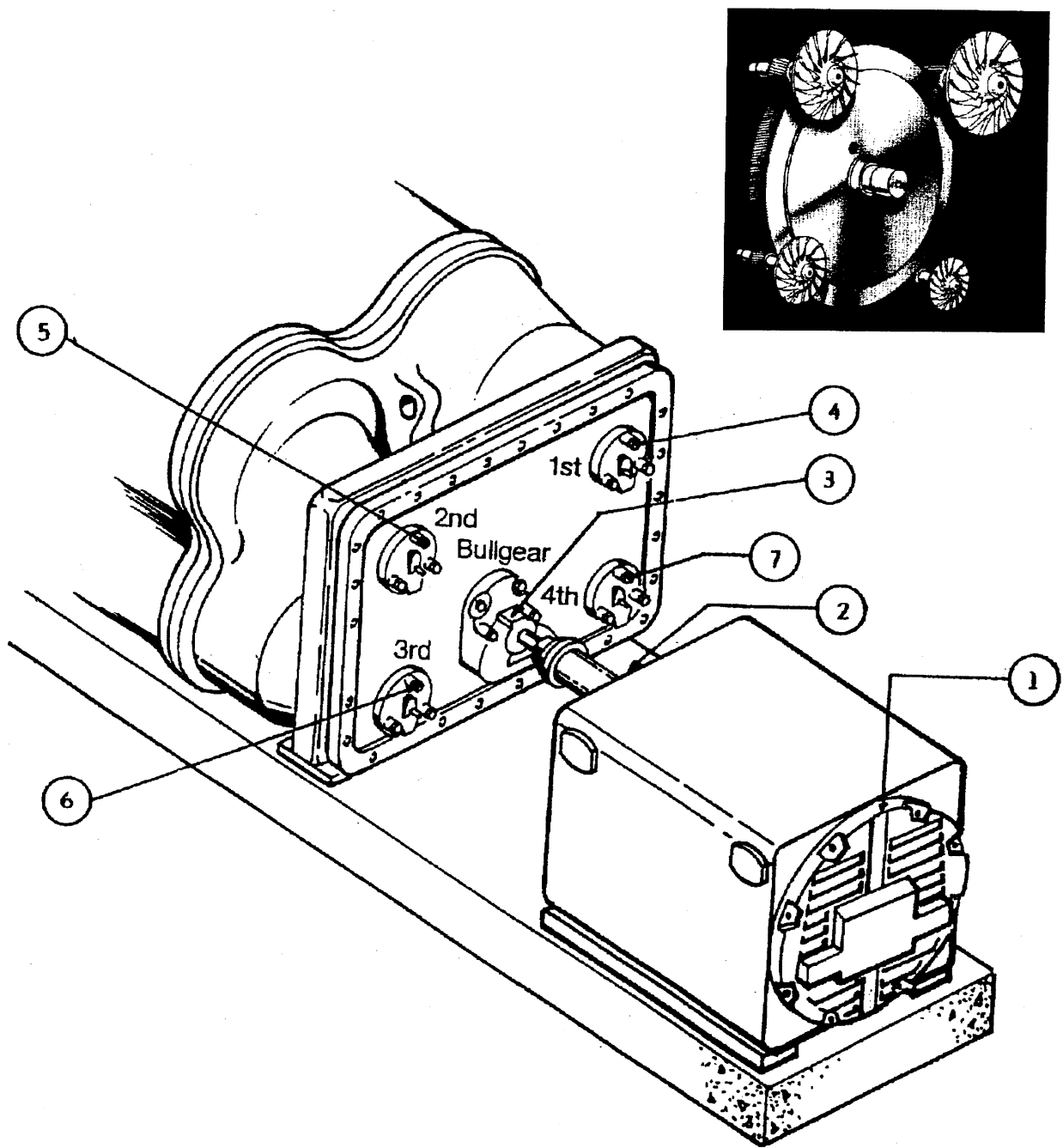
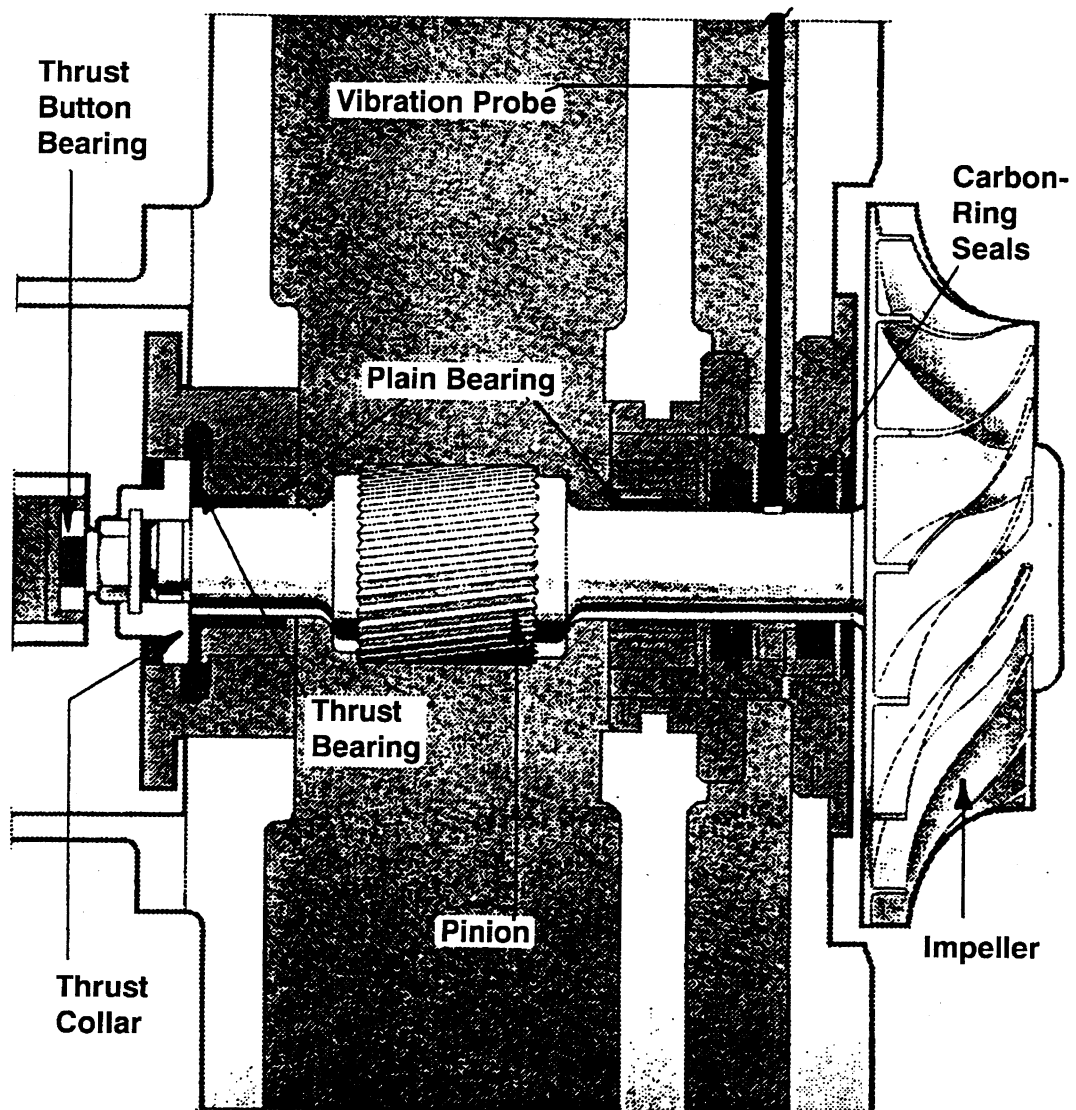


FIGURE 1: ACCELEROMETER LOCATIONS ON AN INGERSOLL-RAND CENTAC CENTRIFUGAL AIR COMPRESSOR (MODEL C-21 2100 ICFM)

FIGURE 2. INGERSOLL-RAND CENTAC IMPELLER ROTOR CROSS SECTION



DETECTION OF SERIOUS WEAR OF A CENTAC BULLGEAR INBOARD BALL BEARING

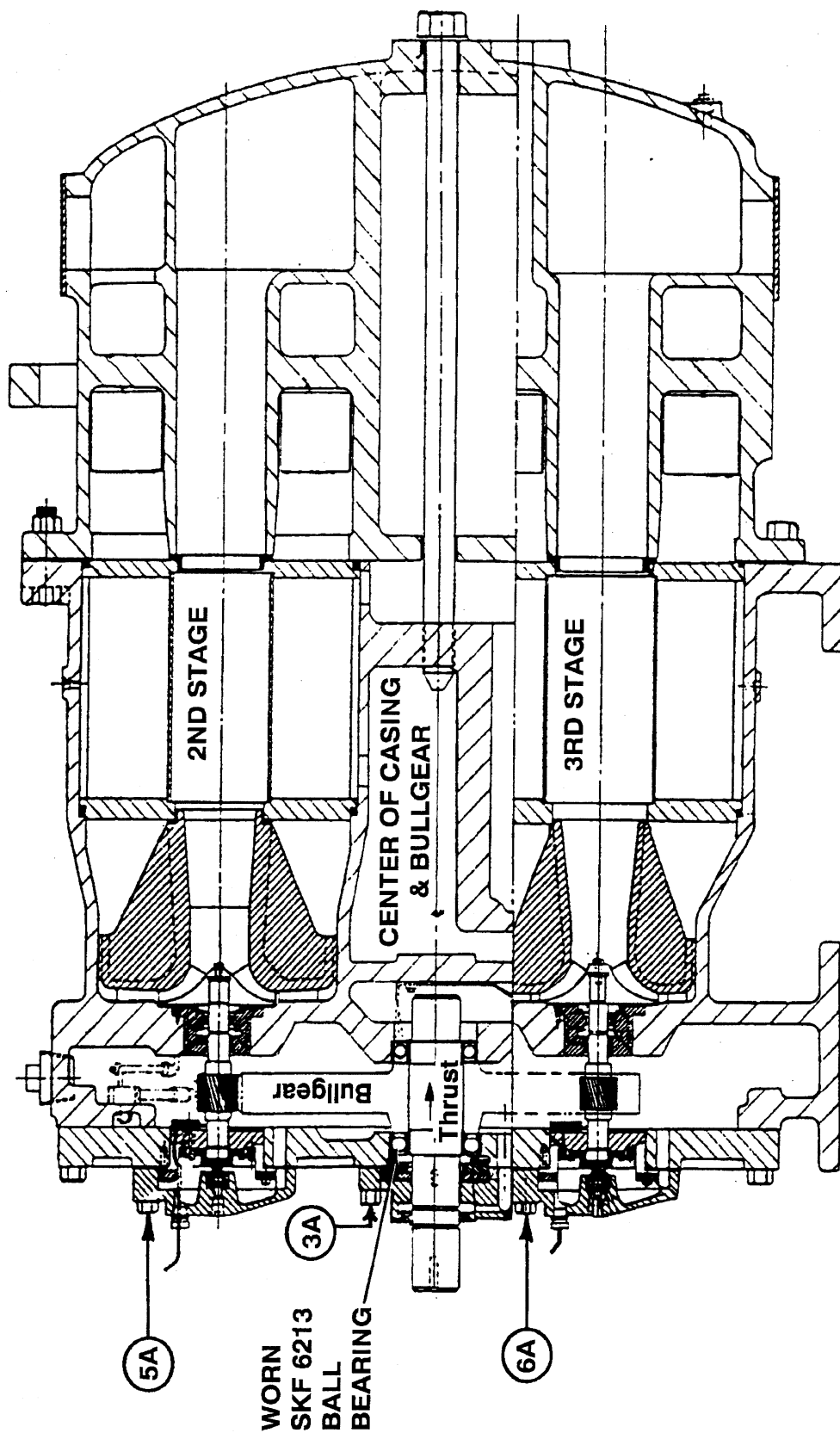


FIGURE 3. SECTION VIEW OF AN INGERSOLL-RAND CENTAC CENTRIFUGAL AIR COMPRESSOR (FRAME 1)

TABLE 1. ANTICIPATED FREQUENCIES FOR MACHINE #6 C-21 CENTAC (HERTZ)

ITEM FREQ.	$\frac{F_4}{F_M}$	1F	2F	3F	4F	5F	6F	7F	8F	9F	10F
GME.....	344FM	20,528	41,056	61,585	82,113	102,641	N/A	N/A	N/A	N/A	N/A
BPF4.....	178.88F	10,675	21,349	32,024	42,699	53,373	64,048	74,723	85,397	96,072	106,747
BPF3.....	165.63F	9,884	19,768	29,652	39,536	49,420	59,304	69,188	79,072	88,956	98,839
BPF2.....	150.50F	8,981	17,962	26,943	35,924	44,905	53,886	62,868	71,849	80,830	89,811
BPF1.....	122.86F	7,331	14,663	21,995	29,326	36,658	43,989	51,320	58,652	65,983	73,315
F4.....	13.760F	821.13	1642.3	2463.4	3284.5	4105.6	4926.8	5747.9	6569.0	7390.2	8211.3
F3.....	12.741F	760.30	1520.6	2280.9	3041.2	3801.5	4561.8	5322.1	6082.4	6842.7	7603.0
F2.....	10.750F	641.51	1283.0	1924.5	2566.0	3207.5	3849.0	4490.5	5132.1	5773.6	6415.1
F1.....	8.190F	488.8	977.5	1466.3	1955.1	2443.8	2932.6	3421.4	3910.1	4398.9	4887.7
F _{ump}	13FM	775.78	1551.6	2327.3	3103.1	3878.9	4654.7	5430.4	6206.2	6982.0	7757.8
7213 BPFI	9.18F	547.81	1095.6	1643.4	2191.2	2739.0	3286.8	3834.6	4382.4	4930.2	5478.0
7213 BPFO	6.84F	408.18	816.36	1224.54	1632.72	2040.9	2449.08	2857.26	3265.44	3673.62	4081.8
7213 BSF	2.71F	161.72	322.4	483.6	644.8	806.0	967.2	1128.4	1289.6	1450.8	1612.0
7213 FTF	0.426F	25.42	50.84	76.26	101.68	127.1	152.52	177.94	203.36	228.78	254.2
6213 BPFI	5.898F	351.96	703.92	1055.88	1407.84	1759.8	2111.76	2463.72	2815.68	3167.64	3519.6
6213 BPFO	4.098F	244.55	489.1	733.65	978.2	1222.75	1467.3	1711.85	1956.4	2200.95	2445.5
6213 BSF	2.680F	159.93	319.86	479.79	639.72	799.65	959.58	1119.51	1279.44	1439.37	1599.3
6213 FTF	0.410F	24.47	48.94	73.41	97.88	122.35	146.82	171.29	195.76	220.23	244.7
* F _M	1FM	59.675	119.35	179.03	238.70	298.38	358.05	417.73	477.40	537.08	596.75

* These calculations are based on a motor running speed of 3580 rpm = 59.675 Hz.

NORMAL OPERATING SPEEDS

COMPONENT	SYMBOL	RPM	Hz
4th Stage	F ₄	49,270	821.13
3rd Stage	F ₃	45,620	760.30
2nd Stage	F ₂	38,490	641.51
1st Stage	F ₁	29,325	488.75
Motor	F _M	3,580	59.68

* NOTE: See "Terminology" on page 22 for definitions of "ITEM FREQUENCIES" in above Table VIII.

FIGURE 4: SEVERE DEFECT CONCENTRATED ON ONE LOCATION OF THE INNER RACE OF THE SKF 6213 BULLGEAR BEARING (9/28/84)

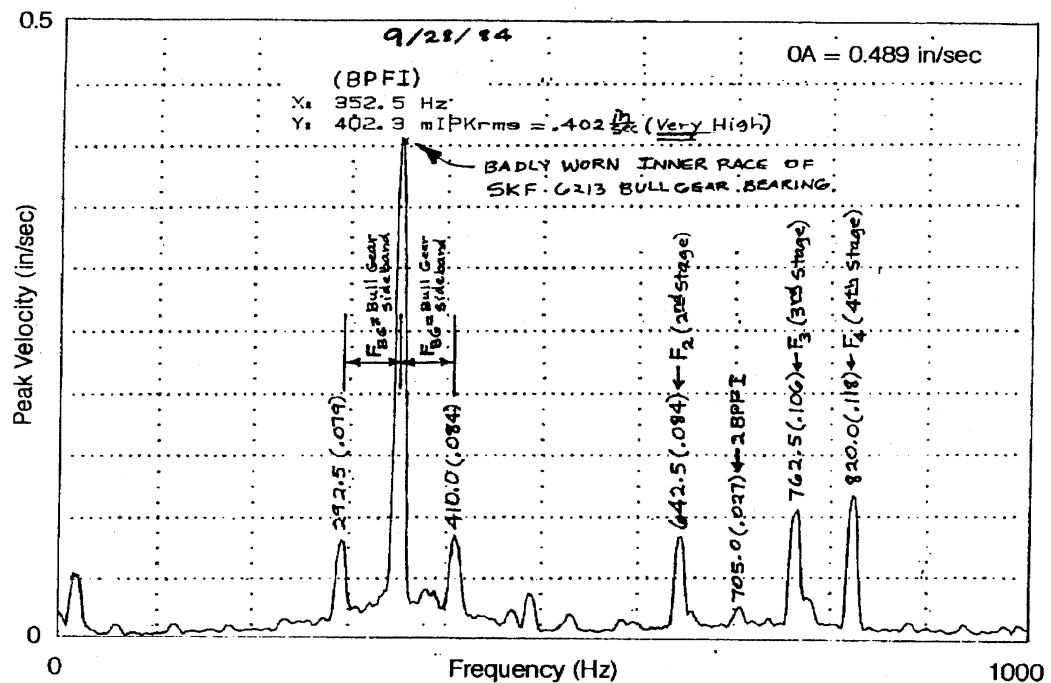
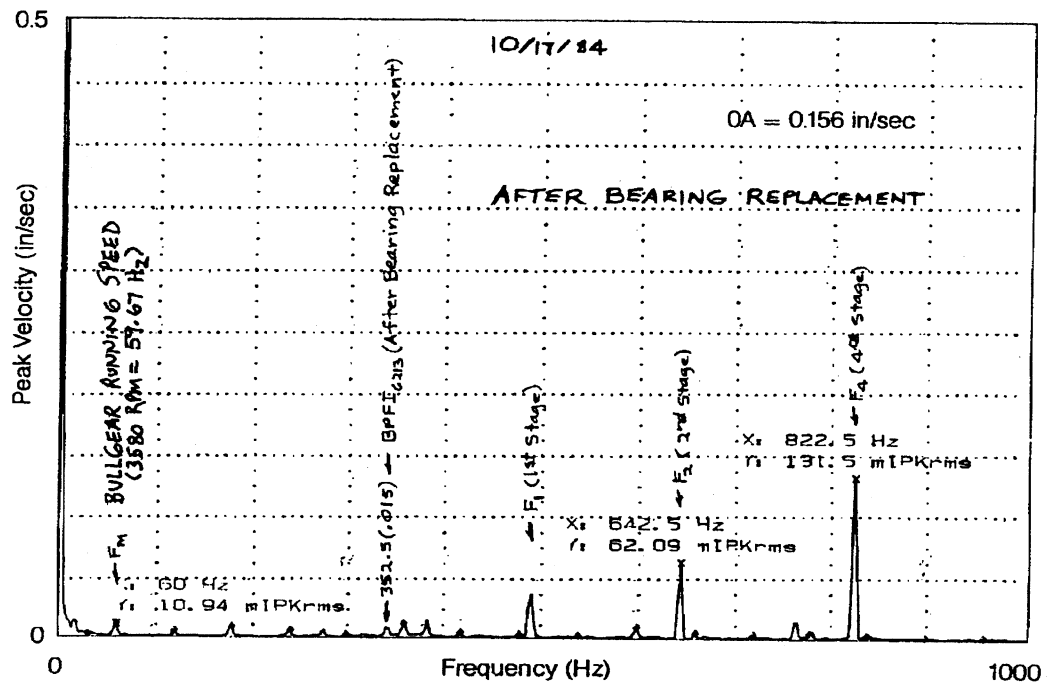
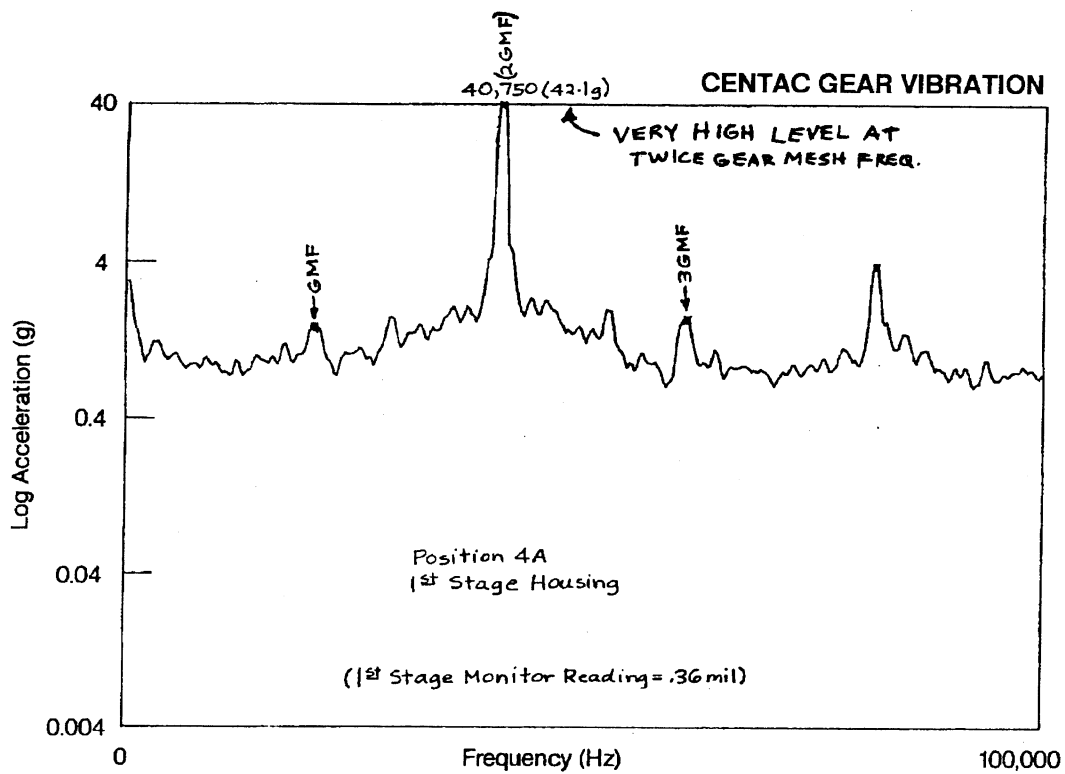


FIGURE 5: AFTER REPLACEMENT OF SKF 6213 BULLGEAR BEARINGS (10/16/84)



**INGERSOLL-RAND CENTAC BULLGEAR BALL BEARING
CAUGHT JUST PRIOR TO FAILURE**



**FIGURE 6: HIGH ACCELERATION DUE TO GEAR WEAR
9 MONTHS AFTER BASELINING**

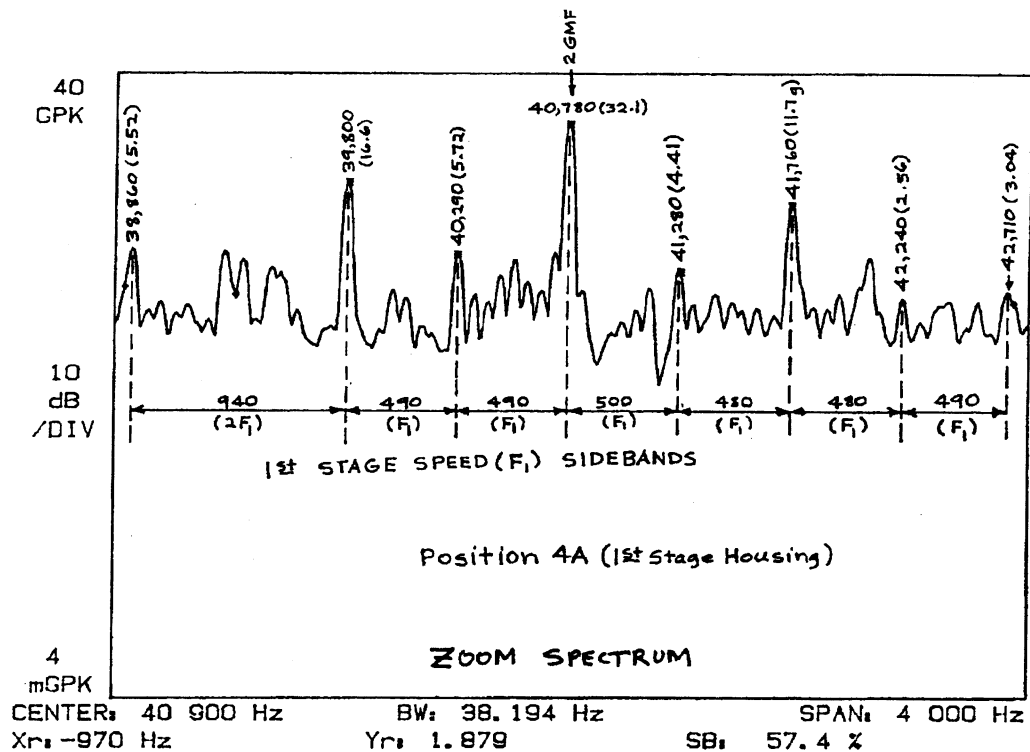
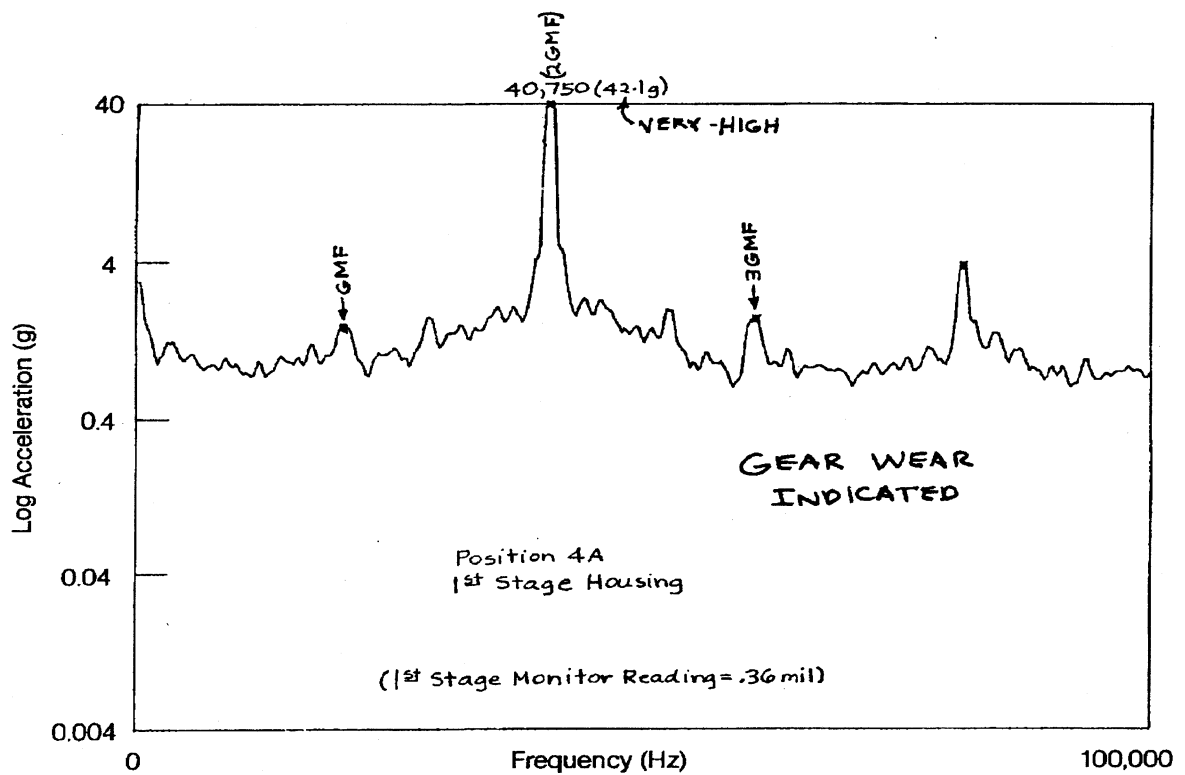
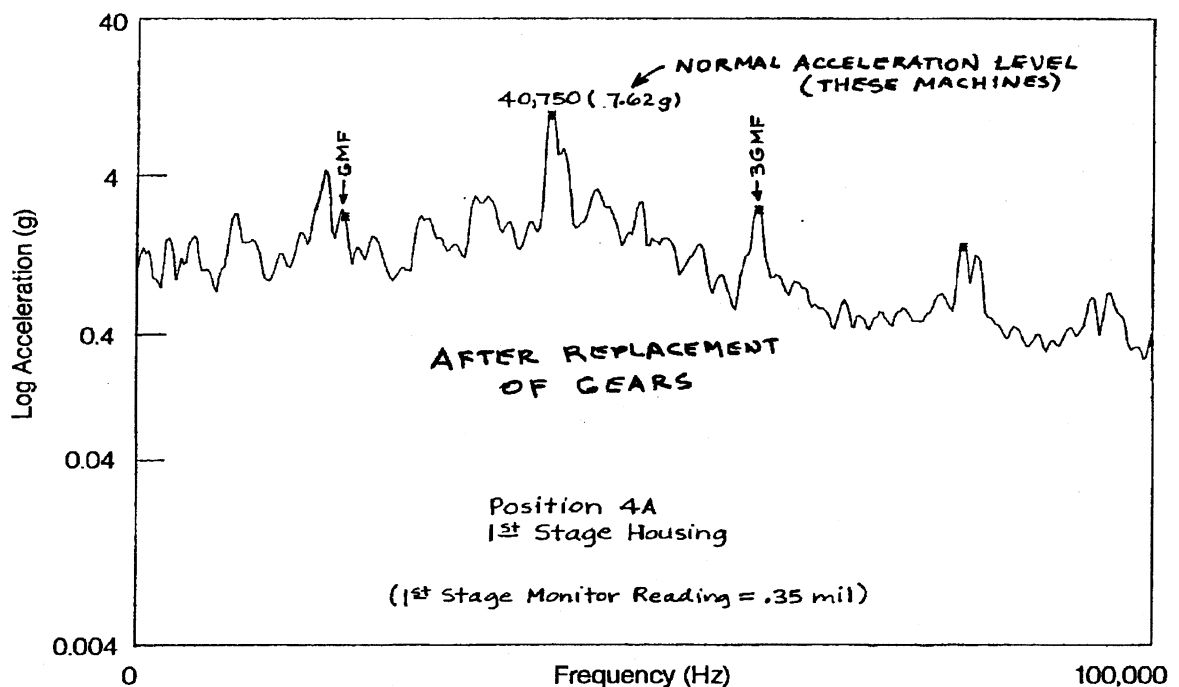


FIGURE 7: FIRST STAGE SPEED SIDEBANDS AROUND 2X GEAR MESH FREQUENCY



**FIGURE 8: HIGH ACCELERATION DUE TO GEAR WEAR
9 MONTHS AFTER BASELINING**



**FIGURE 9: SPECTRUM AFTER REPLACEMENT OF BULLGEAR,
1ST, 3RD & 4TH STAGE PINIONS**

SEVERE VIBRATION TREND OF A TIMKEN ROLLER BEARING INSTALLED IN A CIRCULATING FAN

I. BACKGROUND

F 21 Circulating Fan #4 suffered a severe bearing frequency vibration problem with its inboard fan bearing which is pictured as Position 3 Figure 1. Note that Timken model 34478 bearings are installed on the fan which is overhung from these two bearings. Severe vibration and spike energy levels were measured during the August 3, 1990 survey which mandated close analysis to prevent potentially catastrophic failure. This machine is analyzed on a regularly scheduled basis as part of a Predictive Maintenance Program annual contract.

II. ANALYSIS RESULTS

Figure 2 shows the vibration spectrum measured on Position 3A of the F 21 Circulating Fan #4 taken on August 3. Note that both Bands 5 and 6 were in alarm especially due to frequencies at 12,870 CPM (Band 5) and 25,740 CPM (Band 6). Comparison with the Timken 34478 bearing defect frequencies in Figure 1 confirmed that the frequency at 12,870 CPM was the outer race frequency (BPFO) while that at 25,740 CPM was its second harmonic (2X BPFO).

Figure 3 shows the trend of spike energy and that of the Band 6 velocity level between March 30, 1989 and October 11, 1990. Note the tremendous increase in spike energy from .630 to 2.89 g/SE on August 3. Figure 4 shows two monthly measurement reports which displayed overall vibration at each point both before bearing replacement (8-3-90 report) and after bearing replacement (10-11-90 report). Note the significant decrease in both vibration and spike energy for fan Positions 3A, 3H and 3V which occurred on the October 11 report (decreased from 80% to 93%) after replacement of the fan bearings.

Figure 5 shows a waterfall plot which illustrates how the vibration spectrum at Position 3A changed from March 30, 1989 through October 11, 1990. Note that no bearing frequencies (or any high frequency energy for that matter), was present on either the March 30 or April 25 signatures. However, high amplitude outer race frequencies did appear as of the December 12 spectrum (.270 in/sec at 2X BPFO). Then, both BPFO and 2X BPFO roughly doubled as of the January 18, 1990 spectrum. They then continued to increase with the next spectrum taken August 3 up to the final levels of .150 in/sec at BPFO and an excessive .775 in/sec at 2X BPFO.

III. RECOMMENDATIONS MADE

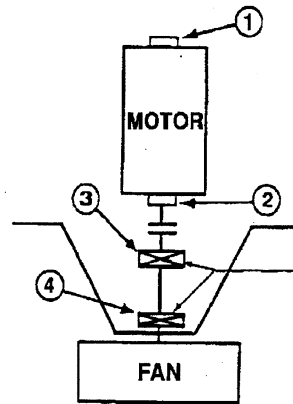
Technical Associates recommended immediate replacement of the fan bearings in order to prevent potential catastrophic failure (recommendations have been made to replace these bearings since the January 18 spectrum was captured). In addition, it was recommended that the plant take steps to balance the fan which had trended from only about .120 in/sec at fan speed up to approximately .320 in/sec as of the August 3 spectrum.

IV. CORRECTIVE ACTIONS TAKEN

The plant did follow through with replacement of each of the fan bearings, but did not schedule the fan balancing at this time. Also, the plant took extra efforts to ensure good alignment between the motor and fan shafts.

V. RESULTS OF CORRECTIVE ACTIONS

Figure 5 shows that all higher frequencies were eliminated in the spectrum of October 11 taken after bearing replacement. In addition, some improvement in fan vibration occurred even without the fan balance. Similarly, Figure 3 shows that both the spike energy and the overall velocity levels dropped significantly following bearing replacement.

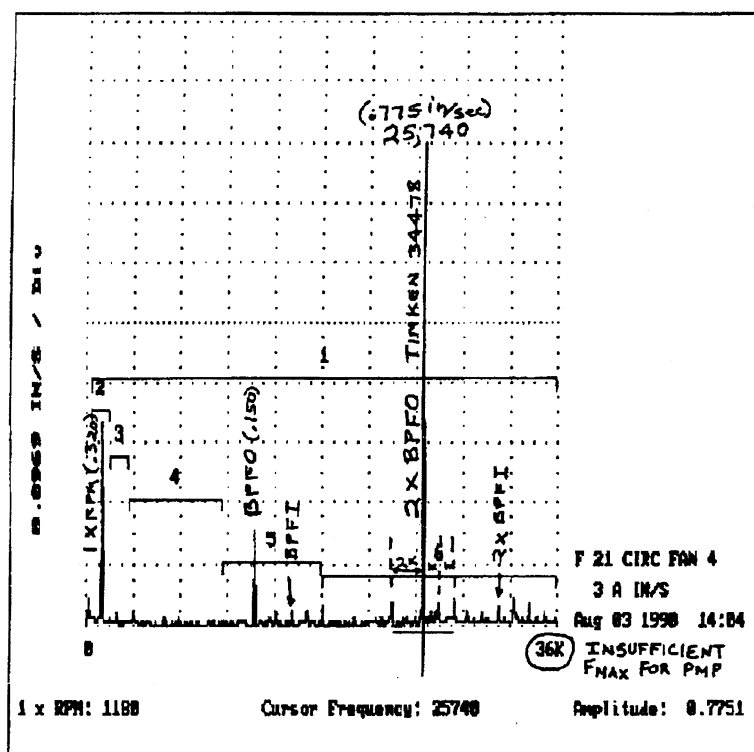
RELIABILITY SURVEY		predictive maintenance program		(Page __ of __)
 TIMKEN 34478 BPI = 13.211X RPM BPFO = 10.789X RPM BSF = 4.734X RPM FTF = 0.450X RPM				
MACHINE IDENT: FF 21 CIRCULATING FANS (1-6)		AREA: _____ ROUTE: _____		
DRIVER	INTERMEDIATE	DRIVEN		
RPM min : 1180 NAME PLATE	RPM in : _____	RPM min : 1180 RPM NAME PLATE		
max : _____	out : _____	max : _____		
MFGR : _____	RATIO: _____	MFGR. : _____		
MODEL : _____	MODEL : _____	MODEL : _____		
SER NO. : _____	SER NO. : _____	SER NO. : _____		
V _____ P _____ A _____	RATING : _____	RATING : _____		
SHEAVE : _____ DIA.	BELT # : _____	#BLADES : _____		
BRG#: _____ (I)	BRG#: _____ (I)	BRG#: TIMKEN 34478 (I)		
BRG#: _____ (O)	BRG#: _____ (O)	BRG#: TIMKEN 34478 (O)		
H.P.: 40	GEAR T: _____ CT.	GEAR T: _____ CT.		
OTHER: _____	OTHER: _____	OTHER: _____		

CF5

FIGURE 1

DCB-8

R-9210-1

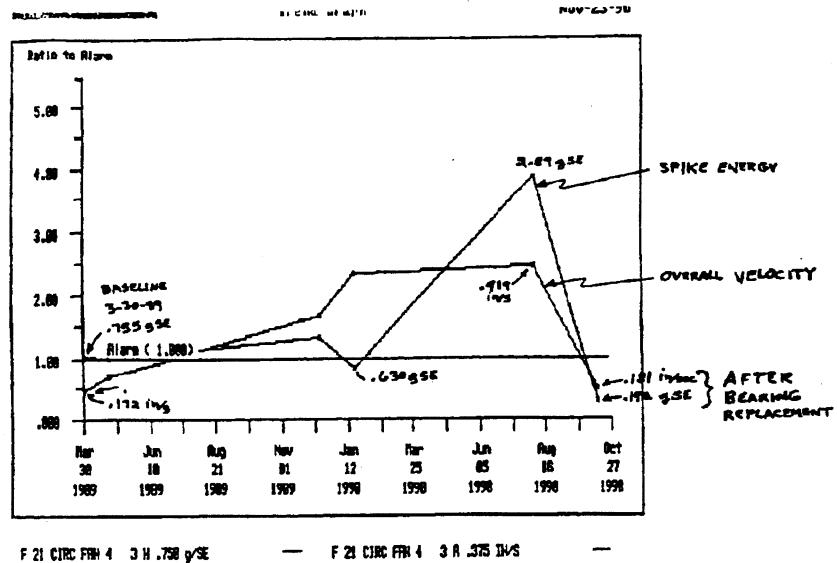


002% x 1.000 = 0.0200 1 x RPM: 1180.00

Machine	Pos	Dir	Units	Date	
F 21 CIRC FAN 4	3	A	IN/S	Aug 03 1990 14:04	
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude
180	0.153	0.0450	31500	26.695	0.0300←2BPFI
1170	0.992	0.3199	32670	27.686	0.0450
2340	1.983	0.0200	33840	28.678	0.0349←2BPFI + 2X
3600	3.051	0.0200	35010	29.669	0.0249
12870	10.907	0.1500←BPFO			
14580	12.356	0.0200			
15750	13.347	0.0249←BPFI			
16920	14.339	0.0249			
18090	15.331	0.0300			
23400	19.831	0.0349←2BPFO - 2X RPM			
25740	21.814	0.7751←2XBPFO			
26370	22.347	0.0249			
26730	22.653	0.0249			
26910	22.805	0.0349←2BPFO + 1X RPM			
28080	23.797	0.0450←2BPFO + 2X RPM			
29070	24.636	0.0249←2BPFI - 2X RPM			
30330	25.703	0.0200←2BPFI - 1X RPM			

T - Threshold adjust Q - Quit

FIGURE 2



IRD Mechanalysis Data Analysis System

Line	F 21 CIRC FAN 4	Pos	3	Dir	H	Units	g/SE	Alarm	.750
Ampl	Insp	Date	Time	Ampl	Insp	Date	Time		
Code	mm/dd/yy			Code	mm/dd/yy				
.755	0	03-30-89	10:57						
.744	0	04-25-89	14:51						
.979	0	12-12-89	11:49						
.630	0	01-18-90	11:33						
2.89	0	08-03-90	14:05						
.192	0	10-11-90	15:07						

IRD Mechanalysis Data Analysis System

Machine	F 21 CIRC FAN 4	Pos	3	Dir	A	Units	IN/S	Alarm	.375
Ampl	Insp	Date	Time	Ampl	Insp	Date	Time		
Code	mm/dd/yy			Code	mm/dd/yy				
.172	0	03-30-89	10:57						
.263	0	04-25-89	14:51						
.628	0	12-12-89	11:49						
.872	0	01-18-90	11:33						
.919	0	08-03-90	14:04						
.181	0	10-11-90	15:07						

FIGURE 3.

IRD Mechanalysis Data Analysis System
Nov-23-90

Machine Report (Latest Data) for: F 21 CIRC FAN 4

Date	Time	Pos	Dir	Ampl	Units	Change	Alarm	Inspection
Oct-11-90	15:05	1	A	.285	IN/S	21%	.375	10-11-90 MEASUREMENT REPORT
Oct-11-90	15:05	1	H	*.536	IN/S	267%	.375	
Oct-11-90	15:05	1	H	.012	g/SE	-94%	.500	
Oct-11-90	15:05	1	V	*.466	IN/S	134%	.375	
Oct-11-90	15:05	2	A	.096	IN/S	-64%	.375	
Oct-11-90	15:06	2	H	.162	IN/S	31%	.375	
Oct-11-90	15:06	2	H	.028	g/SE	-81%	.500	
Oct-11-90	15:06	2	V	.275	IN/S	-6%	.375	
Oct-11-90	15:07	3	A	.181	IN/S	-80%	.375	
Oct-11-90	15:07	3	H	.039	IN/S	-91%	.375	
Oct-11-90	15:07	3	H	.192	g/SE	-93%	.500	
Oct-11-90	15:07	3	V	.077	IN/S	-80%	.375	

12 lines this report

Hit any key to continue

IRD Mechanalysis Data Analysis System
Nov-23-90

Page 1

Machine Report (Latest Data) for: F 21 CIRC FAN 4

Date	Time	Pos	Dir	Ampl	Units	Change	Alarm	Inspection
Aug-03-90	14:24	1	A	.236	IN/S	36%	.375	8-3-90 MEASUREMENT REPORT
Aug-03-90	14:24	1	H	.146	IN/S	-51%	.375	
Aug-03-90	14:24	1	H	.207	g/SE	527%	.500	
Aug-03-90	14:24	1	V	.199	IN/S	2%	.375	
Aug-03-90	14:24	2	A	.266	IN/S	29%	.375	
Aug-03-90	14:24	2	H	.124	IN/S	33%	.375	
Aug-03-90	14:24	2	H	.149	g/SE	51%	.500	
Aug-03-90	14:25	2	V	.293	IN/S	17%	.375	
Aug-03-90	14:04	3	A	*.919	IN/S	5%	.375	
Aug-03-90	14:05	3	H	*.452	IN/S	12%	.375	
Aug-03-90	14:05	3	H	*2.89	g/SE	776%	.500	
Aug-03-90	14:05	3	V	*.383	IN/S	22%	.375	
Aug-03-90	14:06	4	A	*.390	IN/S	-16%	.375	
Aug-03-90	14:06	4	H	*.627	IN/S	44%	.375	
Aug-03-90	14:06	4	H	*2.94	g/SE	167%	.500	
Aug-03-90	14:06	4	V	*.625	IN/S	101%	.375	

16 lines this report

FIGURE 4

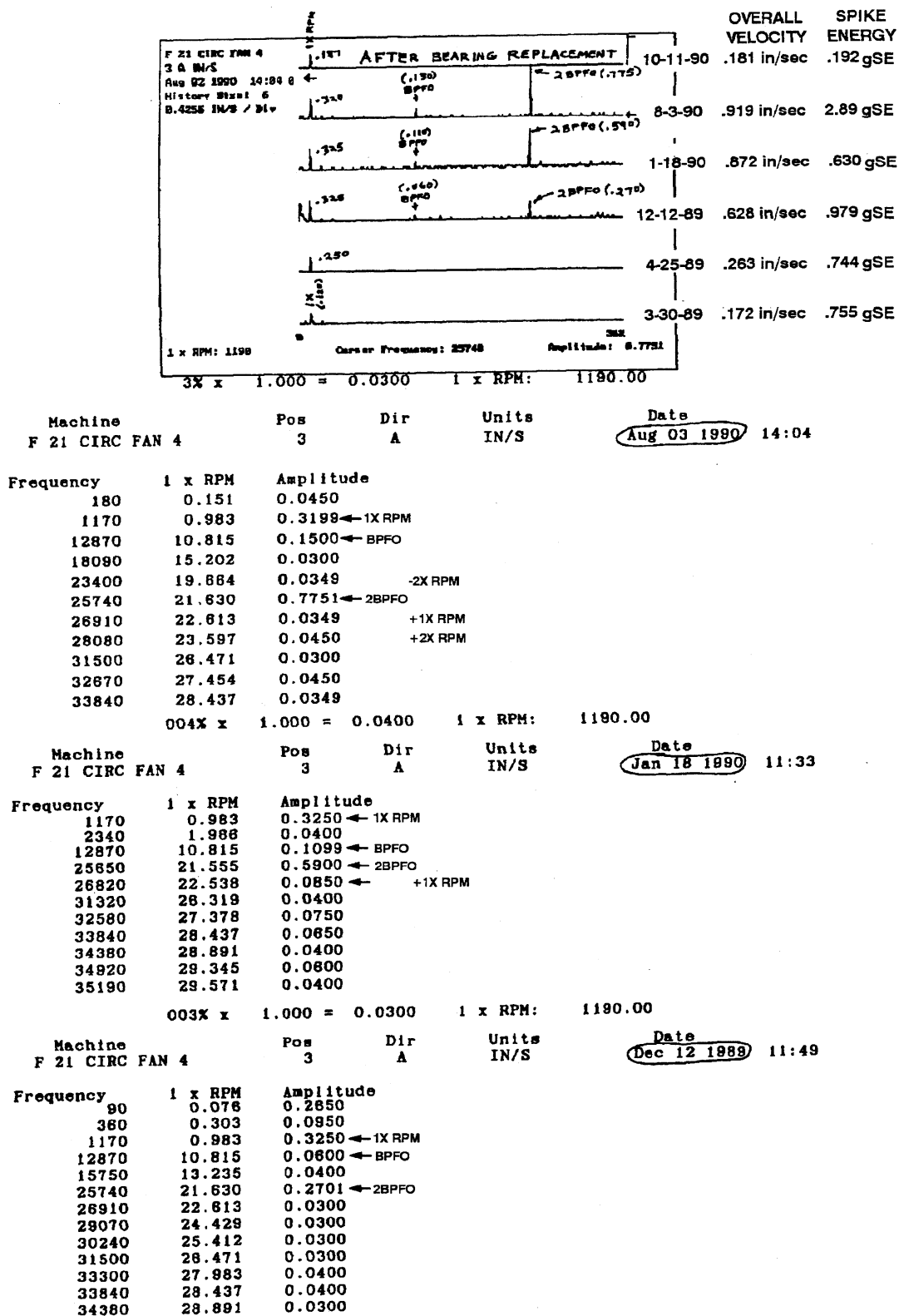


FIGURE 5

DETECTION OF SERIOUS BEARING PROBLEMS WITHIN A 400 HP MOTOR SERVING A HOFFMAN BLOWER

I. BACKGROUND

On January 5, 1988, a client requested the services of Technical Associates to evaluate a potential problem on its 400 HP motor which served a 2-Stage Hoffman blower which was a very critical machine. Measurements by the client with his IRD 880 swept filter analyzer showed a significant increase in vibration, particularly in the higher frequencies of the spectrum. However, due to the wide frequency bandwidth of the IRD 880, it was not possible to separate the bearing frequencies from running speed harmonics. Therefore, the services of Technical Associates using a real-time analyzer were requested to confirm the source of the problem and to make recommendations to correct any problems detected.

II. ANALYSIS RESULTS

Figure 1 shows where measurements were made on the motor and blower bearing housings. Note that each of the motor bearings were Fafnir model 7209 bearings which have each of the following defect frequencies:

Inner Race Freq.	=	BPFI	=	8.185X RPM
Outer Race Freq.	=	BPFO	=	5.816X RPM
Ball Spin Freq.	=	BSF	=	2.486X RPM
Cage Freq.	=	FTF	=	0.415X RPM

Plot 1 is a zoom spectrum that showed the motor running speed (F_m) was 3586 RPM. Plot 2 was taken on motor Position 1H which is shown in Figure 1 on the outboard bearing. It showed low vibration at 1X and 2X RPM, each of which was less than .06 in/sec. However, it also showed a .200 in/sec peak at about 29,700 CPM along with frequencies surrounding this peak. Therefore, a zoom spectrum centering on this 29,700 CPM frequency was captured and is shown in Plot 3. This zoom spectrum showed the actual frequency to be about 29,730 CPM which corresponded to 8.19X RPM. Referring back to the bearing defect frequency list, this confirmed an inner race defect frequency.

Also, referring back to Plot 3, note that it was surrounded by peaks at a spacing of about 1385 CPM (.40X RPM). Therefore, this confirmed the presence of sidebands spaced at cage frequency (FTF) around BPFI. Normally, cage frequencies or cage frequency sidebands are one of the last frequencies to show up in a bearing on its path to failure [the normal order of appearance includes the ball pass frequencies (BPFO and BPFI) followed by the ball spin frequency (BSF) and then the cage frequency of both the inner race

and cage frequencies (FTF)]. Thus, the simultaneous appearance caused a good deal of concern on the health of this outboard motor bearing. Plot 5 next shows the comparison of the spectrum on the inboard motor bearing (Plot 5A) with that captured on the outboard motor bearing (Plot 5B). Note the great difference in vibration amplitude at the bearing frequency on each of these two locations (.001 in/sec on the inboard versus .200 in/sec on the outboard). It was concluded that the likely reason for the great difference was considerably more wear on the outboard motor bearing.

III. RECOMMENDATIONS

Because this simultaneous presence of both inner race frequency itself along with sidebands at cage frequency was confirmed, and due to the fact that this machine was so critical, it was recommended that the plant replace each of the motor bearings at first opportunity to prevent potential catastrophic failure of one or both of these bearings. In addition, recommendations were made on alignment specifications which were 2 mils parallel offset maximum and .3 mil/inch of slope maximum angularity. It was recommended that the manufacturer be consulted on the thermal offset required for cold alignment.

DETECTION OF A SERIOUS PROBLEM WITH A HOFFMAN BLOWER BEARING

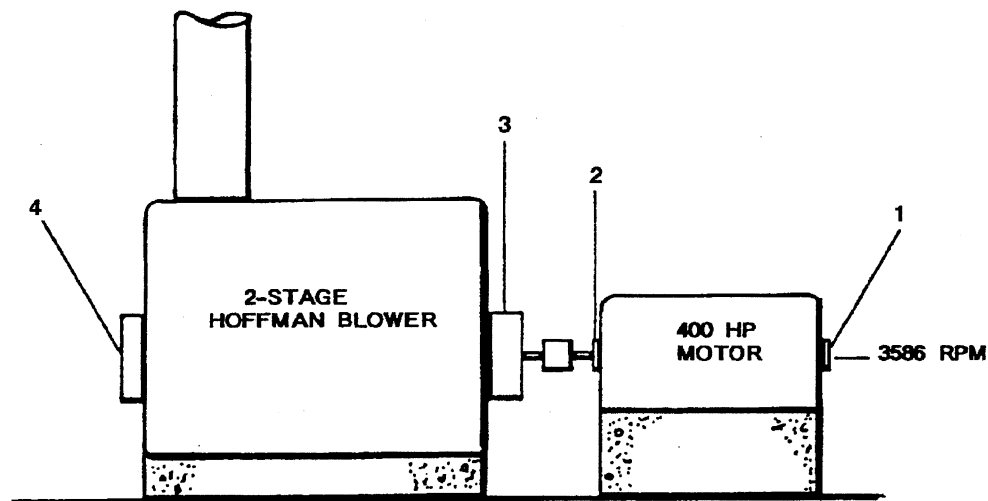
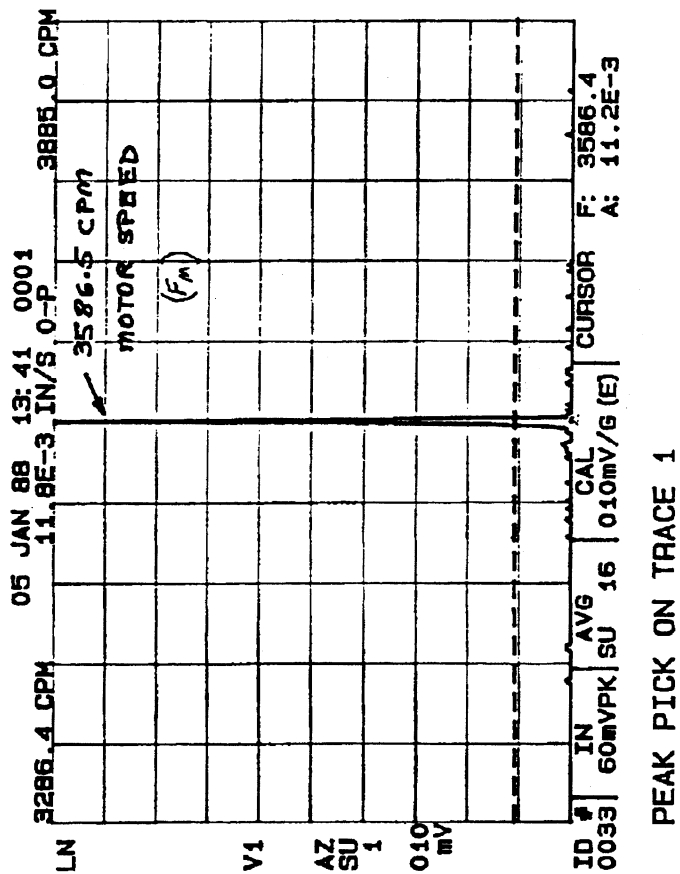


FIGURE 1
HOFFMAN BLOWER
#5
VIBRATION MEASUREMENT LOCATIONS

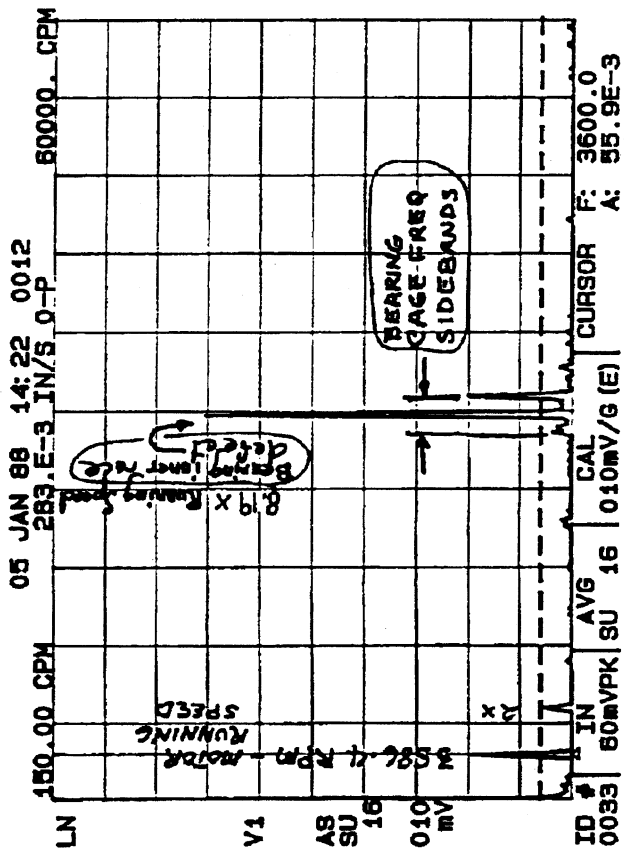


IN/S 0-P	CPM
11.3E-3	3586.5

06-JAN-88

07: 24: 47

PLOT 1



PEAK PICK ON TRACE 1

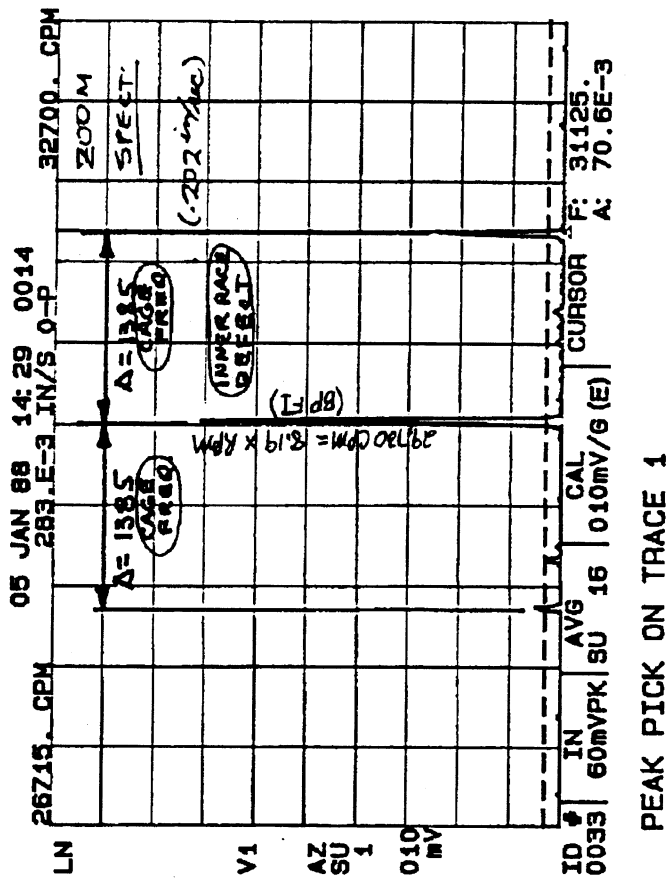
POSITION 1H

06-JAN-88

07: 43: 53

PLOT 2

IN/S 0-P	CPM
55.6E-3	3600.0
200.E-3	29699.
56.9E-3	31199.



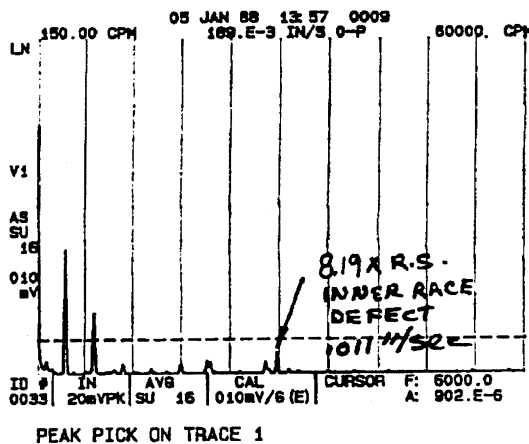
IN/S 0-P	CPM
14.2E-3	28350.
202.E-3	29730.
69.8E-3	31125.

POSITION 1H

06-JAN-88

08: 00: 52

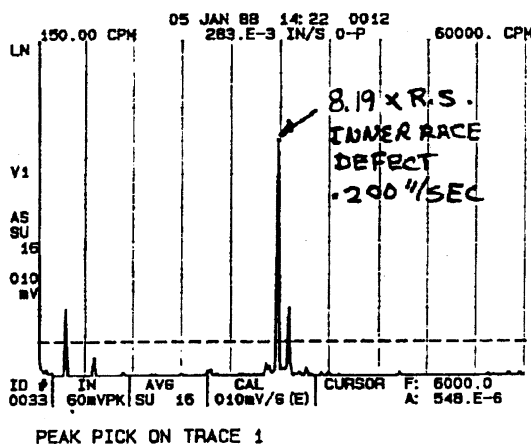
PLOT 3



IN/S 0-P	CPM
70.8E-3	3600.0
34.5E-3	7200.0

PLOT 5A
MOTOR INBOARD BEARING
AXIAL MEASUREMENT
POS. 2A

06-JAN-88



IN/S 0-P	CPM
55.6E-3	3600.0
200.E-3	29699.
56.9E-3	31199.

PLOT 5B
MOTOR OUTBOARD BEARING
AXIAL MEASUREMENT
POS. 1A

GREAT DIFFERENCE IN VIBRATION
AT POSITIONS 1A & 2A

06-JAN-88

10: 02: 47

SURPRISING DETECTION OF A ROLLING ELEMENT BEARING PROBLEM ON A DRAW FRAME INPUT PINION BEARING AFTER REPLACEMENT OF A FALK GEARBOX WHICH HAD BEEN THOUGHT TO BE THE SOURCE OF THE PROBLEM

Subtitle:

(How it Pays to Know the Identity of Components Within Your Critical Machinery)

I. BACKGROUND

Technical Associates of Charlotte, Inc. was contacted with regards to performing **Vibration Diagnostic Testing** on a Draw Frame gearbox shown in Figure 1. The problem was described as a bumping noise and vibration that appeared to be occurring at once per revolution of the gearbox output speed. This draw frame includes each of 30 rolls of 18 inch diameter supported on 9 inch diameter shafts as shown in Figure 2. These rolls are gearbox driven and are outfitted with high cost bearings. There are several individual drives for various sections of the 30 draw frame rolls including one like that shown in Figure 1. This draw frame is used in production of a high strength fiber material called staple ranging in density from 2,000,000 to 3,000,000 denier. From Figure 2, the tow band applies very high forces on the order of 30,000 lbs on each of the bearings supporting the draw frame rolls as the staple wraps around some rolls greater than 180 degrees. This can put great strain on many of the draw frame components as well as their drives.

The Falk gearbox shown in Figure 1 had been replaced at a cost of over \$36,000 which unfortunately had little or no effect on the high vibration problem. Several draw frame roll bearings had also been replaced which likewise showed no improvement. Therefore, the plant requested outside assistance to help identify just what was the source of the problem and what was required to resolve it.

The **Vibration Diagnostic Testing** was performed on June 16, 1989.

The following report contains **CONCLUSIONS**, **RECOMMENDATIONS**, and a **DISCUSSION** of test data obtained.

II. CONCLUSIONS

- A.** Surprisingly, the problem in this machine had nothing to do with components in either the gearbox nor motor, but instead with the bearing supporting the input pinion of the draw frame itself (see Figure 1). An inner race defect frequency and harmonics for the FAG Model #22326ES bearing was detected. This is the bearing closest to the draw frame input pinion. The amplitudes and frequencies associated with the bearing defect indicated that the bearing was in the final stages of failure.

-
- B. Once the data on bearing model numbers and gear tooth counts was made available, the analysis was straightforward, although the vibration spectra had initially appeared very strange when such information was unknown.
 - C. Data obtained showed both the gearbox and motor to be in good condition.
 - D. Due to the inaccessibility of the large draw frame roll bearings mounted deep within the machine, the mechanical condition of these bearings could not be assessed. Recommendations were made by the analyst while on-site how this situation could be remedied by placing permanently mounted accelerometers in key locations which were identified on machine prints given to the analyst. It was important to take the speed of these rolls into account when specifying both hardware and techniques since they normally ranged from only 32 to 104 RPM (see Figure 2).

III. RECOMMENDATIONS

- A. Recommendations were made while on-site to replace the Fag 22326ES Draw Frame input pinion bearings as soon as possible to prevent catastrophic failure which may have been imminent as evidenced by vibration spectra.
- B. Monitor the gearbox utilizing the client's existing IRD 890/7090 **Predictive Maintenance Program** hardware and software to detect similar problems in the future. Also, complete the research on identification of bearing and gear components within the draw frame as well as its driving gearboxes and motors. This data could prove to be invaluable preventing not only future failures of these expensive components, but also helping to prevent costly expenditures replacing gearboxes and/or motors that actually are in good operating condition.

IV. DISCUSSION

A. Initial Analysis Without Available Information on Motor, Gearbox and Draw Frame Components:

Upon arrival at the plant, information concerning the motor, gearbox and draw frame bearing model numbers and gear tooth counts was not yet available. However, after capturing several interesting, high amplitude spectra like those shown in Plots 1, 2 and 3, it became mandatory that this information be supplied since the sources of most of the peaks could not be identified.

Plot 1 showed three peaks, only one of which could be identified due to the lack of information on components within the motor, gearbox and draw frame. Plot 1 is a zoom spectra (taken on position 1H) which included the motor speed of 1229.3 CPM. It also contained unidentified peaks at 1183.5 CPM and 1318.5 CPM. However, even though these two peaks were unidentified, it was noticed that their difference frequency was 135 CPM which equaled the output speed of the gearbox ($1318.5 - 1183.5 = 135$ CPM).

Data captured on the gearbox output shaft at positions 7A and 8A looked very unusual (see Plots 2 and 3). None of the dominant peaks which included frequencies of 1170, 2370 and 3540 CPM could be identified at that

time. These three frequencies could not be related to either fundamentals nor harmonics of either the gearbox input speed (1229 RPM), intermediate speed (660 RPM) nor output speed (133 RPM). Also unusual about Plots 2 and 3 was a whole series of sideband frequencies which were spaced at gearbox output speed surrounding the 1170, 2370 and 3540 CPM peaks (as confirmed by zoom spectra taken around the approximate 1170 and 2370 CPM peaks shown in Plots 4 and 5). Depending on the cause of these spectra generating this unique pattern of frequencies, it was felt that the problem could in fact be quite severe due to amplitudes up to .351 in/sec and sideband levels up to almost .100 in/sec shown in Plots 4 and 5 (bearings, gears, electrical problems?). However, there simply was not sufficient machine component information available.

Therefore, Technical Associates' analyst strongly asserted this data simply must be made available. Despite several arguments claiming this would require an extended time, the client finally dispatched a technician to re-search this information as quickly as possible.

B. Subsequent Analysis Once Gearbox Component Data Was Supplied:

Although it took approximately 2 hours, the plant finally was able to supply information on the gearbox, but advised no information was available at that time concerning components within either the motor nor draw frame.

The following mechanical frequencies were calculated based on the motor speed of 1229.3 RPM as seen in Plot 1 and the component information supplied (the speed was also confirmed by a strobe light):

Motor Speed	= 1,229.3 RPM
Input Shaft Gear Mesh	$1229.3(22T) = 27,044 \text{ CPM}$
Intermediate Shaft Speed	$27,044.6/41T = 659.62 \text{ RPM}$
Intermediate Pinion Gear Mesh	$659.62(20T) = 13,192.5 \text{ CPM}$
Output Shaft Speed	$13,192.4/99T = 133.25 \text{ RPM}$

GEARBOX BEARINGS

1. Input Shaft Bearings	(Timken Nos. 6559 and 6536)
Outer Race (BPFO)	$8.23 (1229.3) = 10,117.14 \text{ CPM}$
Inner Race (BPFI)	$10.77 (1229.3) = 13,239.56 \text{ CPM}$
Ball/Roller (BSF)	$7.14 (1229.3) = 8,777.20 \text{ CPM}$
Cage (FTF)	$.43 (1229.3) = 538.60 \text{ CPM}$

2. Intermediate Shaft Bearings (Timken Nos. HH224335 and HH224310)

Outer Race (BPFO)	$7.11 (659.62) = 4,689.90 \text{ CPM}$
Inner Race (BPFI)	$9.88 (659.62) = 6,517.0 \text{ CPM}$
Ball/Roller (BSF)	$5.87 (659.62) = 3,872.0 \text{ CPM}$
Cage (FTF)	$.42 (659.62) = 277.0 \text{ CPM}$

3. Output Shaft Bearings (Timken No. 99100)

Outer Race (BPFO)	$10.19 (133.25) = 1,357.80 \text{ CPM}$
Inner Race (BPFI)	$12.81 (133.25) = 1,707.0 \text{ CPM}$
Ball/Roller (BSF)	$8.40 (133.25) = 1,119.30 \text{ CPM}$
Cage (FTF)	$.44 (133.25) = 58.60 \text{ CPM}$

Even after compiling all this data, the dominant peaks in spectra shown in Plots 1 through 5 still could not be pinpointed. Each of these spectra were dominated by a frequency of about 1183 CPM or multiples of this frequency. Therefore, information was again requested on the components within the motor and the draw frame which is driven by this gear box. Several technicians now took approximately 4 hours to research this data. Finally, they were able to acquire information on the draw frame components. This finally uncovered the source of the real problem which was never within gearbox components at all. This will next be discussed.

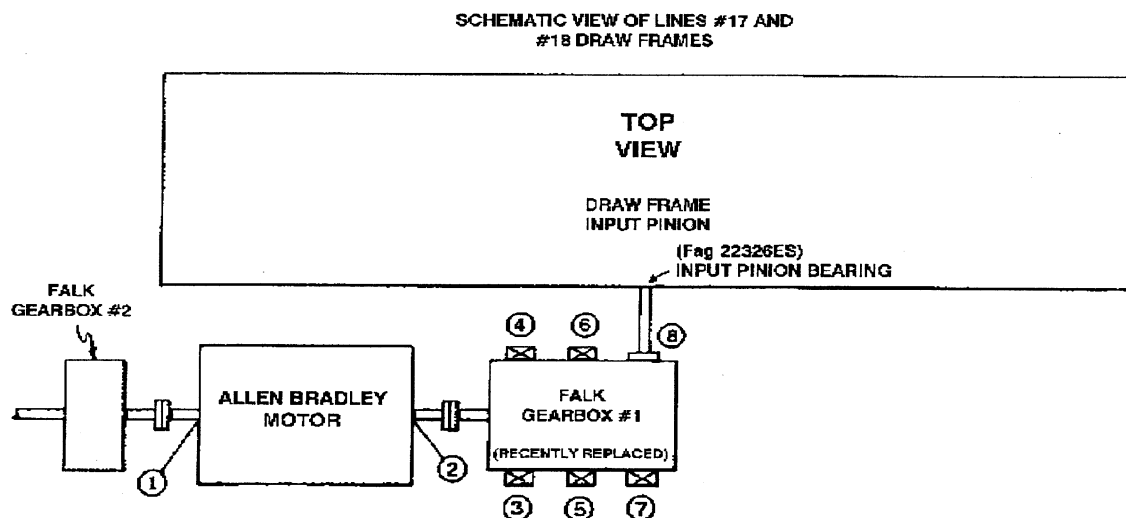


FIGURE 1

DETECTION OF A ROLLING ELEMENT BEARING PROBLEM IN A DRAW FRAME GEARBOX

C. Results After Identification of Motor and Draw Frame Components:

Following below are calculated mechanical frequencies for the draw frame components once this information was finally made available:

DRAW FRAME PINION BEARINGS

1. Inboard Bearing, near Input Pinion (FAG #22326ES)

Outer Race (BPFO) $6.12 (133.25) = 815.50 \text{ CPM}$

Inner Race (BPFI) $8.88 (133.25) = 1,183.3 \text{ CPM}$ **Problem Source**

Ball/Roller (BSF) $2.65 (133.25) = 353.10 \text{ CPM}$

Cage (FTF) $.41 (133.25) = 54.60 \text{ CPM}$

2. Outboard Bearing (Timken #22226ES)

Outer Race (BPFO) $8.22 (133.25) = 1,095.30 \text{ CPM}$

Inner Race (BPFI) $10.80 (133.25) = 1,439.10 \text{ CPM}$

Ball/Roller (BSF) $3.53 (133.25) = 470.40 \text{ CPM}$

Cage (FTF) $.43 (133.25) = 57.30 \text{ CPM}$

Once the information shown above was made available, the solution was straightforward. Please refer to Plots 6 and 7 which are repeats of zoom Plots 4 and 5, but with each of the formerly unidentified peaks now pinpointed. Plots 6 and 7 are zoom spectra with a resolution of 3 CPM. Therefore, the peaks at 1185 and 2367 CPM were confirmed to be 1X BPFI and 2X BPFI for the FAG 22326ES draw frame input pinion bearing, respectively. Then, each of the 135 CPM sidebands surrounding them were found to be spaced at 1X RPM of the draw frame input pinion bearing.

There are normally four (4) stages of rolling element bearing failure as shown in Figure 3. In the early stages of bearing failure, the natural frequencies of bearing components (races, rolling elements, cages, etc.) are excited. These frequencies usually occur between 30,000 and 120,000 CPM. During later stages of failure, bearing defect frequencies appear. Then, as the defect propagates around the periphery of the bearing races, harmonics of defect frequencies appear and develop sidebands normally spaced at running speed when wear becomes clearly visible. In the final stage, 1X RPM will increase in amplitude and the defect frequencies will diminish or disappear completely. At this point, the bearing can fail at any time.

Since the bearing wear was obviously well into late stage 3, it was strongly felt that wear of this bearing was likely quite severe. This was particularly concerning when it was realized that the spectrum of Plot 1 was captured all the way back on position 1H on the motor outboard bearing; even here the dominant vibration was not at 1X motor RPM, but

instead at 1X BPFI of the draw frame pinion bearing remotely located from this position (see Figure 1). Therefore, the analyst recommended replacing the bearing as soon as possible.

V. RESULTS OF CORRECTIVE ACTIONS

NOTE: Plot 8A was taken on the same machine at Gearbox Position 8A after replacement of the draw frame input pinion bearing (FAG 22326ES). It compares directly to Plot 3 taken before (which is shown below Plot 8A as "Plot 8B" to help the reader in spectral comparison). Note the significant drop in vibration and the total elimination of draw frame bearing defect frequencies. As of the writing of this case history almost 2 years since this diagnostic investigation, the draw frame has not experienced any further failures and continues to enjoy low vibration and noise levels.

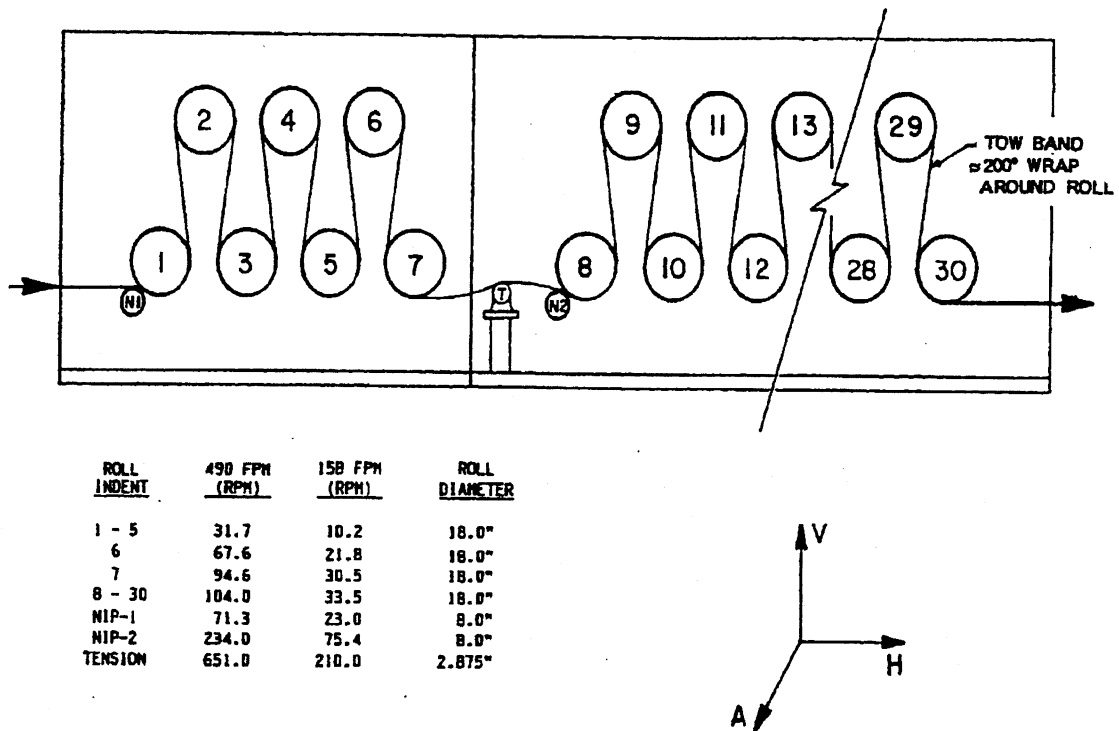
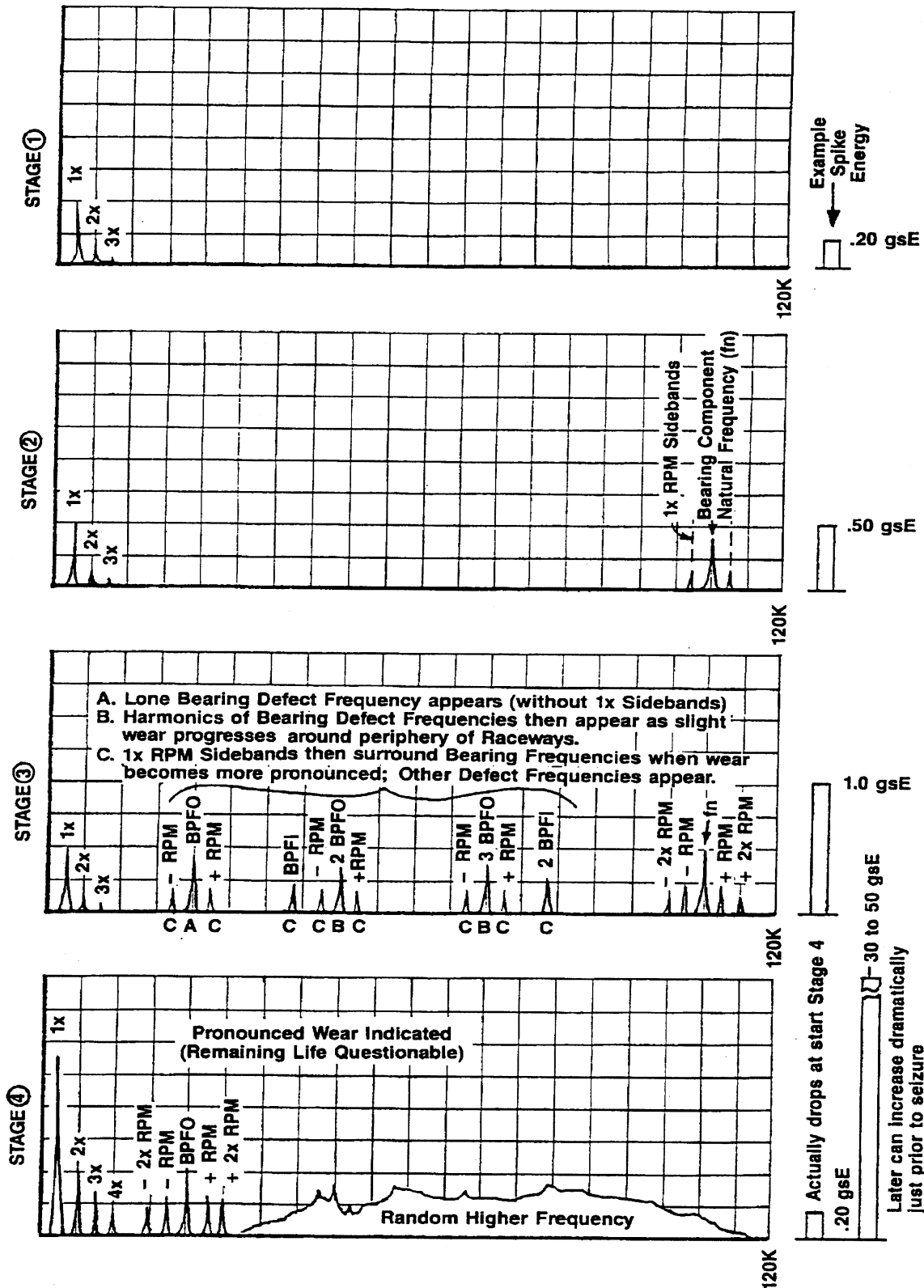


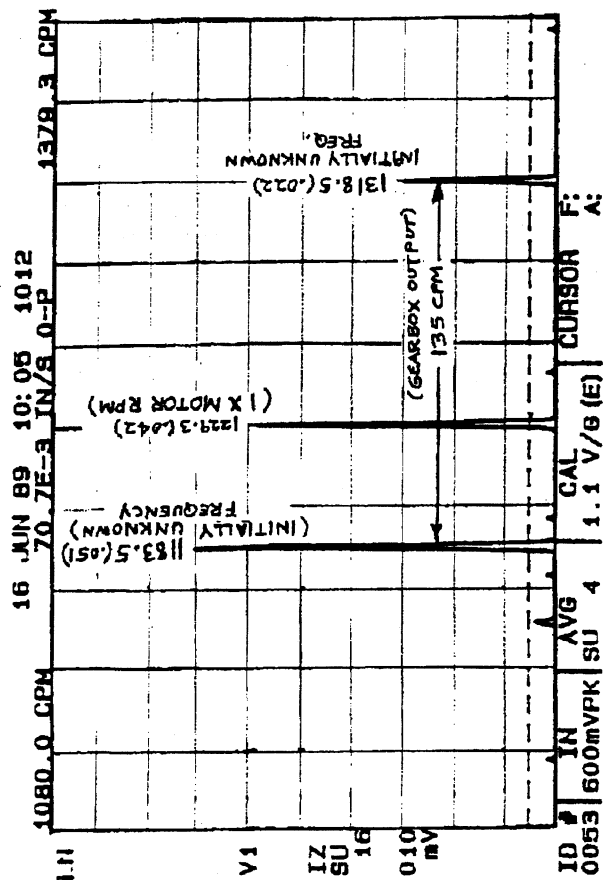
FIGURE 2
TENSION METER SET-UP ON DRAW FRAME

DF1

R-9210-1

FIGURE 3. 4 PRIMARY FAILURE STAGES THRU WHICH MOST ROLLING ELEMENT BEARINGS PASS (VELOCITY SPECTRA)





POSITION 1H

PEAK PICK ON TRACE 1

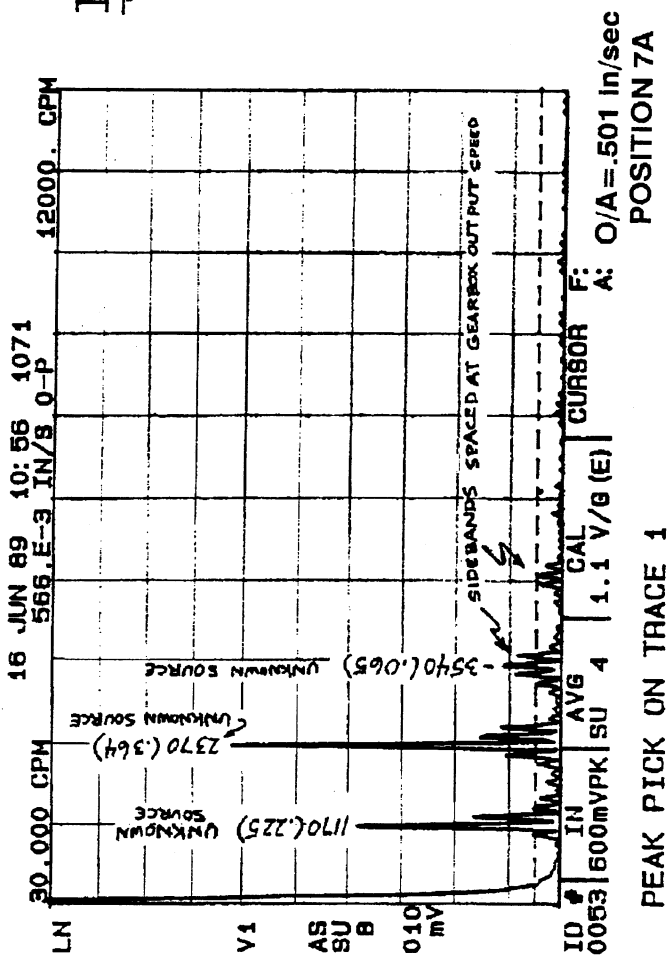
21-JUN-89

07: 28: 04

PLOT 1

LINE 18

IN/S 0-P	CPM
51.3E-3	1183.5
42.3E-3	1229.3
21.6E-3	1318.5



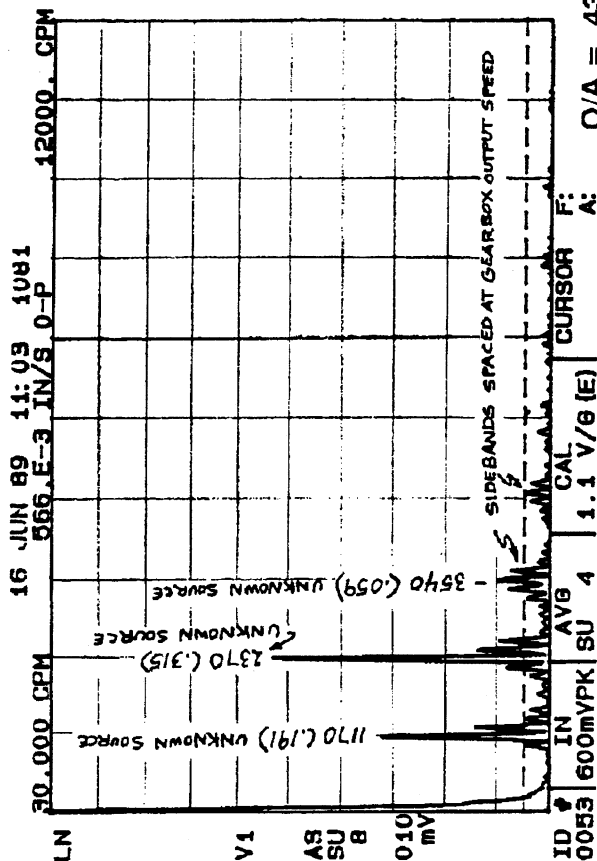
IN/S 0-P	CPM
31.0E-3	1050.0
* 225.E-3	1170.0
98.2E-3	1320.0
31.0E-3	1440.0
62.0E-3	2220.0
* 364.E-3	2370.0
90.4E-3	2490.0
67.2E-3	2640.0
28.4E-3	3269.9
51.7E-3	3420.0
64.6E-3	3540.0
49.1E-3	3690.0

PLOT 2

LINE 18

21-JUN-89

07:52:41



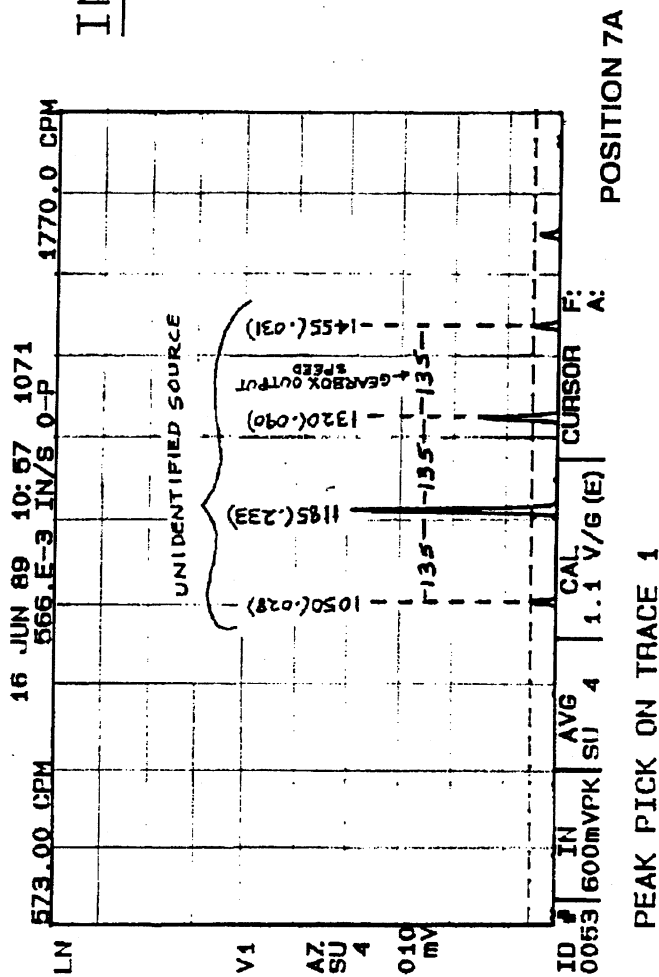
IN/S 0-P	CPM
* 191.E-3	1170.0
85.3E-3	1320.0
49.1E-3	2220.0
* 315.E-3	2370.0
82.7E-3	2490.0
56.9E-3	2640.0
46.5E-3	3420.0
59.4E-3	3540.0
43.9E-3	3690.0

21-JUN-89

08:00:03

PLOT 3

LINE 18



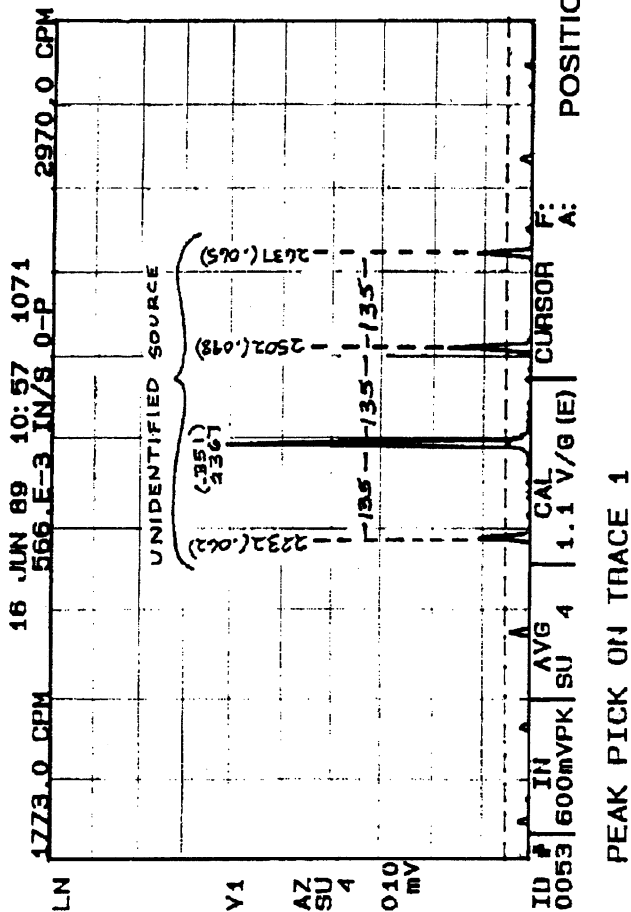
IN/S 0-P	CPM
28.4E-3	1050.0
233.E-3	1185.0
90.4E-3	1320.0
31.0E-3	1455.0

21-JUN-89

07:54:43

PLOT 4

LINE 18



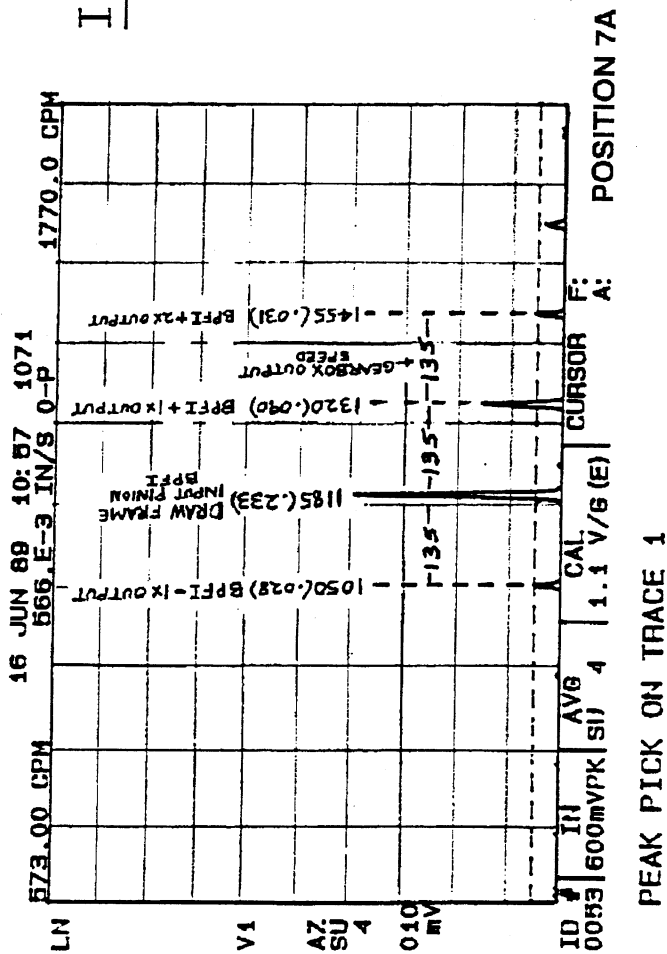
IN/S 0-P	CPM
62.0E-3	2232.0
351.E-3	2367.0
98.2E-3	2502.0
64.6E-3	2637.0

21-JUN-89

07:56:19

PLOT 5

LINE 18



IN/S 0-P	CPM
28.4E-3	1050.0
233.E-3	1185.0
90.4E-3	1320.0
31.0E-3	1455.0

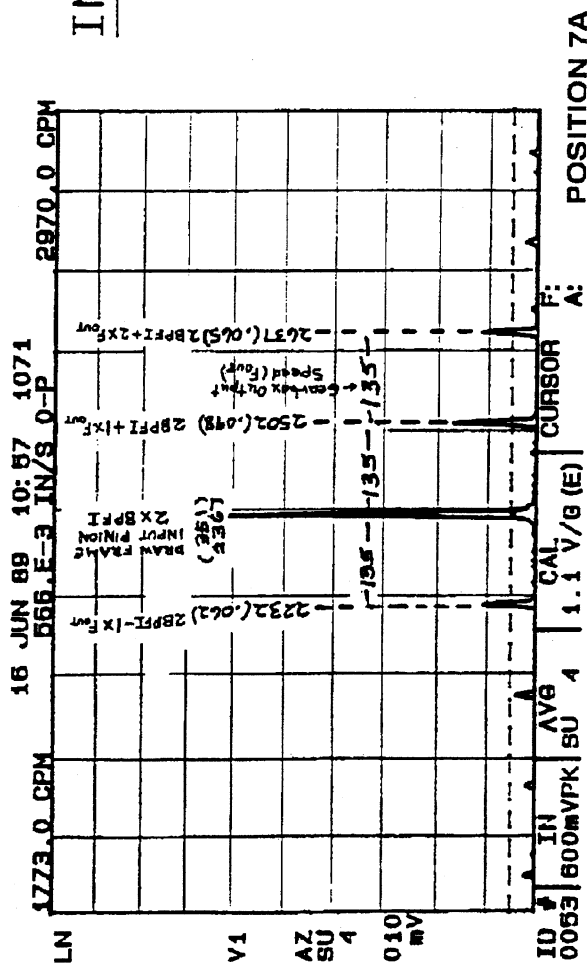
21-JUN-89

07:54:43

SIGNIFICANT WEAR INDICATED OF
FAG 22326ES DRAW FRAME INPUT
PINION BEARING

PLOT 6

LINE 18



PEAK PICK ON TRACE 1

IN/S 0-P	CPM
62.0E-3	2232.0
351.E-3	2367.0
98.2E-3	2502.0
64.6E-3	2637.0

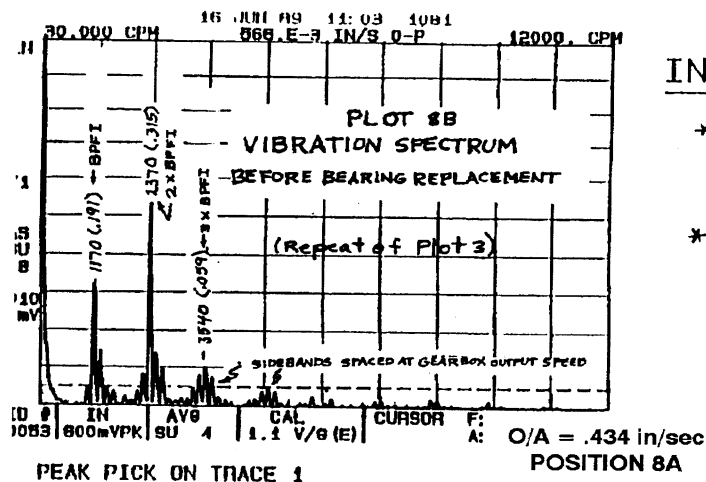
SIGNIFICANT WEAR INDICATED OF
FAG 22326ES DRAW FRAME INPUT
PINION BEARING

21-JUN-89

07:56:19

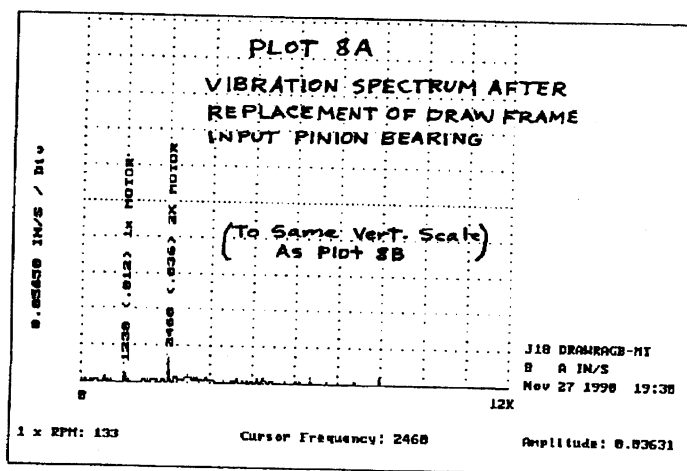
PLOT 7

LINE 18



IN/S 0-P	CPM
* 191.E-3	1170.0
85.3E-3	1320.0
49.1E-3	2220.0
* 315.E-3	2370.0
82.7E-3	2490.0
56.9E-3	2640.0
46.5E-3	3420.0
59.4E-3	3540.0
43.9E-3	3690.0

21-JUN-89



003% x 0.1000 = 0.00300 1 x RPM:

Machine	Pos	Dir	Units	Date	
J18 DRAWRAGB-MT	8	A	IN/S	Nov 27 1990 19:30	
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude
390	-	0.00516	3510	-	0.00718
480	-	0.00405	3600	-	0.00356
660	-	0.00599	3690	-	0.00521
1230	-	0.01231	4230	-	0.00437
1740	-	0.00359	4350	-	0.00589
1980	-	0.00332	4770	-	0.00370
2310	-	0.00451	4920	-	0.00612
2460	-	0.03631	5100	-	0.00751
2670	-	0.00797	5340	-	0.00492
2790	-	0.00443	6150	-	0.00334
2910	-	0.00786	6420	-	0.00902
3000	-	0.01072	8370	-	0.01001
3180	-	0.00720			
3240	-	0.00600			
3330	-	0.00446			
3390	-	0.00426			
3450	-	0.00656			

T - Threshold adjust Q - Quit

SERIOUS PROBLEM OF A BEARING TURNING ON A SHAFT OF ONE MOTOR AND DETECTION OF SEVERE BEARING PROBLEMS ON ANOTHER MOTOR AS EVIDENCED BY NUMEROUS OUTER RACE SUM AND DIFFERENCE FREQUENCIES

I. BACKGROUND

On Saturday March 12, it is understood that the Rolling Mill #4 Main Drive Motor had high vibration and noise amplitudes which could even be felt in the floor and heard in other areas. Technical Associates was contacted by the plant on March 15 and brought in to diagnose the problem. By this time, the noise and vibration had dropped significantly, but there was a peculiar noise pulse present each time the machine was started up under load, particularly at lower speeds up to approximately 300 to 400 RPM. Measurements were made on the #4 Main Drive Motor using a real-time analyzer. In addition, an analysis was conducted on the #4 Rewind Inboard Motor which was experiencing high vibration. Following is a Discussion of the measurements that were made on March 15. Measurements were made with a Rockland 5815A narrowband real-time analyzer.

Later, after corrective actions were completed, a Predictive Maintenance Program was instituted at the client's plant by Technical Associates. This allowed capture of vibration spectra following the corrective actions using an IRD 890 and IRD 7090 software.

II. CONCLUSIONS

A. MILL 4 MAIN DRIVE MOTOR (See Figure 1)

Vibration spectra indicate the following:

1. No vibration spectra on this machine exceeded .300 in/sec alarm levels on this March 15 test date at any of the six (6) positions evaluated on the outboard and inboard motor bearings (See Plots 2 and 3 containing spectra for all 6 positions tested).
2. Bearing Health - No bearing frequencies which would indicate defects were found. Therefore, this indicated motor bearings were likely in good condition.
3. Balance and Alignment - All spectra indicated a well balanced rotor assembly with good alignment.
4. Electrical Condition - No electrical problems were evident from the spectra based on signatures taken in the morning with a lighter gauge metal product, nor were any evident in the afternoon when a heavier gauge, non-annealed product was run which put the rolling mill drive under heavier strain.
5. Mechanical Looseness - Definite mechanical looseness was indicated in spectra taken at all 6 motor positions, based on the presence of a

series of motor speed harmonics (multiples of RPM). On the March 15 test day, it was verbally reported that the source of looseness was likely either between the inboard bearing surface and shaft journal or at the female bearing seat supporting the housing. Since highest overall levels on March 15 were all below .20 in/sec (compared to .30 in/sec Alarm level), there was no indication of imminent failure at that time. However, because plant maintenance engineering reported much higher amplitudes had been present on March 12, and due to the presence of pronounced pulsing noises each time the motor was brought up to speed, concern was expressed on the health of the machine despite the fact that vibration was now, on March 15, below alarm.

B. MILL 4 REWIND MOTOR (See Figure 2)

1. In general, vibration was about 5 times higher on March 15, 1988 than it was when similar spectra were captured on the #4 Inboard Rewind Motor on December 10, 1987. All spectra were dominated by the presence of bearing sum and difference frequencies indicating defects on the outer race. Vibration spectra indicated motor bearings in advanced state of wear. On March 15, it was recommended that both motor bearings be replaced ASAP.
2. The motor still seemed well balanced. There was no evidence of electrical problems at that time. However, slight misalignment was indicated between the outboard and inboard motors and/or there seemed to be a slight problem with the electric clutch assembly located between the two motors (see Figure 2).

III. RECOMMENDATIONS (Made before failure of the Mill 4 Main Drive Motor)

A. MILL 4 MAIN DRIVE MOTOR

Because of the definite mechanical looseness symptoms described in "Conclusions" Section II-A5, it was recommended that the plant take readings each day to monitor possible deterioration of the problem, and also to see if any defect frequencies for the motor bearings would appear. It was not felt that any significant damage had yet been done to the bearings due to the complete absence of defect frequencies (this could be asserted with particular confidence since defect frequencies for this particular SKF 6236 bearing were well removed from integer running speed harmonics - $BPFI = 6.338X$; $BPFO = 4.662X$; $BSF = 3.205X$). In fact, a series of zoom spectra confirmed that no noticeable frequencies other than running speed harmonics and gear mesh frequencies were present. It was further recommended that measurement intervals could be increased to weekly and later monthly if no further damage was evident in future vibration spectra.

UPDATE:

Later, it was reported that the Mill 4 Main Drive Motor did experience failure on March 18. Inspection confirmed the mechanical looseness, and found the source at the inboard motor bearing, revealing that the shaft journal had been turning in the bearing removing .126" from the shaft diameter. In all probability, much of

this damage was done on March 12 when all the high vibration and noise was noticed by the plant. The inspection also confirmed that the bearing itself was still in fact in good shape as had been indicated by the vibration signatures, despite the excessive damage to the shaft.

B. MILL 4 REWIND MOTOR

It was strongly recommended that the plant take immediate steps to replace the motor bearings due to both the amplitude and the presence of a great number of the difference frequencies spaced at the SKF 6218 outer race defect frequency (4.11X RPM). In addition, it was recommended that the plant carefully check the alignment of the rewind drive assembly as well as the condition of the electric clutch assembly located between the out-board and inboard motors (see Figure 2).

IV. DISCUSSION

A. MILL 4 MAIN DRIVE MOTOR

A series of axial, horizontal and vertical measurements were made on the main drive motor bearings in order to assess the condition of the motor (see Figure 1). The vibration spectra derived from these measurements indicated the following:

1. No vibration levels exceeding alarm were present on either motor bearing on this test day.
2. No bearing defect frequencies were present, indicating good bearing condition.
3. Balance and alignment indications were good.
4. No motor electrical problems were present, even when the mill was operating with heavy gage material.
5. The dominant feature of all the spectra taken was the large number of harmonics of running speed, an indication of mechanical looseness in the motor. A verbal diagnosis at the plant was made that the most likely problem was looseness of the inboard bearing seat in which the bearing housing was held (closest to gearbox) and/or looseness between the bearing surface and shaft journal.

A visual inspection of the inboard bearing was made with no concrete conclusions, but there was some indication that the bearing seat could be loose.

Later, it is understood that a shutdown was mandated on March 18 because of sparks and other signs of imminent failure. Disassembly and inspection by the plant revealed that on the inboard end of the motor, the shaft journal had been turning in the bearing and the journal had been eroded to the point of being .126 inch undersized. The journal, normally a press fit on the bearing, now had a very large clearance, leaving the motor shaft loose on the inboard end. This explains the mechanical looseness indicated by the March 15 measurements, an example of which is shown by Plot 1, derived from a measurement at the inboard

bearing of the motor. Note the large number of harmonics of running speed present in the spectrum, a typical spectrum for mechanical looseness. For comparison purposes, Plots 2 and 3 are included showing spectra taken at all 6 motor positions located on the outboard and inboard bearing housings. Note that mechanical looseness is evident in all 6 spectra, based on the large number of 567 RPM running speed harmonics. However, note that all overall levels remained below the alarm level for this machine, which would be about .300 in/sec (overalls ranged from .074 in/sec to .176 in/sec).

The motor was disassembled and the shaft was rebuilt by General Electric Service Shop. It is understood that the motor has been put back into service and is apparently operating satisfactorily. It is recommended that signatures be obtained to check current machine health and to serve as **Baseline Signatures** for future reference.

B. MILL 4 REWIND MOTOR

Measurements made on the Mill 4 Inboard Rewind Motor indicated definite bearing problems with the SKF 6218 outboard bearing. While measurements were being made, a large number of difference frequencies were noticed. The spacing of these difference frequencies was equal to the outer race defect frequency (4.11 F). The upper spectrum of Plot 4 shows a high overall of .544 in/sec, dominated by the outer race difference frequencies. The lower signature is a zoom spectrum showing that the difference frequencies were spaced at 2280 CPM when the motor speed was 555 RPM - equal to 4.11 times the motor speed confirming the presence of bearing wear. For comparison purposes, spectra taken at all 6 positions are provided in Plots 5 and 6. They likewise showed wear of the outboard bearing due to high outer race frequency difference frequencies.

Therefore, on the March 15 test day, it was recommended that both of the bearings be changed in the inboard rewind motor, a recommendation which has been carried out. The motor is now back in service and is apparently functioning well.

V. CORRECTIVE ACTIONS TAKEN

A. MILL 4 MAIN DRIVE MOTOR

Upon investigation, the plant found that one of the problems which may have contributed to the bearing failure was too light a press fit. This bearing had a bore of 180 mm (7.086 in), but a press fit of only .0009 inch. It was decided to increase this to an M6 fit which ranges from .0006 to .0030 inch in the new motor bearings. In addition, an investigation also found that when this variable speed machine ran at certain speeds, the gear mesh frequency transmitting from the gearbox into the motor was exciting an axial natural frequency of the motor. Thus, it was decided not to operate the motor in this particular speed range in which it was resonant.

B. MILL 4 REWIND MOTOR

Each of the motor bearings was first replaced. Then, a slight misalignment problem was corrected.

VI. RESULTS OF CORRECTIVE ACTIONS

Vibration spectra were captured with an IRD 890 analyzer after corrective actions were completed as part of a plant predictive maintenance program.

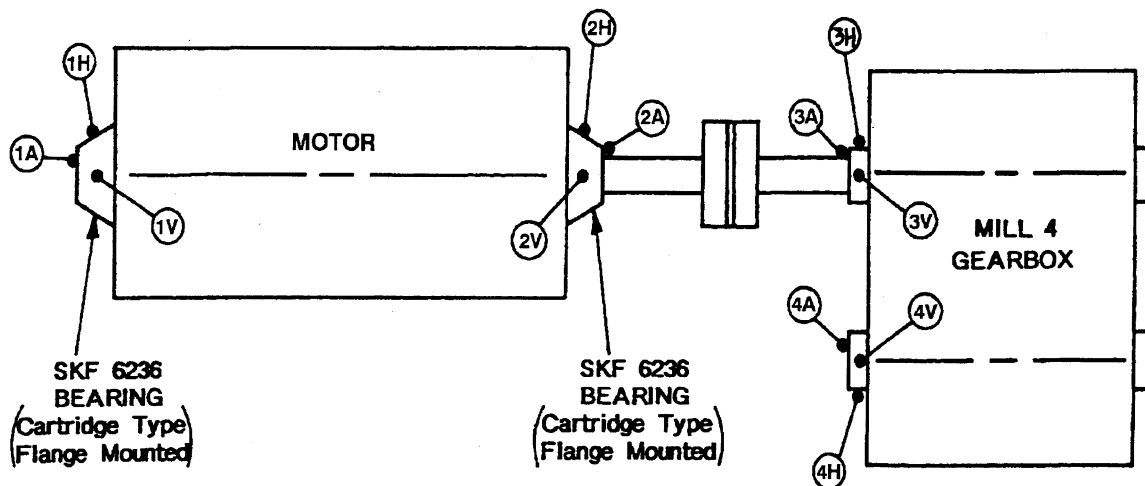
A. MILL 4 MAIN DRIVE MOTOR

Plots 7A and 7B compare spectra taken before and after motor bearing replacement respectively (note that the PMP route spectra had an F_{MAX} of 60,000 CPM compared to 12,000 CPM on the Plot 7A spectrum taken during the earlier diagnostic investigation). Importantly, note that no harmonics at running speed were present in the "After" spectrum of Plot 7B (at least none were present exceeding the .004 in/sec threshold value). Note also that the overall velocity dropped from .176 in/sec to .059 in/sec.

B. MILL 4 REWIND MOTOR

Plots 8A and 8B compare "Before" and "After" spectra on the Mill 4 Rewind Motor. Note the dramatic drop in overall velocity from .544 in/sec to only .018 in/sec - a reduction of over 96%. Also, note the tremendous drop in higher frequency vibration, particularly between 20,000 and 60,000 CPM.

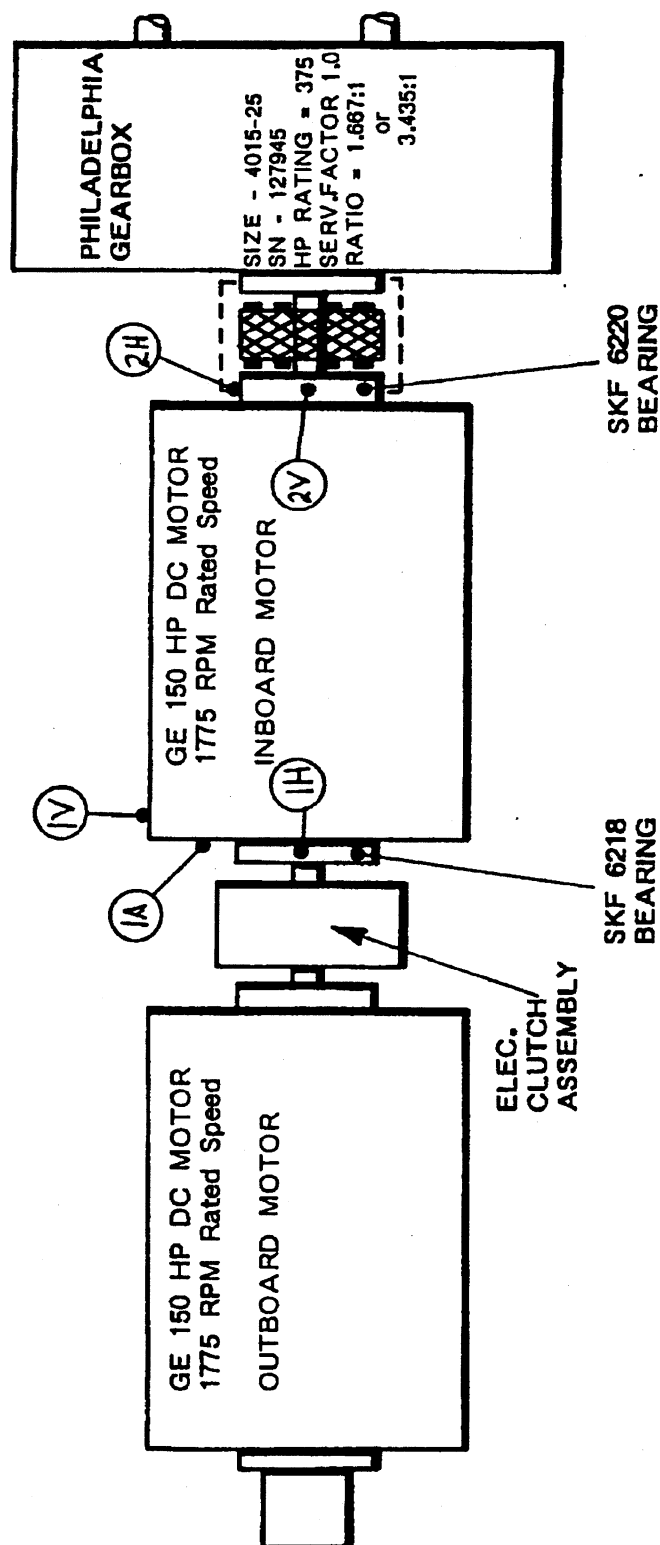
FIGURE 1. MILL 4 MAIN DRIVE ARRANGEMENT



SKF 6236 BEARING DEFECT FREQUENCIES AT 567 RPM:

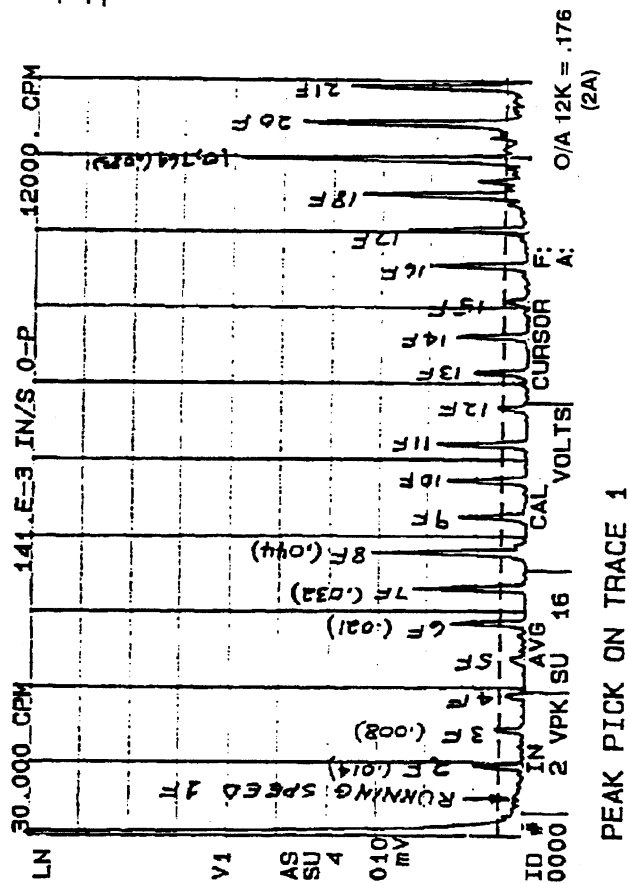
$BPFI = 6.338 \times RPM = 6.338 \times 567 = 3594 \text{ CPM}$
 $BPFO = 4.662 \times RPM = 4.662 \times 567 = 2643 \text{ CPM}$
 $BSF = 3.205 \times RPM = 3.205 \times 567 = 1817 \text{ CPM}$
 $FTF = 0.424 \times RPM = 0.424 \times 567 = 240 \text{ CPM}$

FIGURE 2. MILL 4 REWIND DRIVE ASSEMBLY



MILL 4 REWIND MOTOR BEARING DEFECT FREQUENCIES AT 555 RPM

BEARING	BPFI	BPO	BSF	FTF
SKF 6220	5,907 X = 3278 CPM	4,093 X = 2272 CPM	2,665 X = 1479 CPM	0.409 X = 227 CPM
SKF 6218	5,889 X = 3268 CPM	4,111 X = 2282 CPM	2,723 X = 1511 CPM	0.411 X = 228 CPM

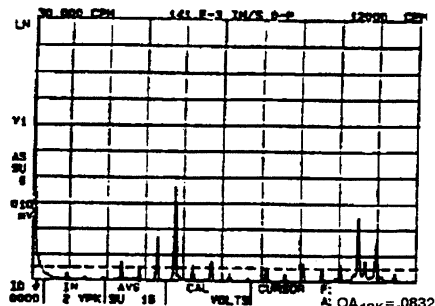


RPM = 567

POINT 2A ON MOTOR
ALL SIGNIFICANT SPIKES
ARE HARMONICS OF
RUNNING SPEED
MECHANICAL LOOSNESS
IS INDICATED

PLOT 1

MILL 4 MAIN DRIVE MOTOR

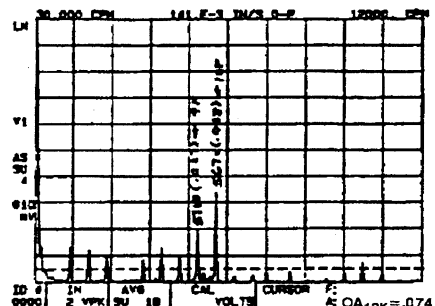


PEAK PICK ON TRACE 1

POSITION IA
(OUTBOARD MOTOR BRG)

IN/S 0-P	CPM
14.2E-3	210.00
7.08E-3	330.00
10.3E-3	2820.0 $\Delta=1140$
23.8E-3	3959.9 $\Delta=570$
51.5E-3	4530.0 $\Delta=570$
8.37E-3	5100.0 $\Delta=570$
10.9E-3	5670.0 $\Delta=2820$
9.66E-3	8490.0 $\Delta=1140$
9.01E-3	9630.0 $\Delta=570$
35.4E-3	10200. $\Delta=210$
11.6E-3	10410. $\Delta=359$
21.2E-3	10769. $\Delta=359$

MIL4 MDMOT

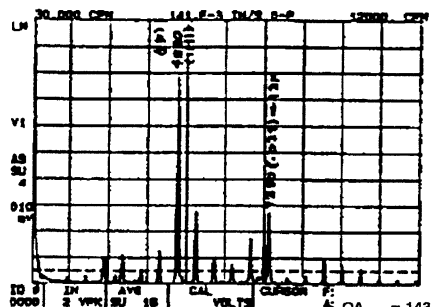


PEAK PICK ON TRACE 1

POSITION IH

IN/S 0-P	CPM
19.3E-3	240.00
7.08E-3	360.00
18.7E-3	1140.0 =2F
17.4E-3	1710.0 =3F
13.5E-3	2250.0 =4F
11.6E-3	3390.0 =6F
18.0E-3	3959.9 =7F
14.2E-3	4530.0 =8F
27.0E-3	5100.0 =9F
47.6E-3	5670.0 =10F
10.3E-3	10200.

MIL4 MDMOT



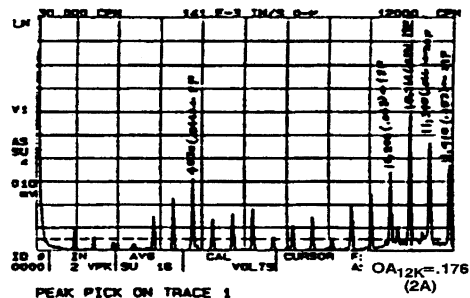
PEAK PICK ON TRACE 1

POSITION IV

IN/S 0-P	CPM
13.5E-3	2250.0 =4F
15.5E-3	2820.0 =5F
7.73E-3	3390.0 =6F
18.0E-3	3959.9 =7F
111.E-3	4530.0 =8F
39.3E-3	5100.0 =9F
13.5E-3	5670.0 =10F
10.3E-3	6240.0 =11F
24.5E-3	6810.0 =12F
34.1E-3	7200.0
38.6E-3	7379.7 =13F
13.5E-3	9060.0 =16F
9.01E-3	9630.0
8.37E-3	10200.

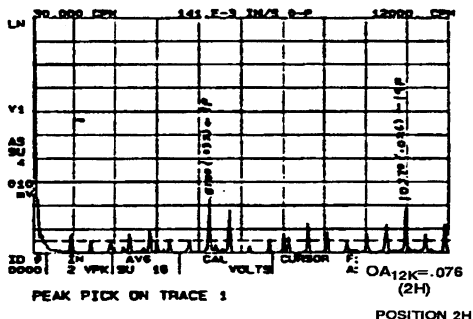
MILL 4 MAIN DRIVE MOTOR POSITIONS IA,IH,&IV PLOT 2

MIL4 MDMOT



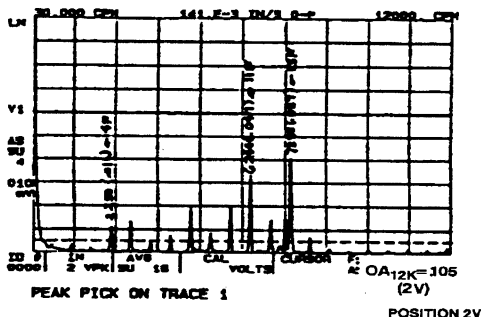
IN/S 0-P	CPM
13.5E-3	1140.0 =2F
8.37E-3	1710.0 =3F
21.2E-3	3390.0 =6F
32.2E-3	3959.9 =7F
44.4E-3	4530.0 =8F
19.3E-3	5100.0 =9F
22.5E-3	5670.0 =10F
25.8E-3	6240.0 =11F
8.37E-3	6810.0 =12F
15.5E-3	7350.0 =13F
20.6E-3	7920.0 =14F
7.08E-3	8490.0 =15F
28.3E-3	9060.0 =16F
34.8E-3	9630.0 =17F
48.3E-3	10200.0 =18F
7.08E-3	10320.0
14.8E-3	10410.0
83.0E-3	10769.0 =19F
7.73E-3	10919.0
11.6E-3	11100.0
9.01E-3	11159.0

MILL 4 MAIN DRIVE MOTOR



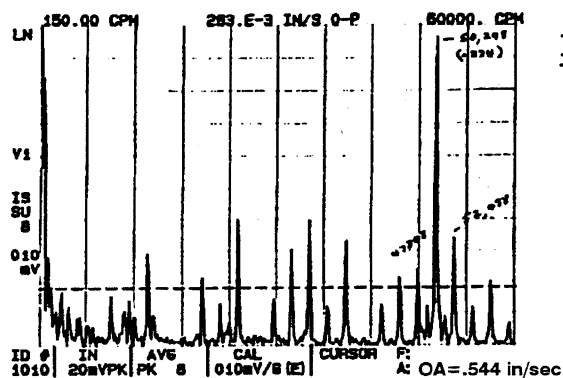
IN/S 0-P	CPM
12.2E-3	240.00
10.3E-3	300.00
7.08E-3	360.00
10.9E-3	1140.0 =2F
10.9E-3	2820.0 =5F
12.9E-3	3390.0 =6F
7.08E-3	3600.0
7.73E-3	3959.9 =7F
7.08E-3	4530.0 =8F
31.5E-3	5100.0 =9F
25.1E-3	5670.0 =10F
7.73E-3	6810.0 =12F
10.9E-3	7200.0
9.01E-3	7350.0 =13F
17.4E-3	7920.0 =14F
12.2E-3	8490.0 =15F
7.08E-3	9060.0 =16F
10.9E-3	9630.0 =17F
16.7E-3	10200.0 =18F
26.4E-3	10769.0 =19F
10.9E-3	11340.0 =20F

MILL 4 MAIN DRIVE MOTOR



IN/S 0-P	CPM
14.2E-3	240.00
15.5E-3	2250.0 =4F
18.7E-3	2820.0 =5F
10.3E-3	3959.9 =7F
28.3E-3	4530.0 =8F
12.2E-3	5100.0 =9F
27.7E-3	5670.0 =10F
47.0E-3	6240.0 =11F
20.0E-3	6810.0 =12F
20.6E-3	7200.0
57.3E-3	7379.7 =13F
9.66E-3	7920.0 =14F

MILL 4 MAIN DRIVE MOTOR POSITIONS 2A,2H,&2V PLOT 3

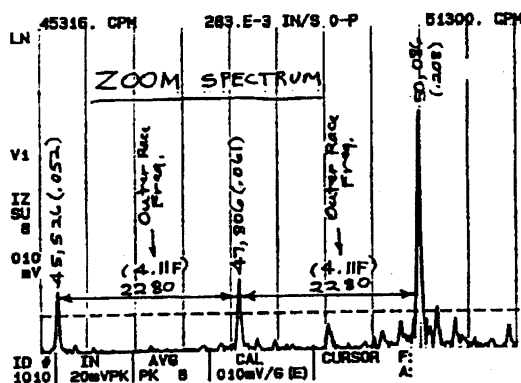


PEAK PICK ON TRACE 1

POSITION IA
OUTBOARD BEARING
(INBOARD MOTOR)
MOTOR SPEED = 555 RPM
(AT THIS INSTANT)

IN/S 0-P	CPM
239.E-3	600.00
80.1E-3	1200.0
53.0E-3	1649.9
82.7E-3	13649.
60.7E-3	20550.
114.E-3	25049.
86.6E-3	31950.
112.E-3	34200.
94.3E-3	38850.
60.7E-3	45600. $\Delta=2248$
68.5E-3	47848. $\Delta=2400$
274.E-3	50248. $\Delta=2250$
96.9E-3	52498.
56.9E-3	57000.

MIL4 REWMT



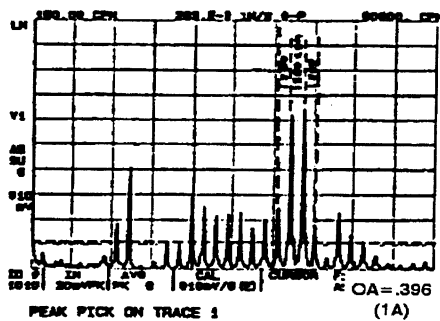
PEAK PICK ON TRACE 1

POSITION IA
CURRENT MOTOR SPEED = 555 RPM (AT THIS INSTANT)
$$2280 \text{ CPM} = \frac{2280}{555} = 4.11 \text{ F} = \text{OUTER RACE BEARING FREQUENCY}$$

IN/S 0-P	CPM
51.7E-3	45526.
60.7E-3	47806.
208.E-3	50086.
34.9E-3	50310.

MILL 4 REWIND MOTOR
POSITION IA
PLOT 4

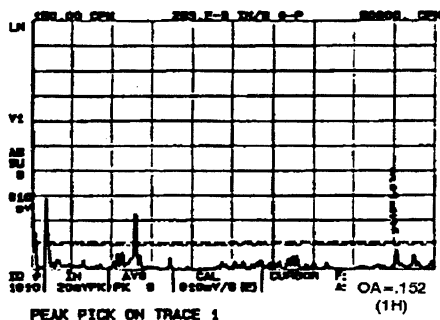
MIL4 REWMT



POSITION IA
MOTOR SPEED $\approx$ 450 RPM
(TOWARDS END OF CYCLE)
(SIDE BANDS $\approx$ 1800 CPM)
MILL 4 REWIND MOTOR

IN/S 0-P	CPM
51.7E-3	12899.
115.E-3	14700.
81.4E-3	23849.
71.1E-3	25649.
60.7E-3	27600.
62.0E-3	29399.
64.6E-3	31199.
46.5E-3	33000.
56.9E-3	34950.
69.8E-3	36750. $\Delta=1800$
176.E-3	38550.
36.2E-3	39000. $\Delta=1350$
182.E-3	40350.
40.1E-3	40800. $\Delta=1350$
50.4E-3	42150. $\Delta=3750$
64.6E-3	45898. $\Delta=1802$
38.8E-3	47700.

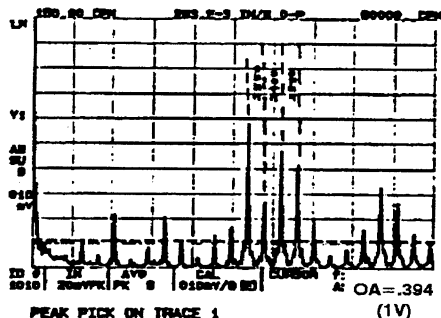
MIL4 REWMT



POSITION IH

IN/S 0-P	CPM
42.6E-3	600.00
81.4E-3	2250.0
63.3E-3	15449.

MIL4 REWMT

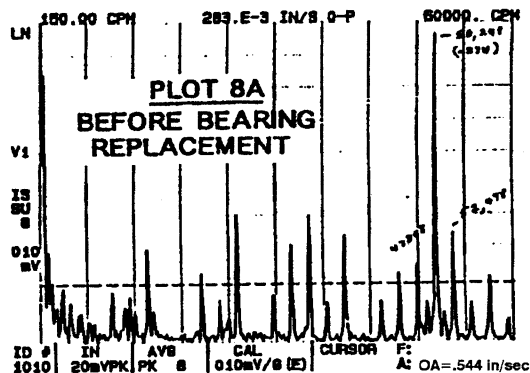


POSITION IV

IN/S 0-P	CPM
96.9E-3	600.00
32.3E-3	1200.0
60.7E-3	12450.
58.1E-3	19800.
37.5E-3	27300.
45.2E-3	29699.
31.0E-3	31650.
164.E-3	32249. $\Delta=2550$
74.9E-3	34800. $\Delta=2400$
133.E-3	37200. $\Delta=2550$
118.E-3	39750.
53.0E-3	42150.
43.9E-3	49650.
93.0E-3	52050.
74.9E-3	54598.
38.8E-3	57000.
28.2E-3	7500
23.8E-3	17400

MILL 4 REWIND MOTOR
(POS. IA,IH,&IV)
PLOT 5

MIL4 REWMT



PEAK PICK ON TRACE 1

(REPEAT OF PLOT 4)

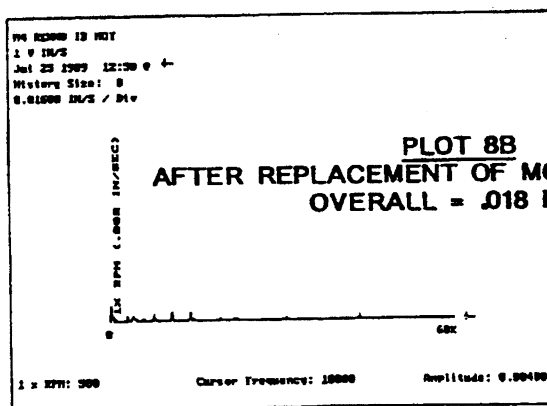
POSITION 1A
OUTBOARD BEARING
(INBOARD MOTOR)
MOTOR SPEED = 555 RPM
(AT THIS INSTANT)

IN/S 0-P

CPM

239.E-3	600.00
80.1E-3	1200.0
53.0E-3	1649.9
82.7E-3	13649.
60.7E-3	20550.
114.E-3	25049.
86.6E-3	31950.
112.E-3	34200.
94.3E-3	38850.
60.7E-3	45600.-
68.5E-3	47848.-
274.E-3	50248.-
96.9E-3	52498.-
56.9E-3	57000.-

△=2248
△=2400
△=2250



Machine M4 REMND IS MOT			Pos 1	Dir V	Units IN/S	Date Jul 25 1989 12:50
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude	
150	0.30	0.00699	43200	86.40	0.00248	
600	1.20	0.00200				
900	1.80	0.00098				
3000	6.00	0.00200				
4050	8.10	0.00248				
5550	11.10	0.00098				
5850	11.70	0.00098				
7200	14.40	0.00149				
7650	15.30	0.00349				
10800	21.60	0.00400				
13950	27.90	0.00400				
14400	28.80	0.00098				
19200	38.40	0.00098				
21600	43.20	0.00098				
22050	44.10	0.00098				
30000	60.00	0.00098				
30450	60.90	0.00149				

T - Threshold adjust Q - Quit

DETECTION OF A BALL BEARING PROBLEM ON A GLAND STEAM EXHAUST FAN WHOSE BEARING MANUFACTURER & MODEL NUMBER WERE UNKNOWN

I. BACKGROUND

A utility company contracted Technical Associates to provide them a complete vibration analysis predictive maintenance program. This program began by acquisition of baseline measurements on all machines selected followed by data analysis and composition of a **Baseline Report**. For the most part, little was known about the machines except for nameplate data. One of the machines tested was the Gland Steam Exhaust Fan. This unit was an integrally mounted fan (shaft mounted) with anti-friction bearings and a nominal running speed of 3500 RPM. The machine arrangement is shown in Figure 1.

II. DISCUSSION OF RESULTS

IRD Mechanalysis Data Analysis System
Dec-04-89

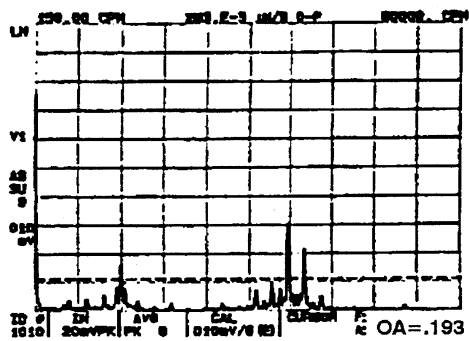
TABLE I.

Machine Report (Latest Data) for: GLND STM EXH 61

Date	Time	Pos	Dir	Ampl	Units	Change	Alarm	Inspection
Aug-24-89	15:00	1	A	.286	IN/S	-----	% .375	
Aug-24-89	15:00	1	H	.171	IN/S	-----	% .375	
Aug-24-89	15:00	1	H	1.14	g/SE	-----	% 1.25	
Aug-24-89	15:00	1	O	.176	IN/S	-----	% .375	
Aug-24-89	15:00	1	V	.211	IN/S	-----	% .375	
Aug-24-89	15:02	2	H	.208	IN/S	-----	% .375	
Aug-24-89	15:02	2	H	*1.32	g/SE	-----	% 1.25	
Aug-24-89	15:02	2	V	*.401	IN/S	-----	% .375	

8 lines this report

Table I shows the list of overall levels of vibration (in/sec) and spike energy (g/SE). Note from Table I that both spike energy (g/SE) measurements on the outboard and inboard bearings were relatively high (1.14 g/SE on the outboard Position 1 bearing and 1.32 g/SE on the inboard bearing Position 2). The Position 2H spike energy exceeded the alarm as did the Position 2 vertical (2V) overall velocity measurement (.401 in/sec compared to an alarm of .375 in/sec). The Position 1V spectrum (Figure 2) had a well defined frequency at 76,200 CPM with running speed (3600 CPM nominal) sidebands. The Position 2V spectrum shown in Figure 3 had a moderately large 1X RPM amplitude of .350 in/sec and the same 76,200 CPM frequency, again with running speed sidebands.



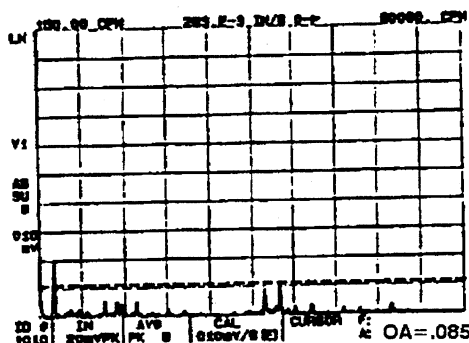
PEAK PICK ON TRACE 1

(2A)

POSITION 2A

IN/S 0-P	CPM
45.2E-3	11850.
86.6E-3	35700.
60.7E-3	38100.
19.9E-3	30900
26.5E-3	33300

MIL4 REWMT



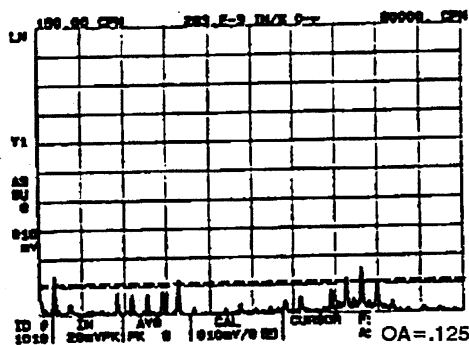
PEAK PICK ON TRACE 1

(2H)

POSITION 2H

IN/S 0-P	CPM
55.6E-3	2250.0
26.4E-3	31800
27.9E-3	34050

MIL4 REWMT



PEAK PICK ON TRACE 1

(2V)

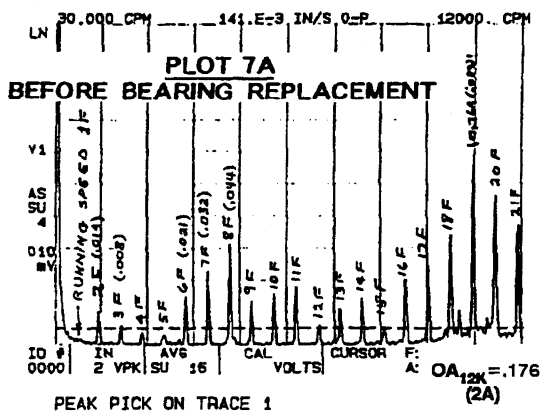
POSITION 2V

IN/S 0-P	CPM
38.8E-3	2250.0
33.6E-3	19650.
33.6E-3	43498.
43.9E-3	45750.
31.0E-3	47848.
22.4E-3	10950
19.4E-3	13050
19.5E-3	15300
22.4E-3	17400
34.1E-3	19650
23.2E-3	41400

**MILL 4 REWIND MOTOR
(POS. 2A,2H,&2V)**

PLOT 6

MIL4 REWMT



(REPEAT OF PLOT 1)
RPM = 567

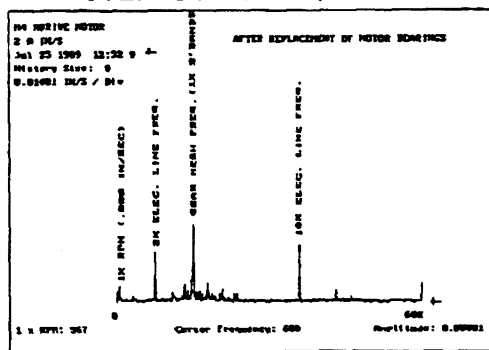
POINT 2A ON MOTOR
ALL SIGNIFICANT SPIKES
ARE HARMONICS OF
RUNNING SPEED

MECHANICAL LOOSENESS
IS INDICATED

IN/S 0-P	CPM
13.5E-3	1140.0=2F
8.37E-3	1710.0=3F
21.2E-3	3390.0=6F
32.2E-3	3959.9=7F
44.4E-3	4530.0=8F
19.3E-3	5100.0=9F
22.5E-3	5670.0=10F
25.8E-3	6240.0=11F
8.37E-3	6810.0=12F
15.5E-3	7350.0=13F
20.6E-3	7920.0=14F
7.08E-3	8490.0=15F
28.3E-3	9060.0=16F
34.8E-3	9630.0=17F
48.3E-3	10200.0=18F
7.08E-3	10320.0
14.8E-3	10410.0
83.0E-3	10769.0=19F
7.73E-3	10919.0
11.6E-3	11100.0
9.01E-3	11159.0

MILL 4 MAIN DRIVE MOTOR

PLOT 7B
AFTER REPLACEMENT OF MOTOR BEARINGS
OVERALL = .059 in/sec



	0028 x 0.2000 = 0.00400			1 x RPM:		567.00	
Machine	Pos	Dir	Units	Date			
M4 MAIN DRIVE MOTOR	2	A	IN/S	Jul 25 1989	12:32		
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude		
150	0.26	0.00550	36000	63.49	0.01050		
600	1.06	0.00801	43200	76.19	0.00648		
7200	12.70	0.02701	59850	105.56	0.01048		
10500	18.52	0.00600					
10950	19.31	0.00400					
12900	22.75	0.00950					
13500	23.81	0.00648					
14700	25.93	0.04249					
15300	26.98	0.00550					
15900	28.04	0.00648					
16500	29.10	0.00448					
17550	30.95	0.01048					
18450	32.54	0.00448					
19950	35.19	0.00448					
20550	36.24	0.00750					
22800	40.21	0.00499					
23400	41.27	0.00499					

T - Threshold adjust Q - Quit

III. CONCLUSIONS

- A. The 76,200 CPM frequency seen on both motor bearings was likely the natural frequency of one of the bearing components (such bearing component natural frequencies are commonly found between approximately 30,000 and 120,000 CPM). When such bearing natural frequencies are sufficiently excited to cause them to appear in the velocity spectrum, this normally indicates the bearing is passing through the second stage of wear (such rolling element bearings usually have four rather well defined wear stages until eventual failure). Normally, when 1X RPM sidebands then appear around the bearing natural frequency, this signals a more serious problem and that the bearing is nearing the end of Stage 2. At this point, the bearing usually has only about 5% remaining life.
- B. The fan wheel is unbalanced.

IV. RECOMMENDATIONS

- A. Since the signatures on both bearings indicated late Stage 2 wear, and because spike energy levels were likewise high on each, it was recommended that both motor bearings be replaced.
- B. It was then recommended the plant reevaluate the unbalance condition after the bearings were replaced. If both signature and phase analysis then still indicated unbalance, the plant was instructed to clean and balance the fan wheel (see Figure 4).



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(704-333-9011)

MACHINERY DATA GUIDE SHEET

RELIABILITY SURVEY

predictive maintenance program

(Page ___ of ___)

UNITS

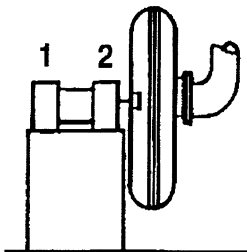
51

52

61

62

DETECTION OF A BALL BEARING PROBLEM ON A GLAND STEAM
EXHAUST FAN WHOSE BEARING MANUFACTURER AND MODEL
NUMBER WERE UNKNOWN



MACHINE IDENT: GLAND STEAM EXHAUST FAN

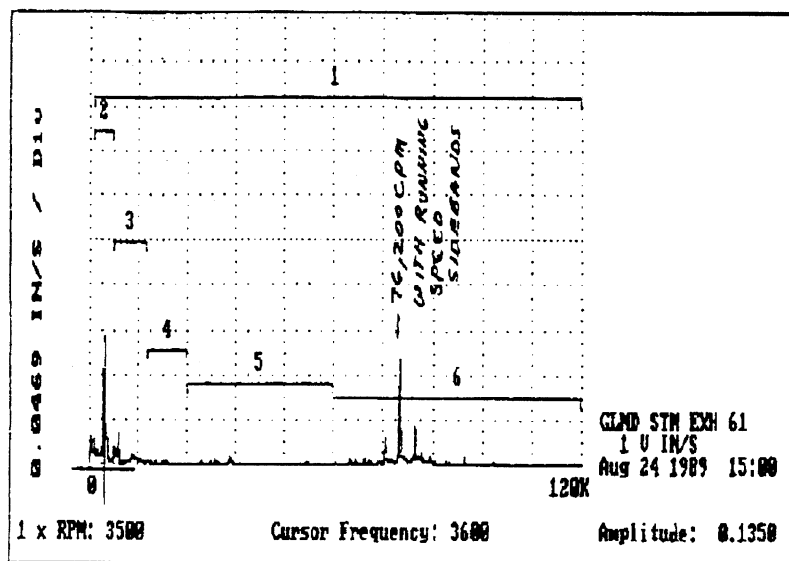
AREA: _____

ROUTE: _____

DRIVER	INTERMEDIATE	DRIVEN
RPM min : <u>3500</u>	RPM in : _____	RPM min : <u>3500</u>
max : _____	out : _____	max : _____
MFGR : _____	RATIO: _____	MFGR. : _____
MODEL : _____	MODEL : _____	MODEL : _____
SER NO. : _____	SER NO. : _____	SER NO. : _____
V _____ P _____ A _____	RATING : _____	RATING : _____
SHEAVE : _____ DIA.	BELT # : _____	#BLADES : _____
BRG#: <u>A</u> (I)	BRG#: _____ (I)	BRG#: <u>A</u> (I)
BRG#: _____ (O)	BRG# _____ (O)	BRG#: _____ (O)
H.P.: <u>10</u>	GEAR T: _____ CT.	GEAR T: _____ CT.
OTHER: _____	OTHER: _____	OTHER: _____

FIGURE 1

IF-08

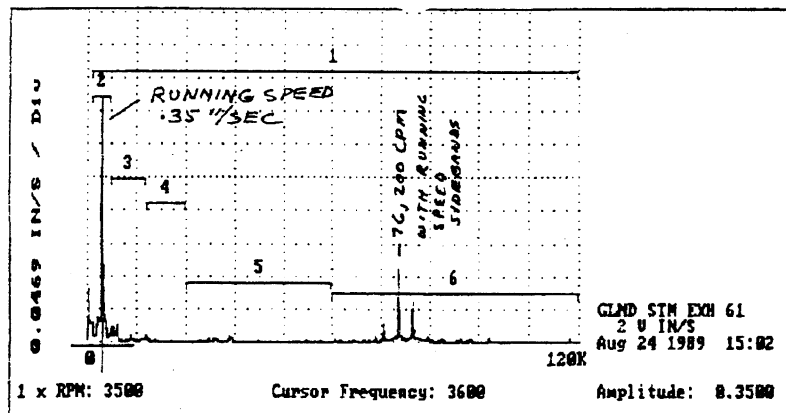


002% x 0.500 = 0.0100 1 x RPM: 3500.00

Machine	Pos	Uir	Units	Date
GLND STM EXH 61	1	U	IN/S	Aug 24 1989 15:00
Frequency	1 x RPM	Amplitude		
300	0.09	0.0300		
1200	0.34	0.0250		
2100	0.60	0.0150		
3600	1.03	0.1350		
6000	1.71	0.0175		
7200	2.06	0.0325		
10800	3.09	0.0100		
72600	20.74	0.0300		
76200	21.77	0.1100		
79500	22.71	0.0400		
81300	23.23	0.0125		
83400	23.83	0.0100		

T - Threshold adjust Q - Quit

FIGURE 2



002Z x 0.500 = 0.0100 1 x RPM: 3500.00

Machine	Pos	Dir	Units	Date
GLND STM EXH 61	2	U	IN/S	Aug 24 1989 15:02
Frequency	1 x RPM	Amplitude		
300	0.09	0.0775		
2400	0.69	0.0350		
3600	1.03	0.3500		
6300	1.80	0.0200		
7200	2.06	0.0225		
14400	4.11	0.0100		
72600	20.74	0.0250-1X RPM S.B.		
76200	21.77	0.1050 ←		
79800	22.80	0.0575-1X		
83400	23.83	0.0100-2X		

T - Threshold adjust Q - Quit

FIGURE 3

SUMMARY OF RESULTS AND RECOMMENDATIONS FOR MACHINES FOUND TO HAVE PROBLEMS
(SORTED FIRST BY PROBLEM PRIORITY; THEN BY PROBLEM SEVERITY WITHIN PRIORITY)

PAGE 2 of 7

AS OF: 8/24/89

AREA	MACHINE COMPUTER NAME	INDICATED PROBLEM(S)																HIGHEST OVERALL VIBRATION	RECOMMENDATIONS & REMARKS	SEE PLOT NO.				
		UNBALANCE	MISALIGNMENT/SHIFT FOOT	BENT SHAFT	LUBRICATION PROBLEM	BEARING PROBLEM	MECHAN. LOOSENESS	BELT/SHAVE PROBLEM	RESONANCE PROBLEM	GEAR PROBLEM	BLADE PASSING PROB.	CAVITATION	MOTOR ELECTRICAL	DR. WHIRL/WHIP	LOOSE FOUNDATION	VIB. MONITOR PROB.	SEAL /MOTOR RUB				OTHER (SEE REMARKS)	PRIORITY OF PROBLEM	POSITION NO.	LEVEL
	GLND STM EXH 61	●			●														①	IV 2V	.2111 .4011	Band 6 alarms at Positions IV and 2V due to bearing defect frequency with running speed (3600 CPM) sidebands. Plot 28 also exhibits unbalance. Replace motor bearings then re-evaluate balance condition. Clean and balance if still required.	27 28	
	TURB DRV BFP 51	●	●																2	1A IV 2H 3A M4 V MILS	.2791 .5101 .3201 .3091 2.30 MILS	Moderate unbalance and misalignment noted at Positions 1 and 2 on the pump and misalignment at Position 3 on the turbine. Suggest checking the alignment between the pump and turbine at Positions 2 and 3 during the next scheduled overhaul. Make any necessary corrections to the alignment. Also, inspect the impeller for foreign debris, removing same, if found, then re-evaluate unbalance and balance if still required. The non-contact monitor displacement (MILS) reading at Position M4 V at the turbine indicates a slight mechanical looseness. Trend all above conditions.	54 55 56 57 58	
	SERVTR VPMP 61	●			●														2	1S-H	.4781	Noticeable one times RPM vibration with harmonics indicating unbalance and mechanical looseness. Mechanical looseness may be from a worn bearing or loosened fasteners. Unbalance could be from foreign material caught in one of the impeller blades. Inspect for above and repair/replace as necessary.	48	
	TURB DRV BFP 61	●																	2	1H 2H	.3701 .4581	Moderate unbalance noted at Positions 1 and 2 on the pump. During the next scheduled outage, suggest inspecting the impeller for foreign debris, removing same, if found. Then re-evaluate unbalance condition and balance if still necessary. A single measurement on the speed increaser showed a relatively high gear mesh amplitude but there is very little sidebanding indicating that the gears are probably OK. The Position 5V measurement shows a gear mesh amplitude of .290 in/sec. Plot 61. This should be carefully trended to insure that no upward trend occurs.	59 60 61	

① = TAKE IMMEDIATE REPAIR ACTION DUE TO SEVERE PROBLEMS NOTED (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)
2 = SCHEDULE REPAIRS TO CORRECT PROBLEMS NOTED AT CONVENIENCE (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)
3 = DESIGNATED PROBLEM DOES NOT WARRANT ACTION AT THIS TIME (CONTINUE ONLY WITH RELIABILITY SURVEY TREND ANALYSIS)



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FIGURE 4.

R-9210-1

SIGNIFICANT IMPROVEMENT IN MOTOR AND BLOWER VIBRATION DUE TO BELT REPLACEMENT

I. BACKGROUND

During a Predictive Maintenance Program vibration survey by Technical Associates on March 22, 1989, a belt-driven air handler, 304 AIRHAND 004, was detected as having an overall vibration of 1.34 in/sec on Position 2H. Position 2H is the inboard motor bearing horizontal measurement as seen in the machine sketch of Figure 1. Note the Figure 1 spectrum for Position 2H on March 22, 1989 shows the dominant vibration was .930 in/sec at 450 CPM. Motor speed was a nominal 1800 RPM and fan speed a nominal 1050 RPM as verified by a hand-held tachometer measurement. It was determined that 2X belt frequency approximated 450 CPM and was the likely cause of the high amplitude. A report of this finding was issued on 3/22/89 warning of the high vibration and its possible causes as seen in Figure 2 which is a copy of the **Rank-Ordered Results and Recommendations Report** issued to the client by Technical Associates. Unfortunately, the client did not follow the recommendations given in Figure 2, but simply aligned and tensioned the belts as seen by the 5/23/89 spectrum taken during the next survey. Of course, the 450 CPM belt frequency second harmonic was still present dominating the spectrum. Again a report was issued recommending replacement of the belt followed by proper tensioning and alignment of the motor and fan sheaves.

II. CONCLUSIONS

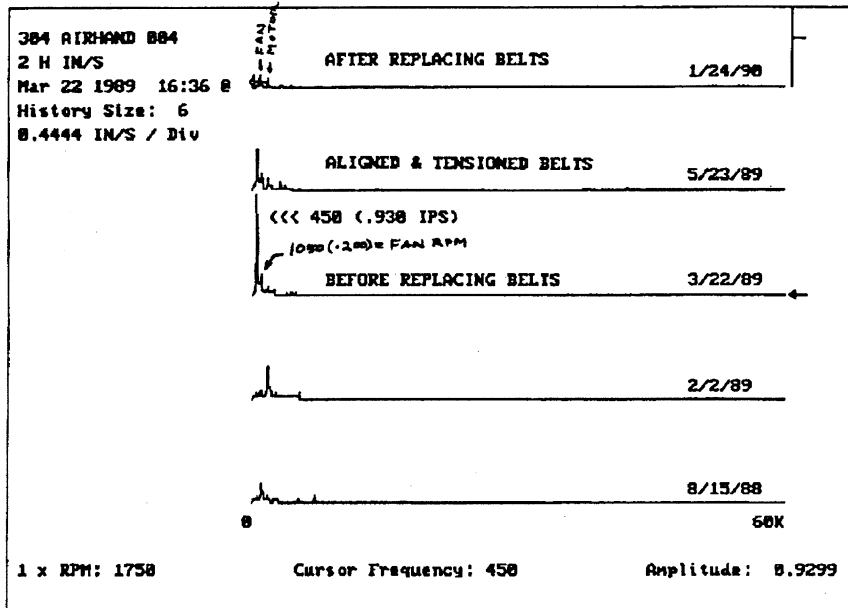
- A. The 450 CPM amplitude was the result of a belt and/or sheave problem.

III. RECOMMENDATIONS

- A. Replace the belts which are the dominant vibration source in this machine.
- B. Check the fan sheave for excessive runout and repair/replace as needed.
- C. Align and tension the belts.

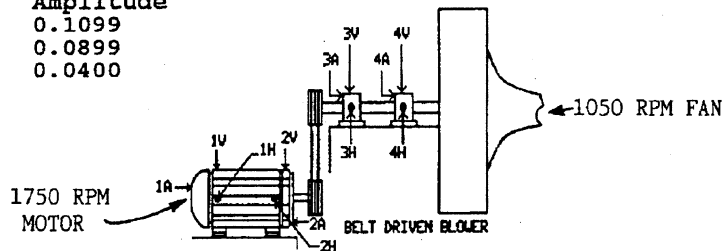
IV. RESULTS OF CORRECTIVE ACTIONS

As seen in the waterfall spectra of Figure 1 and the Position 2H Trend Graph of Figure 3, the alignment and tensioning of the belts reduced vibration significantly, but the 450 CPM vibration remained. Later, the belts were replaced resulting in the spectrum dated January 24, 1990, showing no 450 CPM vibration as noted in the frequency/order/amplitude tabular listing near the middle of Figure 1 that is also dated January 24, 1990. The Trend Graph of Figure 3 shows the improved results which occurred to the Position 2H overall vibration as a result of these corrective actions.



2 1/2 x 2.000 = 0.0400 1 x RPM: 1750.00

Machine	Pos	Dir	Units	Date
304 AIRHAND 004	2	H	IN/S	Jan 24 1990 13:58
Frequency	1 x RPM	Amplitude		
900	0.51	0.1099		
1800	1.03	0.0899		
3300	1.89	0.0400		



T - Threshold adjust Q - Quit

FIGURE 1

SUMMARY OF RESULTS AND RECOMMENDATIONS FOR MACHINES FOUND TO HAVE PROBLEMS
(SORTED FIRST BY PROBLEM PRIORITY; THEN BY PROBLEM SEVERITY WITHIN PRIORITY)

Page 1 of 4

AS OF: 3/22/89

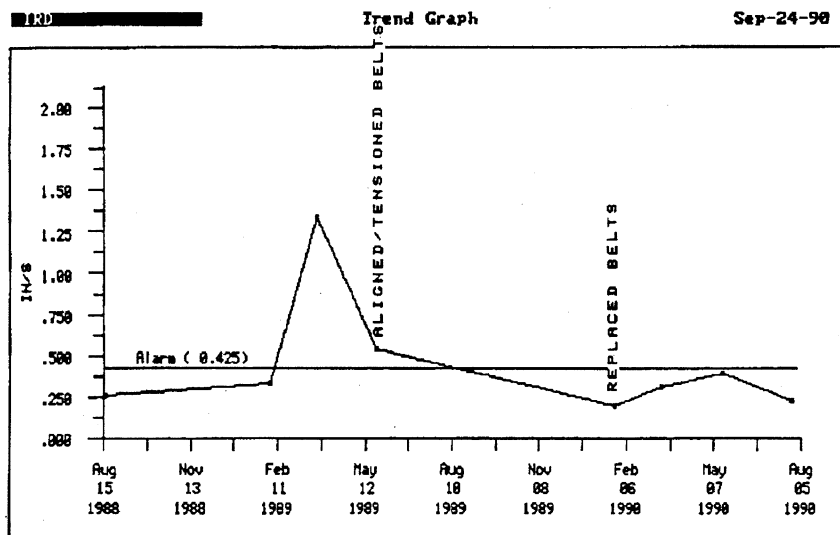
AREA	MACHINE COMPUTER NAME	INDICATED PROBLEM(S)															PRIORITY OF PROBLEM	HIGHEST OVERALL VIBRATION			RECOMMENDATIONS & REMARKS	SEE PLOT NO.	
		UNBALANCE	MISALIGNMENT/SOFT FOOT	BENT SHAFT	LUBRICATION PROBLEM	BEARING PROBLEM	MECHAN. LOOSENESS	BELT/SHAVE PROBLEM	RESONANCE PROBLEM	GEAR PROBLEM	BLADE PASSING PROB.	CAM/TIMING	MOTOR ELECTRICAL	OIL WHIRL/WARP	LOOSE FOUNDATION	VIB. MONITOR PROB.		SEAL MOTOR RUN	OTHER (SEE REMARKS)	POSITION NO.			LEVEL
																							mm = mils g = g's mm/sec
	302 RCPCELL 001	●																	3H	2.20	Extremely high amplitudes measured radially on the motor and chiller with the highest measured at position 3H (2.20 @ 1800 CPM) which indicates severe unbalance. Amplitudes have increased dramatically since September Survey as seen in Plot 37. Unbalance condition is probably caused by a balance weight being thrown off. Recommend shutting compressor off until unit is balanced.	36 37	
	304 AIRHAND 00A																		2 2H	1.34	High amplitude measured at position 2H (.929 in/sec) at 1 times belt RPM measured on the motor indicates a belt/sheave problem. Check fan sheave for excessive runout and repair/replace as needed. Align and tension belts.	40	
	324 AIRHAND 001	●																	2 2H	1.30	Motor unbalance still present at a slightly higher amplitude as seen in Plot 31. Balance motor as needed.	50 51	
	863 RCPOMP 001														●				2 2H	1.17	Possible loose or inadequate compressor foundation. Inspect compressor base for proper mounting that should include an inertie block or a vibration isolation system. Also, look for any loosened fasteners and tighten properly. Will continue to monitor closely.	89	
	302 AIRHAND 001	●																	2 4A	1.07	Belt misalignment and/or excessive sheave runout causing a high axial amplitude at position 1A combined with motor unbalance. Mechanical looseness is also indicated. Recommend checking sheaves for excessive runout and repair/replace as needed. Align and tension belts. Tighten all associated hold down bolts. Unbalance condition may be caused by sheave problem, therefore, trend this in future surveys.	33 34 35	
	318 COOLTRW 001	●																	2 2H	.913	Motor and/or Drive Shaft Unbalance, likely combined with Drive Shaft Misalignment. Check drive shaft for looseness and runout and correct as necessary.	49	

FIGURE 2

① = TAKE IMMEDIATE REPAIR ACTION DUE TO SEVERE PROBLEMS NOTED (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)
2 = SCHEDULE REPAIRS TO CORRECT PROBLEMS NOTED AT CONVENIENCE (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)
3 = DESIGNATED PROBLEM DOES NOT WARRANT ACTION AT THIS TIME (CONTINUE ONLY WITH RELIABILITY SURVEY TREND ANALYSIS)



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304 AIRHAND 004 2 H .425 IN/S

IRB Mechanalysis Data Analysis System

Machine	304 AIRHAND 004	Pos	2	Dir	H	Units	IN/S	Alarm	.425
Ampl	Insp	Date	Time	Ampl	Insp	Date	Time		
	Code	mm/dd/yy			Code	mm/dd/yy			
.255	0	08-15-88	10:50						
.335	0	02-02-89	10:09						
1.34	0	03-22-89	16:36						
.537	0	05-23-89	16:54						
.195	0	01-24-90	13:58						
.313	0	03-13-90	14:23						
.396	0	05-15-90	18:00						
.223	0	07-26-90	15:38						
.223	0	07-26-90	15:38						

FIGURE 3

EFFECT OF TIGHTENING LOOSE BELTS ON MOTOR VIBRATION

I. BACKGROUND

During a Predictive Maintenance Program data collection survey by Technical Associates' Analysts, a belt-driven air handler, 058 AIR HANDLER 002, was noted as having a loose set of belts and an unusual motor vibration of .380 in/sec at 2700 CPM. The .380 in/sec vibration violated the Band 4 alarm of .160 in/sec as seen in Plot 1. A machine diagram showing measurement locations is shown in the upper corner of the spectrum in Plot 1. The motor speed for this unit was nominally 1750 RPM and a hand-held tachometer showed the fan speed was 1530 RPM. A tachometer fan speed measurement is used to identify fan speed due to the possibility of belt slip-page and sheave wheel size changes that could affect fan speed.

Since the 2700 CPM frequency was not related to either the motor or fan speeds (1X, 2X, 3X, etc.), another source needed to be considered. The motor and sheave wheel sizes were estimated, along with their center-to-center distance. Then an approximation of the belt frequency as being around 540 CPM was calculated. This identified the 2700 CPM and other frequencies in Plot 1 as harmonics of belt frequency. A Reliability Survey Report was issued reflecting this result. These reports are given after each Reliability Survey giving a continually updated evaluation of condition of each machine included in the plant Predictive Maintenance Program.

II. CONCLUSIONS

- A. The high amplitude 2700 CPM frequency was found to be due to a 5X belt frequency vibration. The most likely cause of this high vibration as well as that by the other belt frequency harmonics was concluded to be loose belts.

III. RECOMMENDATIONS

- A. Recommendations were made to tighten the loose belts to the proper tension.
- B. It was also recommended the mechanic closely inspect both the motor and fan sheave for signs of wear which can contribute to belt looseness problems as well as accelerate wear of the belts themselves.
- C. It was also recommended the mechanic closely align the motor sheave with respect to the fan sheave.

IV. RESULTS OF CORRECTIVE ACTIONS

After the belts were properly tightened, the 2700 CPM amplitude was reduced almost 80% from .380 in/sec to .078 in/sec as seen in Plot 2. These types of improvements also result in improved performance and reduced energy costs.

CF8

R-9210-1

SUMMARY OF RESULTS AND RECOMMENDATIONS FOR MACHINES FOUND TO HAVE PROBLEMS (SORTED FIRST BY PROBLEM PRIORITY, THEN BY PROBLEM SEVERITY WITHIN PRIORITY)

AS OF: 11/29/88

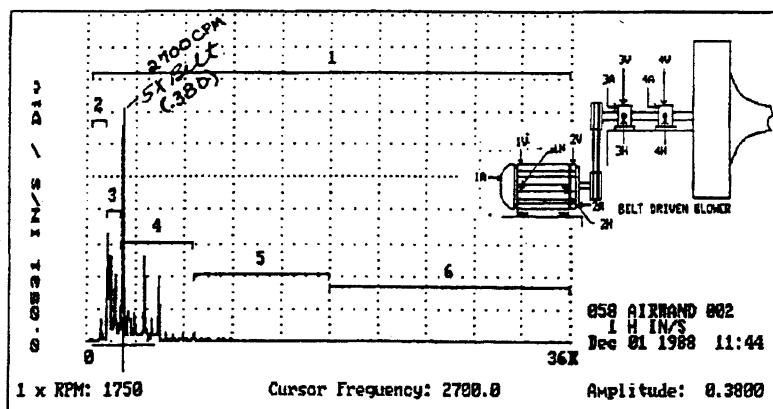
Page 2 of 2

AREA	MACHINE COMPUTER NAME	INDICATED PROBLEM(S)																PRIORITY OF PROBLEM	POSITION NO.	HIGHEST OVERALL VIBRATION			RECOMMENDATIONS & REMARKS	SEE PLOT NO.	
		UNBALANCE	MISALIGNMENT/SHOFT FOOT	BENT SHAFT	LUBRICATION PROBLEM	BEARING PROBLEM	MECHAN. LOOSENESS	BELT/SHEAVE PROBLEM	RESONANCE PROBLEM	GEAR PROBLEM	BLADE PASSING PROBL	CAVITATION	MOTOR ELECTRICAL	OIL WHIRL/WHIP	LOOSE FOUNDATION	VIB. MONITOR PROBL	SEAL/MOTOR RUB			OTHER (SEE REMARKS)					
	096 RCPCOMP 001																		3	1A	.791	Vibration at motor speed on both motor and compressor is likely caused by motor sheave eccentricity with only a slight chance of motor unbalance. Trend to determine if vibration is increasing.	75		
	092 AIRHAND 001																			3	2V	.570	Motor unbalance, possibly caused by sheave, results in moderately high vibration. Trend only.	69	
	941 AIRHAND 001																			3	3A	.556	A 2 X motor speed excitation at fan bearing caused a band 4 alarm of modest amplitude. Trend only.	77	
	058 AIRHAND 002																			3	1H	.549	A belt harmonic (5 X Belt) caused a band 4 alarm at position 1H is likely caused by loose belts. Trend only at this time.	68	
	058 AIRHAND 001																			3	2H	.479	Motor sheave unbalance. Trend only.	67	

① = TAKE IMMEDIATE REPAIR ACTION DUE TO SEVERE PROBLEMS (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)
2 = SCHEDULE REPAIRS TO CORRECT PROBLEMS NOTED AT CONVENIENCE. (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)
3 = DESIGNATED PROBLEM DOES NOT WARRANT ACTION AT THIS TIME. (CONTINUE ONLY WITH RELIABILITY SURVEY TREND ANALYSIS)

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FIGURE 1

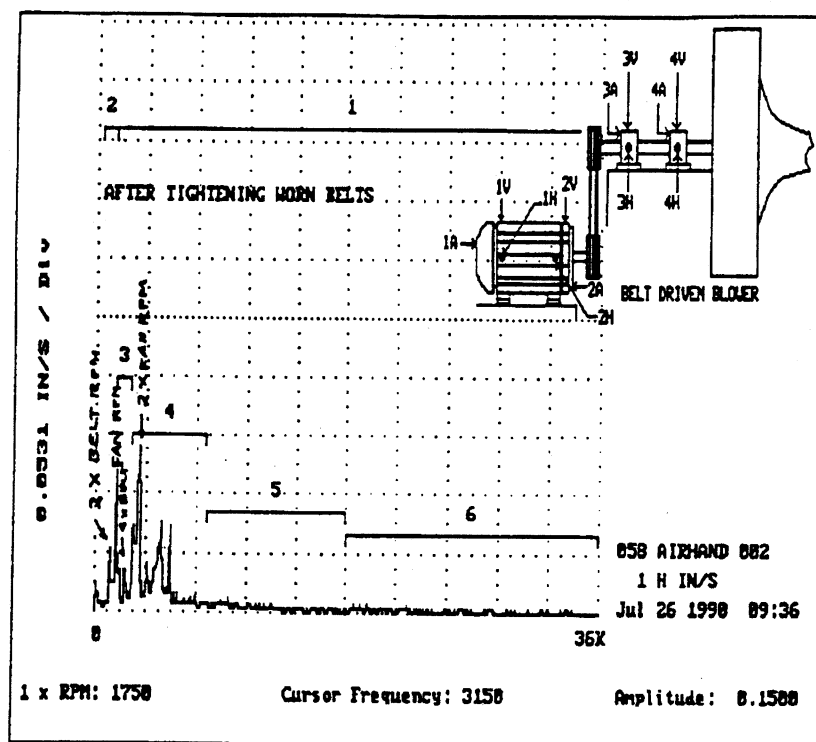


1% x 1.000 = 0.0100 1 x RPM: 1750.00

Machine	Pos	Dir	Units	Date
058 AIRHAND 002	1	H	IN/S	Dec 01 1988 11:44
Frequency	1 x RPM	Amplitude		
1080.0	0.617	0.0349 - 2X BELT		
1530.0	0.874	0.1751 - FAN		
1800.0	1.029	0.1400 - MOTOR		
2160.0	1.234	0.1099 - 4X BELT		
2700.0	1.543	0.3800 - 5X BELT		
3060.0	1.749	0.0500		
3510.0	2.006	0.0450		
4230.0	2.417	0.1400		
4770.0	2.726	0.0349		
5310.0	3.034	0.1050		
5850.0	3.343	0.0149		
6390.0	3.651	0.0100		
7110.0	4.063	0.0100		
8010.0	4.577	0.0149		

T - Threshold adjust Q - Quit

PLOT 1



2 1/2 x 0.500 = 0.0100 1 x RPM: 1750.00

Machine	Pos	Dir	Units	Date	
058 AIRHAND 002	1	H	IN/S	Jul 26 1990 09:36	
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude
90	0.051	0.0250	9540	5.451	0.0125
1080	0.617	0.0575 = 2X BELT			
1530	0.874	0.1250 = FAN RPM			
1800	1.029	0.0375 = MOTOR RPM			
2160	1.234	0.0375 = 4X BELT			
2700	1.543	0.0775 = 5X BELT (WAS .380 in/sec)			
3150	1.800	0.1500 = 2X FAN			
3690	2.109	0.0425			
4680	2.674	0.0800			
5040	2.880	0.0200			
5310	3.034	0.0750			
5580	3.189	0.0150			
5850	3.343	0.0150			
6390	3.651	0.0100			
6840	3.909	0.0125			
7200	4.114	0.0150			
9000	5.143	0.0100			

T - Threshold adjust Q - Quit

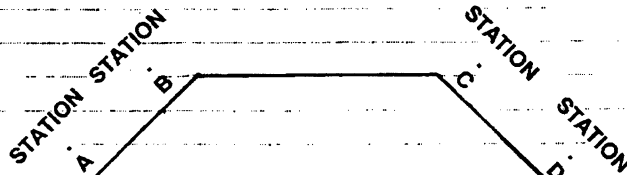
PLOT 2

Technical Associates of Charlotte, Inc.
347 N. Caswell Rd.
CHARLOTTE, NORTH CAROLINA 28204
(704) 333-9011

JOB _____
SHEET NO. _____ OF 100
CALCULATED BY _____ DATE _____
CHECKED BY _____ DATE _____
SCALE _____

WALTER LIEBIG 2A,2B,2C, & 2D

EXCESSIVE MACHINE TOOL SPINDLE VIBRATION
DUE TO BELT FLUTTER



GENERAL ARRANGEMENT SHOWING 4 STATIONS

VIEW OF A SINGLE STATION OF FOUR

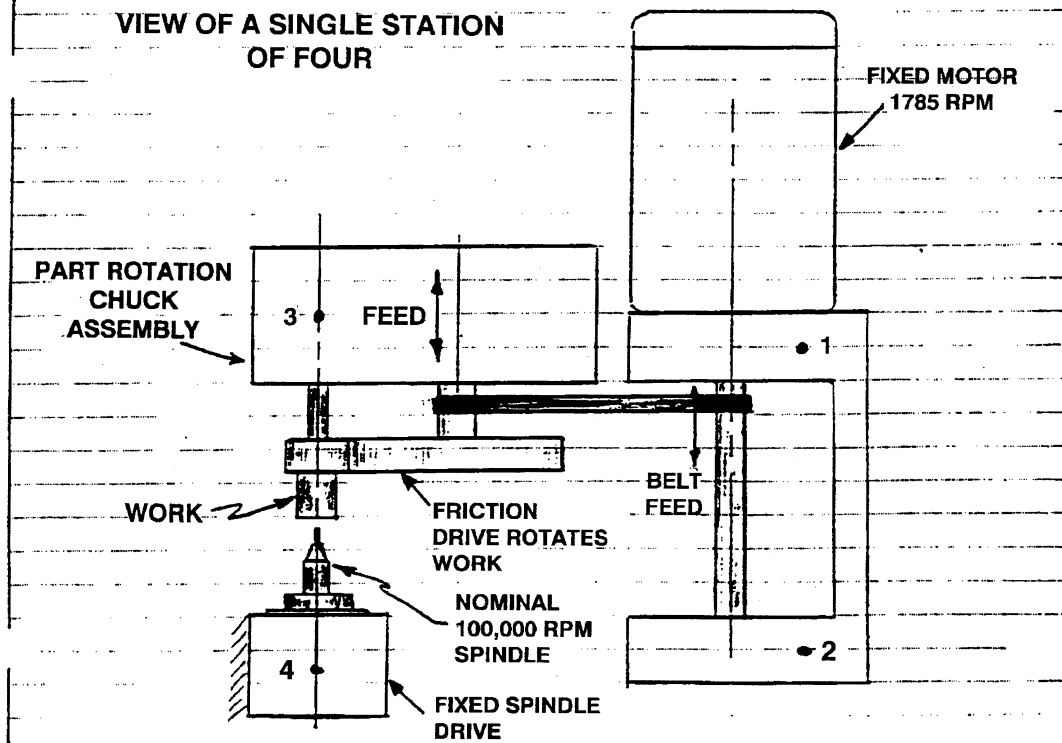


FIGURE 1

BMT-07

SUMMARY OF RESULTS AND RECOMMENDATIONS FOR MACHINES FOUND TO HAVE PROBLEMS
(SORTED FIRST BY PROBLEM PRIORITY, THEN BY PROBLEM SEVERITY WITHIN PRIORITY)

AS OF: MAY 1, 1990

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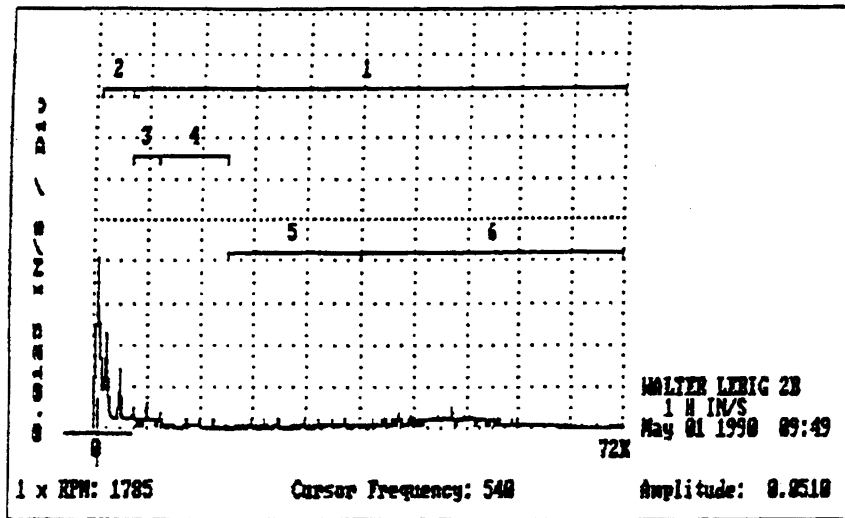
AREA	MACHINE COMPUTER NAME	INDICATED PROBLEM(S)																	HIGHEST OVERALL VIBRATION	RECOMMENDATIONS & REMARKS	SEE PLOT NO.			
		UNBALANCE	MISALIGNMENT/SHOFT FOOT	BENT SHAFT	LUBRICATION PROBLEM	BEARING PROBLEM	MECHAN. LOOSENESS	BELT/SHAVE PROBLEM	RESONANCE PROBLEM	GEAR PROBLEM	BLADE PASSING PROB.	CAVITATION	MOTOR ELECTRICAL	OIL WHIRL/WHIP	LOOSE FOUNDATION	VIB. MONITOR PROB.	SEAL MOTOR RUB	OTHER (SEE REMARKS)						
	EVI ALTIME 2 HP																	3	2271	Multiple harmonics of blade passing frequency (12,600) measured on the pump indicating a slight flow related problem. Recommend checking filters and associated piping for obstructions. Also confirm that gap between impeller vanes stationary diffuser vanes remain constant all the way around periphery.	28			
	COND WTR VAMP22																	3	1AA	Possible slight resonance due to loose foundation. Check foundation bolts for looseness. Trend in future surveys.	20			
	WALTER IEBIG 28																	3	1H 2H	Belt flutter indicated at Positions 1H and 2H (see Plots 9 and 10). Belt was physically observed moving up and down on shaft, causing vibration to increase. Check belt drive and correct as required. Note difference in vibration in WALTER IEBIG 28 & 2D as seen in Plot 11 at the same Position 2H. Trend in future surveys.	9 10 11			
	COOL TUR VAMP21																	3	1A-A	Slightly high vibration at 9 times RPM (16050). Possible blade pass and/or loose foundation. Trend in future surveys. Also, please confirm the number of vanes on impeller and contact Technical Associates.	22			
	44 YORK CHILLER																	3	2H 2H 2H	Incipient wear of inboard motor bearing indicated by increasing level of spike energy which has now exceeded alarm. Closely watch in future surveys and check bearing for proper lubrication.	4			
	SEIRO SET1 1225																	3	3V 4A 7V	Slight motor electrical problem causing an amplitude of .083 and .087 in/sec at 7200 CPM measured at Positions 3V and 4A respectively, on Motor #2. Trend in future surveys. Also, due to the summation of overall vibration it is causing a Band 1 alarm. Trend in future surveys.	29 30 31			

(1) - TAKE IMMEDIATE REPAIR ACTION DUE TO SEVERE PROBLEM(S) NOTED (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)
 2 - SCHEDULE REPAIRS TO CORRECT PROBLEMS NOTED AT CONVENIENCE (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)
 3 - DESIGNATED PROBLEM DOES NOT WARRANT ACTION AT THIS TIME (CONTINUE ONLY WITH RELIABILITY SURVEY TREND ANALYSIS)



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FIGURE 2

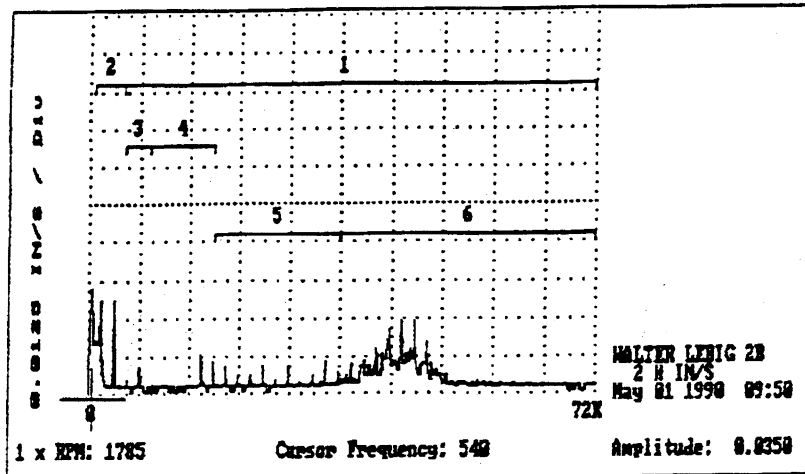


002% x 0.200 = 0.0040 1 x RPM: 1785.00

Machine	Pos	Dir	Units	Date
WALTER LEBIG 2B	1	H	IN/S	May 01 1990 09:4
Frequency	1 x RPM	Amplitude		
540	0.30	0.0510		
1800	1.01	0.0290		
3600	2.02	0.0180		
5400	3.03	0.0060		
7200	4.03	0.0080		
9000	5.04	0.0050		
41220	23.09	0.0050		
42840	24.00	0.0040		
46440	26.02	0.0040		
48240	27.03	0.0060		
50040	28.03	0.0040		
50760	28.44	0.0040		

T - Threshold adjust Q - Quit

PLOT 1

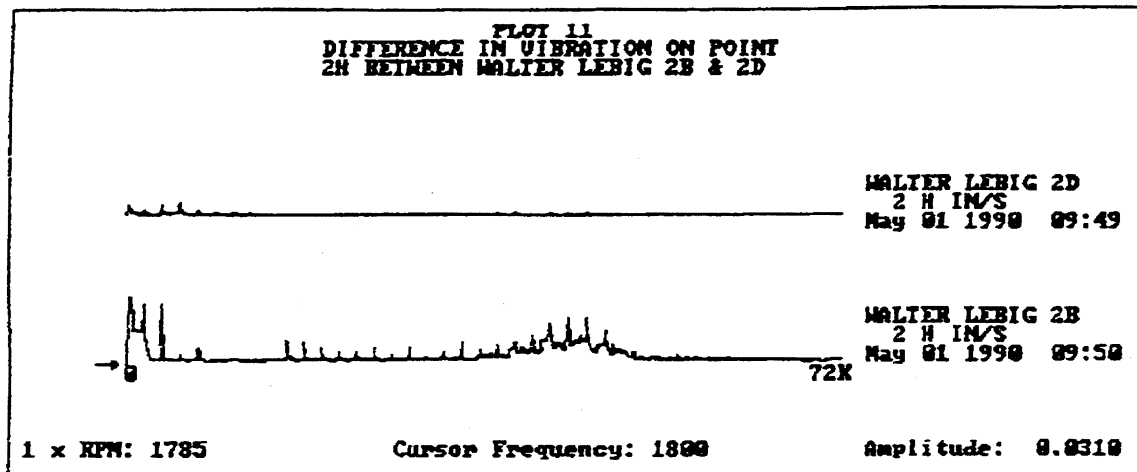


002% x 0.200 = 0.0040 1 x RPM: 1785.00

Machine	Pos	Dir	Units	Date	
WALTER LEBIG 2B	2	H	IN/S	May 01 1990 09:5	
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude
540	0.30	0.0350	39420	22.08	0.0110
1080	0.61	0.0180	40140	22.49	0.0090
1800	1.01	0.0310	41040	22.99	0.0150
3600	2.02	0.0310	41940	23.50	0.0130
7200	4.03	0.0090	42840	24.00	0.0210
16020	8.97	0.0130	43380	24.30	0.0110
17820	9.98	0.0110	43740	24.50	0.0110
19620	10.99	0.0090	44640	25.01	0.0240
21420	12.00	0.0060	45360	25.41	0.0130
23220	13.01	0.0060	45900	25.71	0.0150
25020	14.02	0.0090	46440	26.02	0.0240
28620	16.03	0.0090	48240	27.03	0.0170
32220	18.05	0.0060	48960	27.43	0.0100
34020	19.06	0.0110	49680	27.83	0.0080
35820	20.07	0.0070	51120	28.64	0.0060
36900	20.67	0.0060			
37620	21.08	0.0100			

T - Threshold adjust Q - Quit

PLOT 2



PLOT 3